

PDE Ledger
Archive Ledger Skeleton

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July 10, 2026

Purpose and Scope

TODO. B3 will fill the rebuilt ledger purpose, source hierarchy, and scope boundaries.

0.1 The Role of This Ledger

Placeholder for the rebuilt ledger's role and audience.

0.2 Primary Source Hierarchy

Placeholder for the source hierarchy used by the rebuilt ledger.

0.3 Document Contract

Placeholder for the provenance, status, verification, and downstream-use contract.

0.4 Scope Boundaries

Placeholder for the limits of the rebuilt ledger.

0.5 Completeness Criterion

Placeholder for the completion standard used by later build phases.

How to Read This Ledger

TODO. B3 will fill the reader guide after the rebuilt stage and Part structure is set.

0.6 Layer Structure

Placeholder for the relationship between theorem-block prose, stage cards, notes, and audits.

0.7 Reading Paths

Placeholder for operational, verification, provenance, and authoring reading paths.

0.8 Citation Discipline

Placeholder for result anchors, theorem labels, equation labels, and stage provenance.

0.9 Stage Entries

Placeholder for the expected fields in future stage cards.

0.10 Gap Handling

Placeholder for demotion, splitting, open-gate, and failed-route handling.

Claim-Status Firewall

TODO. B3 will fill the rebuilt ledger's status rules and examples.

0.11 Allowed Status Tags

Status tag	Meaning
EXACT	Placeholder.
EXACT WITHIN CLOSURE	Placeholder.
REDUCED / CONTROLLED REDUCTION	Placeholder.
EFFECTIVE CLOSURE	Placeholder.
NUMERICALLY LOCATED	Placeholder.
OPEN	Placeholder.

0.12 Audit-Status Vocabulary

Placeholder for audit evidence categories.

0.13 Status Inheritance

Placeholder for weakest-essential-input and status propagation rules.

0.14 Citation Language

Placeholder for safe status-aware citation wording.

Notation Firewall

TODO. B3 will fill notation rules after the rebuilt sectors and stage packets are fixed.

0.15 Global Anti-Ambiguity Rules

Placeholder for global notation safeguards.

0.16 Coordinate, Index, and Operator Conventions

Placeholder for coordinate, index, and operator conventions.

0.17 Field and Sector Variables

Placeholder for sector-specific variable conventions.

0.18 High-Risk Symbol Collision Table

Placeholder for symbol collision accounting.

0.19 Notation Checklist

Placeholder for future stage write-up checks.

Result Anchor Map

TODO. B3 will assign rebuilt result anchors after the Part and stage plan is settled.

0.20 Anchor Use Rules

Placeholder for stable citation-anchor rules.

0.21 Anchor Naming Convention

Placeholder for rebuilt anchor naming.

0.22 Top-Level Result Anchors

Anchor	Stage range	Citation target
No anchors assigned yet.		

0.23 Maintenance Checklist

Placeholder for anchor maintenance rules.

Dependency Map

TODO. B3 will fill the rebuilt dependency graph after stages are authored.

0.24 Arrow Types

Placeholder for dependency-arrow types.

0.25 Global Dependency Spine

Placeholder for the rebuilt ledger's dependency spine.

0.26 Top-Level Dependency Table

Block	Inputs	Outputs
No dependency blocks assigned yet.		

0.27 Revision Rules

Placeholder for dependency-map revision rules.

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Part I

Part I – The medium

Chapter 1

Medium Ledger Stages

Stage cards for Part I are collected in Appendix A.

Part II

Part II – Gravity

Chapter 2

Gravity Ledger Stages

Stage cards for Part II are collected in Appendix B.

Part III

Part III – Light

Chapter 3

Light Ledger Stages

Stage cards for Part III are collected in Appendix C.

Appendix A

Part I Stage Cards

A.1 Stage 004: GNLS Action and Dimensional Foundation

Part / anchor. Part I – The medium

Claim status. EXACT WITHIN CLOSURE

Purpose. Fix the field content and $\{L, T, M\}$ dimensional dictionary of the GNLS parent action, derive the pin null-relation and healing scale, and land the base-system verdict `RETAIN_L_T_M`.

Status. EXACT WITHIN CLOSURE. The dimensional dictionary, sound-speed and healing anchors, pin null-relation, and base-system verdict are exact symbolic results; the honest landing carries named residuals (mass non-emergence and the $\{L, T\}$ -conversion gaps), deferred to pathA_21/22 and not repaired here.

Parent action and field content. The 4D-bulk medium is a GNLS condensate with order parameter ψ , number density $\rho = |\psi|^2$, and EOS $P = K\rho^5$ (from $U(\rho) = \frac{K}{4}\rho^5$):

$$\mathcal{L} \sim i\hbar \psi^* \partial_t \psi - \frac{\hbar^2}{2m_{\text{GNLS}}} |\nabla \psi|^2 - U(\rho).$$

Dimensional dictionary (two tiers). **Primitive inputs.** $[\hbar] = ML^2T^{-1}$, $[m_{\text{GNLS}}] = M$, $[\rho_0] = L^{-4}$ (state datum) — posted and consistency-checked against action usage. **Derived by composition.** $[\psi] = L^{-2}$, $[\rho] = L^{-4}$, $[K] = ML^{18}T^{-2}$, forced by action homogeneity across all 17 harness checks (14 foundation, including the bulk continuity equation, plus 3 $\{L, T\}$ -representation gates).

Derived anchors. Sound speed and healing length.

$$c_{s0}^2 = \frac{5K\rho_0^4}{m_{\text{GNLS}}}, \quad [c_{s0}] = LT^{-1}; \quad h_0 = \frac{5K}{4}\rho_0^4 = \frac{m_{\text{GNLS}}c_{s0}^2}{4}, \quad \xi_h = \frac{\sqrt{2}\hbar}{m_{\text{GNLS}}c_{s0}}.$$

Pin null-relation. Four pins $\{a, c_{s0}, \hbar, m_{\text{GNLS}}\}$ on three base dimensions give rank 3, nullity 1:

$$a = \frac{\hbar}{m_{\text{GNLS}}c_{s0}}, \quad a/\xi_h = 1/\sqrt{2}.$$

Verdict. **RETAIN_L_T_M.** Computed from two conjuncts: the $\{L, T\}$ -rejection gate (dropping M leaves ≥ 1 non-dimensionless target, needing M -bearing conversion factors LT^2M , L^2T^2M , L^2T^2) and the mass fork (m_{GNLS} explicit of dimension M ; m_{defect} not derived, **INFLOW_MASS_SOURCE_MISSING**). The $\{L, T\}$ projection is a natural-unit representation only, not a base-system change.

Carried residuals (first-class, not repaired). **INFLOW_MASS_SOURCE_MISSING** (m_{defect} emergence deferred to pathA_21; $\hbar J/c_\gamma^2 = M$ is dimensional-only), **LT_R_norm_gate_fails_without_new_conversion_factor** (**REJECTS_TRUE_LT_BASE**) and the two 4D/3D R -norm conversion gaps (pathA_22), plus **A_PIN_IS_BRANCH_MOMENT_NO**, **EOS_FROM_GNLS_FACTOR**, **NO_NET_ACCRETION_BC_UNDERIVED**, **M_TO_G_UNIFICATION**, **SCALE_MAP_INPUTS**.

A.2 Stage 005: EOS Sound Speed and Light/Sound Ratio as a Free Knob

Part / anchor. Part I – The medium

Claim status. REDUCED / CONTROLLED REDUCTION

Purpose. Derive the phonon sound speed from the imposed EOS, pin the three velocity scales and the $c = c_\gamma$ wave-sector ceiling, and land the light/sound ratio $\lambda_\gamma = c_\gamma/c_s$ as a free calibration input (**C_GAMMA_RATIO_UNDERDETERMINED**).

Status. REDUCED / CONTROLLED REDUCTION. The sound-speed law and the $c = c_\gamma$ ceiling are exact symbolic results *within the imposed EOS closure*; the honest headline is the **CALIBRATED** landing: $\lambda_\gamma = c_\gamma/c_s$ is *unpinned by the parent action* – a free calibration input, with the tail $(c/c_s)^3 = \lambda_\gamma^3$ carried symbolic (not set to 1). Consuming stage: it cites **ledger_stage004** (I-1)’s dictionary and **EOS_FROM_GNLS_FACTOR** handoff (h_0, ξ_h) and owns the sound-speed derivation I-1 deferred here.

Sound speed from the EOS. From the imposed $P = K\rho^5$ (**EOS_CLOSURE_IMPOSED**) and the barotropic definition, by symbolic differentiation:

$$c_s^2(\rho) = \frac{1}{m_{\text{GNLS}}} \frac{dP}{d\rho} = \frac{5K\rho^4}{m_{\text{GNLS}}}, \quad [c_s] = LT^{-1}, \quad \frac{d \ln c_s}{d \ln \rho} = 2.$$

The factor 5 falls out of the derivative. **EARNED** relative to the imposed EOS.

Three speeds and the $c = c_\gamma$ ceiling. Velocity scales. $v_b = (\hbar/m_{\text{GNLS}})\nabla\theta$, c_s , c_γ , each of dimension LT^{-1} (v_b ’s pin consumed from I-1). **Ceiling (derived from the wave sector).** For the trapped mode $\omega = c_\gamma\sqrt{k^2 + k_\perp^2}$,

$$\frac{d\omega}{dk} = \frac{c_\gamma k}{\sqrt{k^2 + k_\perp^2}}, \quad c_\gamma^2 - \left(\frac{d\omega}{dk}\right)^2 = \frac{c_\gamma^2 k_\perp^2}{k^2 + k_\perp^2} \geq 0,$$

strictly positive for $k_\perp \neq 0$: the group speed is bounded by c_γ and approaches it. The bound-mode clock $\omega_0 = c_\gamma k_\perp$ gives the boosted internal rate ω_0/γ (time dilation) with no $E = mc^2$ premise.

Coupled principal symbol (pathA_20b support). On the neutralized homogeneous background the coupled symbol block-diagonalizes: the Madelung phonon determinant $\det M = \hbar(\omega^2 - c_{s0}^2 k^2)$ with $c_{s0}^2 = 5K\rho_0^4/m_{\text{GNLS}}$ falling out via $h' = 5K\rho_0^3$, the gauge block $C_E\omega^2 - C_B k^2 = C_E(\omega^2 - c_{\text{bulk}}^2 k^2)$ ($c_{\text{bulk}}^2 = C_B/C_E$), and $\det P = \det M \cdot P_T^2$. The off-diagonal current/London terms are lower order and do not set the cone (BULK_PRINCIPAL_TRANSVERSE_BRANCH_ESTABLISHED); $[c_s] = [c_\gamma]$ is labeled *non-evidentiary* for equality.

The free-knob landing and its negative control. C_GAMMA_RATIO_UNDERDETERMINED.

Computed, not asserted. With $C_E, C_B, K, \rho_0, m_{\text{GNLS}}$ independent, the residual $C_B/C_E - 5K\rho_0^4/m_{\text{GNLS}}$ is not identically zero and no source equation is present, so `forced_equals_valid = False` $\Rightarrow$ `FORCED_EQUALITY_REJECTED_WITHOUT_SOURCE_EQUATION`. It is *reversible*: inserting the identity $C_B \rightarrow 5K\rho_0^4 C_E/m_{\text{GNLS}}$ flips it to `C_GAMMA_EQUALS_C_S` – so `EQUALS` is reachable only via an actually-inserted source equation the action does not supply. Two-layer sharpening: bulk `C_GAMMA_BULK_UNDERDETERMINED`, brane `C_GAMMA_RATIO_STILL_UNDERDETERMINED`; if c_{bulk} is ρ_0 -independent then $\lambda_\gamma \propto \rho_0^{-2}$ ($d \ln \lambda_\gamma / d \ln \rho_0 = -2$).

Carried residuals (first-class, not repaired). `EOS_CLOSURE_IMPOSED`; `C_GAMMA_RATIO_UNDERDETERMINED`; `STATIONARY_PROFILE_UNDERDETERMINED_BY_BRANCH_DATA`; `NO_NET_ACCRETION_BC_UNDERIVED`; `HBAR_PROVENANCE_U`; `HBAR_FREE_SUBSTRATE_RELATION_MISSING`; `H_2PI_RATE_CLASSIFICATION_UNDERDETERMINED`; `BULK_METRIC_SPEED`; `PARENT_METRIC_ACOUSTIC_IDENTIFICATION_MISSING`; `BRANE_ZERO_MODE_REDUCTION_UNDERIVED`; `BRANE_PHOTON_CONE_REQUIRES_PROFILE`. The mass bridge $\hbar J/c_\gamma^2 = M$ is a labeled dimensional candidate only, not a m_{defect} derivation.

A.3 Stage 006: Two-Phase Material-State Ontology (Order Field χ_B)

Part / anchor. Part I – The medium

Claim status. EXACT WITHIN CLOSURE

Purpose. Formalize the brane/bulk ontology as a POSTULATED two-phase action – one conserved medium, order field $\chi_B \in [0, 1]$; brane = ordered shear-supporting phase, bulk = the same medium de-structured, throat = phase-conversion site – dimensionally classified (`ACTION_SPECIFIED_CLASSIFIED`), with the recovery reduction to the frozen old-ledger S_{leak} verified assert-zero and the θ -as-Maxwell- φ no-go carried able-to-fail.

Status. EXACT WITHIN CLOSURE. **The wall is made explicit as a postulated field – it is NOT derived from the one medium.** Every term of the free-energy functional and order-state balance is POSTULATED and labeled (13 pinned modeling choices P1–P13); the EARNED content is exact *within that imposed closure*: structural-closure identities, single-kink wall admission, and the recovery reduction against the frozen firewall. Cross-engine agreement proves consistent encoding of the same postulated F , not derivation. Consuming stage: cites `ledger_stage004` (dictionary, single-well U), `ledger_stage005` (c_{s0}^2), and `ledger_stage003` ($c_\gamma^2 = \mu_R/\rho_{br}$, $C_J = -J\rho_{B0}$, $B_{\text{eff}} = \rho_{B0}^2/\chi_c$, the second-class $\Rightarrow$ stray-DOF classification rule) with exact-value citation-integrity checks.

The postulated action and balance.

$$F = \int d^4X \left[\frac{1}{2} m_{\text{GNLS}} n |\mathbf{u}|^2 + U(n) + f_B(\chi_B) + \frac{\kappa_B}{2} |\nabla_4 \chi_B|^2 + \chi_B f_{\text{shear}} + f_{\text{throat}} + f_{\text{mix}} \right],$$

$$\partial_t n + \nabla_4 \cdot (n \mathbf{u}) = 0, \quad \partial_t (\chi_B n) + \nabla_4 \cdot (\chi_B n \mathbf{u} + \mathbf{J}_\chi) = n \Gamma_B, \quad [\Gamma_B] = T^{-1}.$$

Key pins: kinetic term carries the constituent mass (KINETIC_MASS_FACTOR_PINNED; $[n] = L^{-4}$); double-well $f_B = a_B \chi_B^2 (1 - \chi_B)^2$ with minima $\{0, 1\}$, n -independent (SAME_DENSITY_DEGENERACY_POSTULATED) – required because the cited $U(\rho) = (K/4)\rho^5$ is *single*-well; MacCullagh-form shear gate on the displacement curl; $\chi_B \perp P$ (POSTULATED_ANISOTROPY dims-only); $\mathbf{J}_\chi = 0$ default; global returns $R_0 = -M_0$, $R_1 = -D_1$ printed as postulates, not asserted locally; throat = phase-conversion ontology, not a solve. Every integrand term audited to $ML^{-2}T^{-2}$; adjunct rows ($\mu_\chi, M_\chi, P_{\text{order}}, M_n$) dimension-pinned, with the energy ledger fixed to the variationally consistent $P_{\text{order}} = \int \mu_\chi D_t \chi_B d^4 X$ (HANDOFF_P_ORDER_N_PLACEMENT_CORRECTED).

Structural closure and wall admission (EARNED within closure). The advective form $D_t \chi_B = \Gamma_B - (1/n) \nabla_4 \cdot \mathbf{J}_\chi$, the disorder complement for $n_D = (1 - \chi_B)n$, and the Γ_B -cancelling sum reproducing total conservation are derived by real derivative expansion in both engines. The static EL equation $\kappa_B \chi_B'' = f_B'$ admits the kink $\chi_B = \frac{1}{2}(1 + \tanh(w/2\delta))$ with solved width $\delta = \sqrt{\kappa_B/2a_B}$ and surface tension $\sigma_{\text{wall}} = \sqrt{2a_B \kappa_B}/6$, $[\sigma_{\text{wall}}] = ML^{-1}T^{-2}$. **Slab caveat (carried).** Bulk on both sides makes the brane a finite kink–antikink *slab*; the double-well selects no slab width – W_{slab} is an un-earned input mapping onto the old ledger’s L/a self-selection item (sim-deferred). Wall admission $\neq$ slab stability.

Recovery reduction (EARNED within closure; the frozen firewall). Projection with unit-normalized $W(w)$ and in-engine IBP gives the two-source law $\partial_t \rho_B + \nabla_3 \cdot \mathbf{J}_B = S_{\text{flux}} + S_{\text{convert}}$ with the general $\chi_B(w), \Gamma_B(w), J_\chi^w(w)$ computationally live; the in-engine limit $\chi_B \rightarrow 1, \Gamma_B \rightarrow 0, \mathbf{J}_\chi \rightarrow 0$ reduces it assert-zero to the *independently constructed* frozen target $S_{\text{leak}} = -[W j^w]_\partial + \int W' j^w dw$, $j^w = n u^w$ (stage_243), including the stage_244 Gaussian one-mode closed form $S_{\text{leak}} = \sqrt{2} \mu_w \rho_0 \mathcal{E}_0 / (2\sqrt{\pi} \lambda^3)$ re-derived by direct integration (RECOVERY_REDUCTION_VERIFIED). Conditionality computed: $\chi_B = c \neq 1$, $\Gamma_B \neq 0$, $J_\chi^w \neq 0$ each leave nonzero computed residuals – the limit does real work.

The θ -as-Maxwell- φ no-go (carried FATAL_FLAW). A stable order-parameter phase has positive gradient energy ($\kappa_{\text{phase}} > 0$) $\Rightarrow K_\theta = -\kappa_{\text{phase}} < 0$ in the Lagrangian; Maxwell’s electric square needs $K_\theta = C_J^2/\rho_{br} > 0$ – opposite signs (exact predicate). The constraint bracket $\{\Phi_1, \Phi_2\} = k^2(J^2 \rho_{B0}^2 + \kappa_{\text{phase}} \rho_{br})/\rho_{br}$ has its zero locus solved in-engine at exactly $K_\theta = +J^2 \rho_{B0}^2/\rho_{br}$: via the consumed stage_003 rule the provenance branch lands FAIL_CAUCHY_STRAY_LONGITUDINAL (finite $B_{\text{eff}} = \rho_{B0}^2/\chi_c$), the $B_{\text{eff}} = 0$ branch FAIL_C5_LONGITUDINAL_ZERO_MODE, and the tuned fixture ($K_\theta = J^2 \rho_{B0}^2/\rho_{br}, B_{\text{eff}} = 0, m_\theta^2 = 0$) C5_RESOLVED_MAXWELL_BY_TUNING – flagged BY_TUNING, never WITH_PROVENANCE (flag computed from provenance booleans; detunings re-fire the pathA_36 control tokens). The only provenance sign-flip ($\kappa_{\text{phase}} < 0$) is a Lifshitz instability, $k_*^2 = -\kappa_{\text{phase}}/2\kappa_4 > 0 \iff \kappa_{\text{phase}} < 0$ – the already-killed pathA_25 wall (FAIL_LIGHT_STARVED lineage). The transverse sector is untouched: the ε -parametrized dispersion derived in-engine gives $\omega^2 = (\mu_R/\varepsilon)k^2$; $\varepsilon = \rho_{br}$ reproduces the consumed c_γ^2 (PASS_TRANSVERSE_UNDISTURBED), $\varepsilon \neq \rho_{br}$ fires FAIL_TRANSVERSE_DISTURBED. **This stage does not earn light** – the no-go is the point.

Drift and carried items (first-class, not repaired). DRIFT(6) computed: $\{\chi_B; a_B; \kappa_B; \alpha_{\text{aniso}}; \Gamma_B$; gating struct (rung-W reconciliation printed; THETA_BRANCH_DEAD_NOT_ADMITTED; ρ_{B0}, χ_c cross-referenced to the pathA_41 Part-VI drift trio, not double-counted). Carried: SECOND_MEDIUM_DRIFT lineage, the slab width W_{slab} , Gate-L $\delta w = u_w$ translation Goldstone (deferred), $\mathbf{J}_\chi \neq 0$, $f_{\text{throat}}/f_{\text{mix}}$ placeholders, dynamics/energy-ledger adjunct. Two source-prose dimensional defects pinned and ablation-documented (kinetic mass factor; P_{order} n -placement).

A.4 Stage 007: Shear-Surface G0 Freeze (Frozen Medium Action, Drift Ledger, DOF)

Part / anchor. Part I – The medium (closing Part-I stage)

Claim status. EXACT WITHIN CLOSURE

Purpose. Formal home of the pathA_35 G0 shear-surface *constitutive* freeze: the frozen brane/light action ($L_{\text{Mac}} + L_{Pu} + L_{uw}$ under the Gaussian profile $g_\ell(w)$), hash-guarded (T0_SHEAR_FROZEN(d9520d3819c3f7...), with the honest drift ledger SECOND_MEDIUM_DRIFT_AT_FREEZE(11) computed from an enumerated table and the flat-brane linear DOF = 8 rank-computed.

Status. EXACT WITHIN CLOSURE. **The 11 freeze inputs are POSTULATED/CALIBRATED – the freeze freezes *terms*, not gate answers.** The EARNED content is exact within that declared closure: (i) *freeze fidelity* – both engines re-extract the canonical **freeze-action** blocks by byte-exact fence-parsing (never fixed offsets) and assert SHA-256 equality (G0 d9520d3819c3f7..., T0 8fa41ac51e88... byte-embedded); (ii) the *flat-brane linear DOF* = 8 computed from rank/nullity bookkeeping (in-plane u^a curl-transverse 2 + kinetic-curl 1; T0 P^i tangent 3 + radial 1; u_w 1; C5 φ 0 – the absent longitudinal partner the later C5 crux attacks); (iii) the *dimensional firewall* over every frozen term. This stage computes **no** gate verdict (Gate-L excluded; exposure names carried as prose provenance only) and does **not** earn light (Stage 003 derives the two transverse photons at $c_\gamma^2 = \mu_R/\rho_{br}$ from this stage's frozen L_{Mac}).

The frozen action. Kept (0 new): the GNLS parent action and the T0 polar-OP action (byte-for-byte, hash-embedded). Frozen brane blocks under $S_{\text{brane}} = \int dt d^4X g_\ell(w) [L_{\text{Mac}} + L_{Pu} + L_{uw}]$:

$$L_{\text{Mac}} = \frac{1}{2}\rho_{br}(\partial_t u^a)^2 - \frac{1}{2}\mu_R \Omega_u^a \Omega_u^a, \quad L_{Pu} = -\lambda_{Pu} \varpi_a \Omega_u^a, \quad \varpi_a := (\hat{w} \times P_\parallel)_a,$$

$$L_{uw} = \frac{1}{2}\rho_{br}(\partial_t u_w)^2 - \frac{1}{2}\rho_{br}\Omega_w^2 u_w^2, \quad g_\ell(w) = \frac{e^{-(w/\ell_g)^2}}{\sqrt{\pi}\ell_g}, \quad \int g_\ell dw = 1 \text{ (derived)}.$$

Preserved verbatim: L_{Pu} *re-admits the ϵ -contracted/chiral class excluded by T0* and requires the conceded axis $\hat{w}$ – a structural-postulate cost, not a free choice.

The drift ledger (computed). Both engines enumerate the 11 and *compute* the subcounts and the verdict string (the source gate's hardcoded literals are retired): **4 constants** $\rho_{br} [ML^{-3}]$, $\mu_R [ML^{-1}T^{-2}]$, $\lambda_{Pu} [ML^{-1}T^{-2}]$, $\Omega_w [T^{-1}]$ (table dims asserted against the firewall targets); **1 function** $g_\ell(w; \ell_g)$ (fixed Gaussian, one width, target-blind admission); **6 structural postulates** (conceded $\hat{w}$ +wall; free-slip u^a ; P^i reused as Cosserat reservoir; massless spin-wave baseline; the parity-even P - u operator; no C5 φ). New-field content (u^a 3 + u_w 1 = 4 DOF) is counted *separately*. **Erratum (2026-07-04), carried first-class: the 11 STANDS** – no ρ_{br} overcount (pathA_25's $\varrho_{br}[\rho]$ belongs to the closed density-smectic candidate, OUT_OF_ACTIVE_NG5); $\{\rho_{br}, \mu_R\}$ carry the registered-pending Route-A reduction (ROUTE_A_UNDERDETERMINED_MISSING_NONLINEAR_THROAT, register R10); the honest cross-sector drift $\{\rho_{B0}, \chi_c, C_{hu}\}$ is a Part-VI (pathA_41) item, guarded in-engine against absorption into the 11.

Supersession and firewalls. Both facts asserted: the Stage 006 χ_B wall superseded the fixed-shape $g_\ell(w)$ as the *material-state* closure, while G0 *remains* the light-sector *constitutive* freeze (Stage 003 consumes L_{Mac} as-is) – a scope split (register R21), not a reduction. Notational firewall (register R22): μ_R (3D brane, $ML^{-1}T^{-2}$) and Stage 006’s $\mu_R^{(4)}$ (4D density, $ML^{-2}T^{-2}$) are asserted dimensionally distinct, related only by the *pending* R17 projection $\mu_R = \int \chi_B \mu_R^{(4)} dw$ (dim-consistency asserted).

Verification. Dual-engine, independent routes (the source pair’s `-compare` JSON payload mirror is retired): SymPy 96 PASS / Mathematica 94 PASS, both exit 0, CWD-independent. Able-to-fail teeth all fire: the five source-gate ablations (2 dimensional, 3 DOF each recomputing $7 \neq 8$), two hash teeth (in-block byte corruption; missing fence), five enumeration teeth (drop entry; miscategorize; inject ρ_{B0} ; corrupt n pre-verdict; corrupt table dim), and the μ_R firewall tooth. Full tri-review on fresh agents: arbiter re-run, `FIDELITY_CLEAN` (block-by-block coverage diff, no dropped check), `ADVERSARIAL_CLEAN` (26-run mutation matrix: hash genuinely fence-parsed – one byte inside the block fails, one byte outside passes; dual-corruption of both engines fails both; no cross-engine comparison anywhere).

Appendix B

Part II Stage Cards

B.1 Stage 001: Solid-Angle and Second-Moment Primitives

Part / anchor. Part II – Gravity

Claim status. EXACT

Purpose. Derive the unit-sphere solid angles and normalized second moments used by the pathA_21c matter-stress force assembly.

Status. EXACT. This stage is closed geometry: no profile, normalization, or field-equation residual remains inside the stated claims.

Derivation. For $S^2 \subset \mathbb{R}^3$, take

$$n(\theta, \phi) = (\cos \theta, \sin \theta \cos \phi, \sin \theta \sin \phi), \quad dS = \sin \theta \, d\phi \, d\theta.$$

Then **Omega2 solid angle**.

$$\Omega_2 = \int_0^\pi \int_0^{2\pi} \sin \theta \, d\phi \, d\theta = 4\pi.$$

With $n_1 = \cos \theta$ and $n_2 = \sin \theta \cos \phi$, **S2 second moments**.

$$\langle n_1^2 \rangle_{S^2} = \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} \cos^2 \theta \sin \theta \, d\phi \, d\theta = \frac{1}{3}, \quad \langle n_1 n_2 \rangle_{S^2} = 0.$$

The cross moment vanishes because $\int_0^{2\pi} \cos \phi \, d\phi = 0$.

For $S^3 \subset \mathbb{R}^4$, take

$$n(\chi, \theta, \phi) = (\cos \chi, \sin \chi \cos \theta, \sin \chi \sin \theta \cos \phi, \sin \chi \sin \theta \sin \phi), \quad dS = \sin^2 \chi \sin \theta \, d\phi \, d\theta \, d\chi.$$

Then **Omega3 solid angle**.

$$\Omega_3 = \int_0^\pi \int_0^\pi \int_0^{2\pi} \sin^2 \chi \sin \theta \, d\phi \, d\theta \, d\chi = 2\pi^2.$$

With $n_1 = \cos \chi$ and $n_2 = \sin \chi \cos \theta$, **S3 second moments**.

$$\langle n_1^2 \rangle_{S^3} = \frac{1}{2\pi^2} \int_0^\pi \int_0^\pi \int_0^{2\pi} \cos^2 \chi \sin^2 \chi \sin \theta \, d\phi \, d\theta \, d\chi = \frac{1}{4}, \quad \langle n_1 n_2 \rangle_{S^3} = 0.$$

The cross moment vanishes because $\int_0^\pi \cos \theta \sin \theta \, d\theta = 0$.

Physical use. $\Omega_2 = 4\pi$ and $\Omega_3 = 2\pi^2$ set the reduced r^{-2} and bulk R^{-3} Gauss normalizations. The second moments give the δ_{ij}/d angular average used in the convective factor $-(1 + 1/d)$ and the pressure factor $1/d$.

B.2 Stage 002: Matter-Stress Inter-Defect Force Assembly

Part / anchor. Part II – Gravity

Claim status. REDUCED / CONTROLLED REDUCTION

Purpose. Assemble the inter-defect force between two drain defects from the matter (Noether) stress, earning the bilinear $Q_1 Q_2$ structure, the reduced-3D r^{-2} and bulk-4D R^{-3} power laws, and the attractive like-drain sign, consuming the Stage 001 geometry primitives.

Status. REDUCED / CONTROLLED REDUCTION. Three verdict layers are carried from the earned source: the leading matter-stress sign `FORCE_ATTRACTIVE_DERIVED` (target-blind); the full sign `SIGN_RESIDUAL_QUANTUM_VCONF_MAXWELL_PROFILE` (honest residual); acceptance `PASS_WITH_NAMED_RESIDUALS`. The bilinear structure, the power laws, and the attractive sign are earned (form + sign); the overall magnitude is a calibration knob; the full sign carries named, profile/sim-deferred residuals.

Derivation. The medium obeys $P = K\rho^5$, $h = (5K/4)\rho^4$, with the barotropic identity $dP/d\rho = \rho(dh/d\rho)$. The matter (Noether) stress in Madelung form is

$$\Pi_{ij} = m\rho v_i v_j + \delta_{ij} P(\rho) + \sigma_{ij}^Q, \quad \sigma_{ij}^Q = \frac{\hbar^2}{4m} \left[\frac{(\partial_i \rho)(\partial_j \rho)}{\rho} - \partial_i \partial_j \rho \right],$$

with the quantum-stress divergence contributing no net far-field force at this order. A defect draining flux Q sets up, by Gauss's law, the inflow

$$v = -\frac{Q \hat{n}}{\Omega_{d-1} r^{d-1}},$$

where $\Omega_2 = 4\pi$ ($d = 3$) and $\Omega_3 = 2\pi^2$ ($d = 4$) are the Stage 001 solid angles. The Bernoulli cross-pressure of the two overlapping inflows is $\delta P = -m\rho(v_1 \cdot v_2)$. Integrating the traction $\Pi_{ij} n_j$ over a control sphere around defect 2 and using the Stage 001 second moment $\langle n_i n_j \rangle = \delta_{ij}/d$ gives the convective factor $-(1 + 1/d)$ and the pressure factor $+1/d$, whose sum is the total flux $-mNQ_2 v_1$ (the $1/d$ pieces cancel). The stationary force $F_{12} = -\oint \Pi_{ij} n_j dS$, with defect 1's own inflow substituted, yields the two lanes.

Reduced-3D force law.

$$F_{12} = -\frac{m N_3 Q_1 Q_2}{4\pi r_{12}^2} \hat{r}_{12}.$$

Bulk-4D force law.

$$F_{12} = -\frac{m \rho_4 Q_1 Q_2}{2\pi^2 R_{12}^3} \hat{R}_{12}.$$

Attractive sign. Since $F \propto -Q_1 Q_2$, like drains ($Q_1 Q_2 > 0$) attract and opposite signs repel; the sign is derived from the Gauss-drain / Bernoulli / surface-integral chain, not assumed.

Physical use and named residuals. Target-blind, the inter-defect force is bilinear in the drain strengths, falls as r^{-2} (reduced 3D) and R^{-3} (bulk 4D), and is attractive between like drains. Honest scope: the overall normalization ($I_F/\Theta_Q/\text{branch-profile}$) is a *calibration* knob, not derived; and the full sign carries the QUANTUM_STRESS_FAR_FIELD, VCONF_BODY_FORCE, and MAXWELL_Z_GAUGE_JEXT residuals, which are profile/sim-deferred and first-class.

B.3 Stage 008: Monopole/Dipole Return Constraint Specification

Part / anchor. Part II – Gravity (first return/radiation stage)

Claim status. OPEN

Purpose. Render the old ledger’s open-item #9 as an exact constraint specification: the raw $\ell = 0/1/2$ outgoing c_s -wave amplitudes and radiation orders, and the precise source moments any brane $\leftrightarrow$ bulk return must cancel (MONOPOLE_DIPOLE_RETURN_CONDITIONAL). The formal home of the targets $R_0 = -M_0$, $R_{1,i} = -D_{1,i}$ consumed by Stages 009/010 (pathA_29) and 022/023 (pathA_34); register edge R23.

Status. OPEN – deliberately. **This is a verified CONSTRAINT-SPECIFICATION, not a falsifiable test of no-monopole-radiation** (the source gate’s own scope caveat, carried verbatim-class). GR forbids monopole/dipole gravitational radiation; the drain breaks brane mass conservation, so suppression is conditional on a return that cancels the moments below. The cancellation condition is algebraic bookkeeping (an $x - x$ identity, labeled as such in both engines); **cancellation_possible** is a literal parameter flag printed as *SCOPE: parameter, not computed – track-3 decides*. The falsification lives downstream (Stages 009/010; the Gate-6 $Z_{0,\text{ret}}/Z_{1,\text{ret}}$ selector).

The DtN ladder (earned; scanned, not typed). From spherical Hankel closed forms $h_\ell = j_\ell + iy_\ell$, both engines form $\Lambda_\ell = k h'_\ell/h_\ell|_{z=ka}$, normalize the admittance by its static value, and *scan* for the first purely-imaginary k -coefficient:

$$p_{\text{raw}} = 1, 3, 5; \quad \text{ kernels } i\frac{\omega}{c_S}a, \quad i\frac{\omega^3}{c_S^3}\frac{a^3}{2}, \quad i\frac{\omega^5}{c_S^5}\frac{a^5}{27},$$

$$A_{\text{raw}}^{(\ell 0)} = \frac{iM_0a\omega}{c_S}, \quad A_{\text{raw}}^{(\ell 1)} = \frac{iD_1a^3\omega^3}{2c_S^3}, \quad A_{\text{raw}}^{(\ell 2)} = \frac{iQ_2a^5\omega^5}{27c_S^5},$$

all symbolically nonzero; steady control $\lim_{\omega \rightarrow 0} A^{(\ell 0)} = 0$; dominance $p(\ell=0) < p(\ell=2)$ from the scanned powers.

Cancellation targets (bookkeeping, labeled). $R_0(\omega) = -M_0(\omega)$ with $M_0 = \int_{\text{brane}} S_{\text{leak}} d^3x$ (cancels $O(\omega^1)$); $R_{1,i}(\omega) = -D_{1,i}(\omega)$ with $D_{1,i} = \int x_i S_{\text{leak}} d^3x + \int j_i d^3x$ including the carried odd wake (cancels $O(\omega^3)$). Conditions, not closure. The derivative outlet vertex $g_{W0} = \eta\omega$ is recorded as a **branch_assumption**, not verdict-bearing. Live-source consistency is anchored by direct integration of the frozen Stage-243/244 closed forms ($\int W' j^w dw = -\sqrt{2} \ell_w j_0/4$; $S_{\text{leak}} = \sqrt{2} \mu_w \rho_0 E_0/(2\sqrt{\pi} \lambda^3)$) with the strict-recovery limits computed but *not* taken as a basis; Stage 006 owns the recovery-reduction derivation (boundary printed).

Verdict machinery and guards. The verdict is computed from computed booleans plus the literal flag; the synthetic control (True, False, False) $\rightarrow$ MONOPOLE_RADIATION_UNAVOIDABLE proves the rung reachable. The nine source controls are carried honestly: four computed, five printed as declarations (not counted checks), with the source’s framing – guards against obvious tautologies, not evidence of suppression-vs-unavoidable. Consumed from Stage 005: $c_s^2 = 5K\rho^4/m_{\text{GNLS}}$ (cited, exact-value integrity check $5 \cdot \frac{1}{500} \cdot (\sqrt{10})^4 = 1$); frozen slice $G = c = c_s = 1$, $K_{\text{eos}} = 1/500$, (a^*, L^*) exact rationals, cited. Q_2 is exported as a **free anchor** (not derived $\ell = 2$ physics) for Stage 026.

Verification. Dual-engine, independent routes, zero file I/O (the source pair’s JSON/YAML engine bridge is retired): SymPy 54 PASS / Mathematica 57 PASS (3 = arity self-checks), both exit 0, CWD-independent. Eight able-to-fail teeth fire (Hankel-form corruption trips the scan; kernel-coefficient, moment-kill, bookkeeping-integrity, verdict-machinery, anchor-prefactor, steady-limit, consumed-law corruptions). Full tri-review on fresh agents: FIDELITY_CLEAN (coverage-diff, math re-derived by hand) and ADVERSARIAL_CLEAN (14/14 mutation matrix; dual-corruption fails both engines; arity scan incl. planted-mismatch detection), followed by a nit remediation (tally hygiene, restored recovery-slice observations, pipeline-routed tooth) re-verified clean.

B.4 Stage 009: Flat-Slab Return Residual (Check A)

Part / anchor. Part II – Gravity (brane $\leftrightarrow$ bulk return, Check A)

Claim status. EXACT WITHIN CLOSURE

Purpose. Derive, on the postulated flat-slab family, what a finite DC-sink brane $\leftrightarrow$ bulk return actually delivers against the Stage-008 cancellation targets: the return transfer $T_\ell(\omega) = \alpha_\ell e^{i\omega 2d/c_s}$, the bounded residual orders $p_{\text{res}}(\ell=0) = 1$, $p_{\text{res}}(\ell=1) = 3$, and the signed local source accounting $Z = -M_0\varepsilon_0/(1 + \varepsilon_0)$ – the Check-A component of RETURN_RESIDUAL_PREDICTION (Check-B localization = Stage 010).

Status. EXACT WITHIN CLOSURE – exact *within the postulated slab family* (brane at $w = 0$, return/absorber at $w = d$; the geometry is postulated, the return response is derived on it). The v3 labeling is carried exactly: $Z < 0$ (drain admissibility) is the **premise** (Z_is_premise=true); the formula $Z = -M_0\varepsilon_0/(1 + \varepsilon_0)$ is **accounting** (channel sum $Z_{\text{throat}} + Z_{\text{return}}$, sign certificate computed). Open-item #9 is *sharpened, not closed*; the headline is printed only as the Check-A component with the Stage-010 qualifier.

The derived return response. From the bidirectional Helmholtz basis $\Phi_\ell = A_\ell e^{i\omega w/c_s} + B_\ell e^{-i\omega w/c_s}$, the round-trip transit time is *solved* ($\tau = 2d/c_s$, asserted via $e^{i\omega\tau} \equiv \text{phase}$). DC continuity fractions $\alpha_\ell = 1/(1 + \varepsilon_\ell)$ with independent per-channel transmissions $\varepsilon_0, \varepsilon_1 > 0$; steady circulation balances ($J_{\text{leak}} = J_{\text{return}} + J_{\text{neighbor}}$, both moments) assert-zero. A genuine ω -order scan computes $\nu_\ell = \text{ord}_\omega(1 - T_\ell) = 0$ – a finite DC sink leaves the ω^0 deviation $1 - T_\ell(0) = \varepsilon_\ell/(1 + \varepsilon_\ell)$ – so the residual amplitudes $A_{\text{res}} = \text{kernel}_\ell \cdot \text{moment} \cdot (1 - T_\ell)$ sit at **computed** orders $p_{\text{res}} = p_{\text{raw}} + \nu_\ell = 1, 3$: a *bounded monopole/dipole c_s -radiation residual tied to the drain strength* – the falsifiable departure from GR. The Check-A classification booleans are computed from the orders.

Reversibility (per-channel, computed). At $\varepsilon_0 \rightarrow 0^+$: $T_0 \rightarrow e^{i\omega\tau}$, strict $\nu_0 = 1 \Rightarrow p_{\text{res},0} = 2$, and $Z \rightarrow 0$ (a perfect-return channel has no drain). At $\varepsilon_1 \rightarrow 0^+$: strict $\nu_1 = 1$ computed *directly from the $\ell = 1$ limit* (the source reused ν_0 ; the reshape computes it independently) $\Rightarrow p_{\text{res},1} = 4$. The

prediction is contingent on the transmissions, not baked in. Consumed from Stage 008 (cited, dual-site integrity-checked): the three radiation kernels; Q_2/kernel_2 inert (**T2_applied=false**). Register: rows $\{d, \varepsilon_0, \varepsilon_1\}$ (slab-family parameters; ε 's FREE-UNREDUCED on the Gate-6 route) + edge R24.

Verification. Dual-engine, independent routes, zero file I/O – the source pair's JSON-digest bridge, its runtime YAML reads of pathA_28 outputs, and its SHA-256 trace bookkeeping are all retired. SymPy 48 PASS / Mathematica 52 PASS, exit 0, CWD-independent. Eight able-to-fail teeth fire. Full tri-review on fresh agents: FIDELITY_CLEAN (all values hand-re-derived; 009/010 boundary clean); the adversarial leg (16-mutant matrix) **caught a genuine rig-class defect** – the consumed-kernel citation-integrity block was set-then-compare-to-self (a single-source corruption passed) – remediated to dual-site integrity (acceptance: all four one-site corruptions now fail in both engines) plus tally-hygiene nits, then re-verified clean. The `Off[Limit::alimv]` suppression was unmasked and shown benign (every silenced limit's result independently asserted).

B.5 Stage 010: Slab Localization $p = 2$ with Reachable NOGO (Check B)

Part / anchor. Part II – Gravity (brane $\leftrightarrow$ bulk return, Check B; completes the pathA_29 fold)

Claim status. EXACT WITHIN CLOSURE

Purpose. Show, on the postulated flat-slab family, that Newtonian $1/r^2$ gravity SURVIVES the finite slab: both admissible DC-sink completions carry a normalizable $m=0$ transverse zero mode whose 3D-radial Green function gives flow $\propto 1/r^2$ ($p = 2$); the anti-localizing warp control gives $p = 3$ and the same classifier returns RETURN_NOGO (the able-to-fail companion). This computes Check B and emits the COMPLETED joint verdict RETURN_RESIDUAL_PREDICTION = (Check A cited, Stage 009) $\wedge$ (Check B computed here).

Status. EXACT WITHIN CLOSURE – exact *within the postulated slab family* (brane at $w = 0$, return/absorber at $w = d$; the geometry is postulated, localization is derived on it). The gravity-range ($1/r^2$) leg passes inside the localizing flat-slab family because both DC-sink completions give $p = 2$; the joint verdict RETURN_RESIDUAL_PREDICTION is completed here, with RETURN_NOGO genuinely reachable. Open-item #9 is *sharpened, not closed*: localization holding for the FAMILY is not the family being SELECTED by dynamics (R19/ W_{slab}); the return-cancellation targets (R23) remain deferred.

The localizing zero mode and $p = 2$. On $0 \leq w \leq d$ each admissible completion is a transverse eigenproblem $f'' + m^2 f = 0$ with a normalizable constant $m=0$ zero mode ($f_0 = 1/\sqrt{d}$, $\int_0^d f_0^2 dw = 1$): the *destructuring/absorbing* (compact cell, Neumann) and *Bloch* ($q=0$ periodic) cells, with gapped first modes at $m_1 = \pi/d, 2\pi/d$. The 3D-radial zero-mode equation $g'' + (2/r)g' + ((\omega/c_s)^2 - m^2)g = 0$ is solved by genuine `dsolve` (dynamic route: solved first, outgoing spherical branch selected via C_1/C_2 – a boundary SELECTION, normalization = compact overlap $1/d$ – then $\omega \rightarrow 0$); the limit Green $1/(4\pi dr)$ gives flow $1/(4\pi dr^2)$ and *extracted* exponent $p = 2$. The static route ($\omega = 0$ first) gives the same $1/(4\pi dr)$ and $p = 2$; the two routes agree on both the exponent and the Green ($\text{dynamic}_{\omega \rightarrow 0} = \text{static}$). The massive/gapped solve gives a Yukawa $e^{-m_1 r}/r$ – only the $m=0$ zero mode sets the far field. Both completions agree $p_{\text{abs}} = p_{\text{bloch}} = 2$.

Able-to-fail: the counterfactual and the NOGO warp. Multiplying the solved Green by r^{-4} ($\rightarrow 1/r^5$) yields radial-operator residual $5/(\pi dr^7) \neq 0$ – REJECTED, proving the solve has teeth. The mandatory anti-localizing warp $\mu(w) = e^{2k_{\text{warp}}w}$ has a *non-normalizable* zero mode ($\int_0^\infty \rightarrow \infty$, computed) and a continuum Green $1/(4\pi r^2)$ giving $p_{\text{delocalizing}} = 3$; the same classifier returns RETURN_NOGO. The classifier verdict is a function of the computed exponents (and the cited `quadrupole_survives`), never a stamp; RETURN_NOGO is reachable if the return delocalizes OR the quadrupole channel does not survive.

Consumed (cited, dual-site integrity + exact anchor). From Stage 008: $p_{\text{raw}}(\ell=2) = 5 \Rightarrow \text{quadrupole_survives}$ (finite quadrupole order; T_2 not applied, `T2_applied=false`, no $\ell=2$ recompute). From Stage 009: the Check-A component `A_residual_pass=true` (joint-composition print only). Register: *zero new counted knobs*; k_{warp} is a control-construction symbol (tracked, not counted); edge R25 (localization $p = 2$ EARNED-within-family, able-to-fail via the warp NOGO – does not discharge R19 or R23).

Verification. Dual-engine, independent routes, zero file I/O – the source pair’s JSON-digest bridge and all SHA-256 trace/`structure_id/expr_digest` bookkeeping are retired; the `.w1` solves the radial ODEs, the normalizations, the warp divergence, and the classifier natively (own `DSolve` basis, own `Limit/Integrate`), with an arity self-check and no silenced messages. SymPy 71 PASS / Mathematica 81 PASS (81 = 71 shared + 10 arity self-checks), exit 0, CWD-independent. Eight able-to-fail teeth fire. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; 009/010 boundary clean – Check-A cited not recomputed) and `ADVERSARIAL_CLEAN` (22-mutant matrix; a hardcoded-wrong-Green mutant confirmed the counterfactual guard is genuine, and the classifier genuinely reaches RETURN_NOGO). The adversarial leg’s one substantive find – a *coordinated both-sites* corruption of the Stage-008 citation to another finite integer escaped – was folded to Stage-009 parity by an exact-value anchor, and re-verified (`REVERIFY_CLEAN`) as the sole gate closing that escape.

B.6 Stage 011: Frozen-Reduction Certificate (L_s Helmholtz) (Check II-G1a)

Part / anchor. Part II – Gravity (brane $\leftrightarrow$ bulk return; the frozen-throat longitudinal reduction; opens the pathA_30 D/N fold, completed by Stage 012)

Claim status. REDUCED / CONTROLLED REDUCTION

Purpose. Certify that, on the postulated straight finite throat with the wall FROZEN ($R = R_0$, $\eta = 0$) and the matter perturbation left dynamic ($\propto e^{-i\omega t}$), the reduced longitudinal operator collapses to the constant-coefficient Helmholtz resonator operator $L_s\psi = \psi''(s) + (\omega/c_s)^2\psi(s) = 0$, with $c_s^2 = 5K\rho_*^4/m$ on $s \in [0, L_0]$ (cap $R_0(L_0) = 0$) — *produced* by a per-term reduction (projection measure plus every intruding coefficient computed with its vanishing/deferral condition), NOT typed — together with the reduction certificate: the validity window under which the collapse is exact.

Status. REDUCED / CONTROLLED REDUCTION — a controlled reduction with an explicit validity certificate. This stage carries the reduction-certificate component of the joint pathA_30 verdict `DN_UNITTEST_BC_DEPENDENT`; the D/N determinant, the DtN $-(\omega/c_s)\tan(L_0\omega/c_s)$, the half-shifted pole ladder, the Robin counterfactual, and the `BC_DEPENDENT` landing (`bc_derivation_emitted = false`, a banked calibration input) are Stage 012’s; the cut is drawn at the general-solution `dsolve` (never

called here). The wave speed is *banked* (Part I edge R1, Stage 005, at the straight-reference density ρ_*), so this stage adds NO reduction of its own: its contributions are the two *structural* edges R26 (the validity record) and R27 (the $\xi \neq \ell_c$ firewall), which discharge no knob.

The produced operator (the de-rig). The reduced operator is *assembled* from dimensionally-consistent terms,

$$L_s = \psi'' + M\psi' + N\psi - B \frac{\hbar^2}{4m^2 c_S^2} \psi'''' ,$$

with $M = \frac{d}{ds} \log \sqrt{\gamma_0}$ ($[M] = L^{-1}$), the ψ -coefficient $N = (\omega/c_s(s))^2$ with $c_s(s)^2 = 5K\rho_0(s)^4/m$ ($[N] = L^{-2}$), and $B \in \{0, 1\}$ the Bogoliubov inclusion flag — the k^4 correction being a genuine *fourth-derivative* intrusion (from the identity $\psi'' + (\omega/c_S)^2\psi - (\hbar^2/4m^2 c_S^2)\psi'''' = 0$ for the full BdG dispersion), NOT a coefficient shift. The three source checks are replaced by genuine computations that can each fail: `operator_is_helmholtz` is `expr_equal(L_s - ideal, 0)` on the *assembled* L_s (not two hand-typed twins); `speed_is_cs` is *extracted* from the ψ -coefficient of L_s ($= (\omega/c_S)^2$, bulk, not wall/healing-renormalized, not literal `True`); `domain_is_L0` is *solved* from the pinch-off $R_0(s) = 0$ on a POSTULATED monotone taper $R_0(s) = R_{\text{mouth}}(1 - s/L_0)$ (root = L_0 , R_{mouth} -independent; a labeled SELECTION, not `L0==L0`); and `unsuppressed_operator_intrusion` is *computed* from the assembly (not hardwired `False`) and gates the verdict.

The reduction certificate (validity window). Under the frozen background ($\rho_0(s) = \rho_*$, $\sqrt{\gamma_0} = A_{\perp 0}$ constant, $A_{M0} = 0$) every intruding term is computed to vanish or is deferred: the projection-measure coefficient $M = 0$; $\rho'_0 = 0 \Rightarrow c'_s = 0 \Rightarrow N = (\omega/c_S)^2$ (the bulk sound speed); $\delta V_{\text{conf}} = 0$; the background quantum potential $Q(\rho_0) = -\frac{\hbar^2}{2m} \partial_s^2 \sqrt{\rho_0}/\sqrt{\rho_0}$ gives $\partial_s Q = 0$; and the BdG term is deferred ($B = 0$) only under $k\xi \ll 1$ ($\xi = \hbar/(mc_s)$), with smallness witness $\hbar^2 k^2/(4m^2 c_S^2) = (k\xi/2)^2$. Thus L_s is constant-coefficient Helmholtz *conditional on* $\{\rho'_0/\rho_0 = 0, \sqrt{\gamma_0} \text{ const}, \delta V_{\text{conf}} = 0, \nabla Q = 0, k\xi \ll 1\}$ (edge R26); the BdG k^4 term is DEFERRED, not dropped unconditionally. The healing length $\xi = \hbar/(mc_s)$ and the confinement length ℓ_c (in $V_{\text{wall}}(\Sigma/\ell_c)$) are kept as DISTINCT symbols (edge R27, the firewall analogous to R22's $\mu_R \neq \mu_R^{(4)}$); ℓ_c is INERT here ($\delta V_{\text{conf}} = 0$).

Able-to-fail teeth. Eight teeth, mutations on copies: a nonconstant $\sqrt{\gamma_0}$ makes $M \neq 0$; a retained BdG ($B = 1$) injects the ψ'''' term; a nonuniform $\rho_0(s)$ makes N s -dependent (each $\rightarrow$ `FAIL_OPERATOR_INTRUSION`); a wall/healing renormalization $\rightarrow$ `FAIL_WRONG_SPEED`; a corrupted taper root $\rightarrow$ `FAIL_WRONG_DOMAIN`; an $\ell_c \rightarrow \xi$ conflation trips the firewall; a one-site R1 exponent corruption trips the dual-site integrity (a coordinated both-site drift is caught by the explicit frozen-export anchor); and a corrupt- $[K]$ mutation flips $[c_S^2]$ to $(3, 0, -2) \rightarrow$ `DN_UNITTEST_FAIL_DIMENSIONAL`. A dedicated witness-path tooth (nonzero δV_{conf} with $M = 0$, $N = \text{bulk}$, $B = 0$) drives the verdict to `FAIL_OPERATOR_INTRUSION` through the intrusion flag alone, making the flag independently load-bearing.

Consumed + dimensional leg. The speed law $c_s^2 = 5K\rho^4/m_{\text{GNLS}}$ is CITED (Part I edge R1, Stage 005) evaluated at ρ_* , integrity-checked dual-site (literal `5*K*rho**4/m` vs. the EOS route $\partial_\rho(K\rho^5)/m$) plus an explicit frozen-export anchor; the EOS exponent-5 closure $P = K\rho^5$ stays IMPOSED. The c_S^2 dimensional leg uses the $[K] = [P] - 5[\rho]$ chain in four spatial dimensions: $[K] = (18, 1, -2)$, $[c_S^2] = (2, 0, -2)$ (order L, M, T); the corrupt- $[K]$ probe flips it to $(3, 0, -2)$. L_0 (throat depth) is POSTULATED ACTION-geometry (like Stage 009's d , not a medium constant); zero new counted knobs.

Verification. Dual-engine, independent routes, zero file I/O — the source pair’s scratch-YAML/_sympy_exprs.wl export, the Mathematica-YAML re-read, and the `expression_digest/engine_agreement` plumbing are retired; the `.wl` derives the certificate, the L_s assembly, the speed extraction, the domain solve, and the dimensional leg natively (own `D/Solve/Coefficient/Simplify`), with an arity self-check and no silenced messages; the transfer matrix, if present, is only an operator-nature cross-check (no DtN/pole/Robin — Stage 012). SymPy 61 PASS / Mathematica 71 PASS (71 = 61 shared + 8 arity self-checks + 2 native trig-basis confirmations), exit 0, CWD-independent. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; the three booleans confirmed genuinely `PRODUCED`; the 011/012 boundary clean) and `ADVERSARIAL_CLEAN` (the de-rig proven able-to-fail — operator via $M/N/B$, speed via renormalization, domain via taper — and the BdG confirmed a real fourth-order term). Three teeth/label nits were remediated (the firewall tooth made genuine, a witness-path tooth added so the intrusion flag is independently load-bearing, the arity labels corrected) and re-verified `REVERIFY_CLEAN`.

B.7 Stage 012: DtN Pole Ladder & Robin Falsifier ($Z_{00} = -(\omega/c_S) \tan$) (Check II-G1b)

Part / anchor. Part II – Gravity (brane↔bulk return; the frozen-throat resonator response; COMPLETES the pathA_30 D/N fold opened by Stage 011)

Claim status. EXACT WITHIN CLOSURE

Purpose. On Stage 011’s cited frozen constant-coefficient Helmholtz operator $L_s\psi = \psi'' + (\omega/c_S)^2\psi = 0$ on $s \in [0, L_0]$, pose and solve the Dirichlet/Neumann boundary-value problem: derive the outward-mouth Dirichlet-to-Neumann map $Z_{00} = -(\omega/c_S) \tan(L_0\omega/c_S)$, its half-shifted resonance ladder $\omega_n = \pi c_S(n + \frac{1}{2})/L_0$, its round-trip $R_{rt} = 1$, and the Robin counterfactual that proves the D/N determination is not hardcodable — landing the joint verdict at `DN_UNITTEST_BC_DEPENDENT` because the boundary condition itself is imposed (`bc_derivation_emitted = false`, a banked calibration input).

Status. EXACT WITHIN CLOSURE — the DtN, ladder, round trip, and Robin counterfactual are exact algebra *within the imposed-BC closure*. This stage COMPLETES the joint pathA_30 verdict `DN_UNITTEST_BC_DEPENDENT`: Stage 011 carried the reduction-certificate component (`REDUCTION_CERTIFIED`), and Stage 012 adds the D/N ladder plus the `BC_DEPENDENT` landing. It is a near-pure *bridge-strip* (contrast Stage 011’s de-rig): the physics is already genuinely computed in the source and the `.wl` transfer-matrix route already carries it, so the reshape severs the scratch-YAML bridge and keeps the native content. The operator, domain, and speed are *consumed* from Stage 011 (dual-site citation-integrity); the stage opens at the general-solution `dsolve`. Its register contribution is the single *imposed* edge R28 (the D/N boundary is a banked calibration input), which discharges no knob.

The D/N boundary-value problem and the DtN. The `dsolve` of the cited L_s gives $\psi = C_1 \sin(\omega s/c_S) + C_2 \cos(\omega s/c_S)$. Imposing the Dirichlet mouth trace $\psi(0) = \psi_M$ and the Neumann cap $\psi'(L_0) = 0$, the 2×2 coefficient matrix (determinant $-\omega \cos(L_0\omega/c_S)/c_S$) is solved by `LUsolve`, and the outward-mouth admittance is

$$Z_{00} = -\frac{\partial_s \psi|_0}{\psi|_0} = -\frac{\omega}{c_S} \tan\left(\frac{L_0\omega}{c_S}\right).$$

`dtn_matches_target` compares this *derived* Z_{00} (from the `LUsolve`) against the typed target $-k \tan(kL_0)$ — a genuine derived-vs-typed comparison, not an $X \equiv X$: perturbing the Neumann RHS recomputes the derived DtN and the comparison fails.

Ladder, static limit, round trip. The DtN denominator's $\cos(L_0\omega/c_S)$ factor gives the pole equation $\cos(L_0\omega/c_S) = 0$, whose roots are the *half-shifted* ladder $\omega_n = \pi c_S(n + \frac{1}{2})/L_0$; `halfshift` is computed as `pole_residual = 0`. The small- ω series $-L_0\omega^2/c_S^2 - L_0^3\omega^4/(3c_S^4) + O(\omega^6)$ is the leading static compliance (distinct from the DC limit $\lim_{\omega \rightarrow 0^+} Z_{00} = 0$; no separate static solve). With Dirichlet/Neumann reflections $r_D = -1$, $r_N = +1$, the round-trip factor $-\exp(2iL_0\omega/c_S)$ evaluates to $R_{rt} = 1$ ($\varphi_0 = 0 \bmod 2\pi$) on the ladder.

The Robin falsifier. Without a counterfactual the D/N determination is hardcodable. The Robin cap $\partial_s\psi - \alpha\psi = 0$ interpolates Neumann ($\alpha \rightarrow 0$) and Dirichlet ($\alpha \rightarrow \infty$); its DtN reads off `robin_denominator_core = k cos(kL0) - α sin(kL0)`. The `counterfactual_guard` has exactly six members, each a *computed residual* (not a stamped truthiness): `robin_determinant_emitted` tied to the computed Robin determinant `robin_det ≡ -robin_denominator_core`; `recovers_DN_at_alpha0` ($\alpha \rightarrow 0$ recovers $-(\omega/c_S)\tan$); `recovers_DD_at_alpha_inf` ($\alpha \rightarrow \infty$ recovers the D/D DtN $k \cot(kL_0)$); `halfshift_destroyed_for_DD` (the D/D denominator $\sin(kL_0)$ gives the *integer* ladder $\pi c_S j/L_0$, not the half-shifted one); `numeric_alpha_distinct` ($\alpha = 2/L_0$ distinct from both limits, exact rationals); and `dtn_mismatch`. The artifact `dd_zero_mode_removable = 1/L0` is held out of the guard. Breaking `robin_denominator_core` flips the recomputed `recovers_DD_at_alpha_inf` and is caught ($\rightarrow$ FAIL_COUNTERFACTUAL).

Imposed BC (honest scope) and the completed joint verdict. The D/N boundary pair is IMPOSED (`bc_provenance = imposed`); the explicit mouth/cap V_{wall} gradient derivation is *not* emitted (`bc_derivation_emitted = false`), so the verdict lands at `DN_UNITTEST_BC_DEPENDENT` rather than `DN_UNITTEST_PASS` (edge R28; earning PASS is a deferred upgrade, not fabricated here). The 012-scoped verdict is a function of the computed rungs (`FAIL_DIMENSIONAL` $\rightarrow$ `FAIL_POLE_LADDER` $\rightarrow$ `FAIL_COUNTERFACTUAL` $\rightarrow$ the `BC_DEPENDENT` landing), and the joint verdict is printed as `DN_UNITTEST_BC_DEPENDENT = (011: REDUCTION_CERTIFIED, cited) ∧ (012: DtN ladder + BC_DEPENDENT landing, computed here)`, not typed as 012-earned.

Consumed + dimensional leg. The frozen operator L_s , domain $[0, L_0]$, and speed c_S are CONSUMED from Stage 011 with dual-site citation-integrity: for L_s , site A the typed export form and site B the *null-space reconstruction* from the `dsolve`'d pair $\{\sin, \cos\}$ (posit $y'' + ay' + by$, solve for $(a, b) = (0, (\omega/c_S)^2)$) — a genuinely independent construction (a $\sin/\cos \rightarrow \sinh/\cosh$ corruption recovers $b = -(\omega/c_S)^2$ and fires); for $c_S^2 = 5K\rho^4/m$ (R1 at ρ_*), the literal vs the EOS route $\partial_\rho(K\rho^5)/m$. The 012 dimensional legs use the $[K] = [P] - 5[\rho]$ chain: $[\tan_argument] = [kL_0] = 1$ (dimensionless), $[Z_{00}_prefactor] = [-k] = L^{-1}$; the corrupt- $[K]$ probe propagates $[c_S^2] \rightarrow [c_S] \rightarrow [k]$, flipping both legs $\rightarrow$ `DN_UNITTEST_FAIL_DIMENSIONAL`. α (Robin cap admittance, $[\alpha] = L^{-1}$) is a control-construction symbol; zero new counted knobs.

Verification. Dual-engine, independent routes, zero file I/O — the source pair's scratch-YAML/`_sympy_exprs.wl` export, the Mathematica-YAML re-read, and the `expression_digest/engine_agreement` plumbing are retired; the `.wl` KEEPS its transfer-matrix route as the independent engine (propagate the mouth state through `transferMatrix[L0]`, impose the Neumann cap, read Z_{00} — a different decomposition from the `.py` `LUsolve`, computed agreement), preserving its native

`robinAlpha0==dtnTransfer/robinAlphaInf==ddTransfer` self-checks, with an arity self-check and no load-bearing message silenced. SymPy 84 PASS / Mathematica 90 PASS (the +6 = the .w1 arity block net of two SymPy-only DtN-detail checks), exit 0, CWD-independent. Full tri-review on fresh agents: **FIDELITY_CLEAN** (all values hand-re-derived; the six-member Robin guard each a genuine residual; `dtn_matches_target` genuine derived-vs-typed; the L_s null-space dual-site independent; the consume-from-011 boundary clean) and **ADVERSARIAL_CLEAN** (16-mutant matrix — the guard un-stampable, the $\sin/\cos \rightarrow \sinh/\cosh$ corruption fires in both engines, even a hardcoded DtN target is caught, all four verdict rungs reachable, arity clean). Eight able-to-fail teeth, mutations on copies.

B.8 Stage 013: Breathing Harmonic Profiles & M/K Operator Projection (Check II-G2a)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 0$ breathing mode; the (a, L) collective-coordinate closure; the M/K -projection component of the joint **BREATHING_CALIBRATED**, with truncation consistency = Stage 014 and the legacy-Hessian + Hellmann–Feynman force = Stage 015)

Claim status. EXACT WITHIN CLOSURE

Purpose. Reduce the $\ell = 0$ linearized breathing of Gate-1’s frozen constant-coefficient wall operator $\mathcal{L}_0 = \mu_\eta^{-1}(-\partial_w(T_w \partial_w \cdot) + K_\eta \cdot)$ on $w \in [0, L_0]$ to a two-mode collective closure $Q = (\delta_a, \delta_L)$: solve the harmonic-lift profiles α_a, α_L (from $\mathcal{L}_0 \alpha = 0$ under the collective BCs), project the wall inertia and stiffness onto them by *real $\int dw$ operator projection* to build the mass/stiffness matrices M_{AB}, K_{AB} , and assemble the dynamical-EOM left-hand side $M_{AB} \ddot{Q}^B + K_{AB} Q^B = F_A^{(\text{HF})}$. The first Part-II *calibrated* stage: the structure is earned, the wall constants are calibration inputs.

Status. EXACT WITHIN CLOSURE — the harmonic-lift profiles, the M_{AB}/K_{AB} operator projection, and the EOM left-hand side are exact algebra *within the calibrated wall closure*. The verdict is the joint **BREATHING_CALIBRATED**; this stage carries its M/K -projection + (a, L) -closure component. It is the **first Part-II calibrated stage**: the gravity-sector algebra was knob-free through Stages 008–012, and the calibration enters here with the breathing-mode wall response. *The structure is EARNED; the values are CALIBRATED*. The frozen throat packet is *consumed* from Gate-1 (Stages 011/012) with dual-site citation-integrity; c_S is *not* consumed (the matter-sector $c_s/\text{BdG } k^4$ is deferred under $k\xi \ll 1$).

The harmonic-lift profiles. The general solution of $\mathcal{L}_0 \alpha = 0$ ($\Leftrightarrow \alpha'' - \beta^2 \alpha = 0$, with $K_\eta = T_w \beta^2$) is $\alpha = C_1 \sinh(\beta w) + C_2 \cosh(\beta w)$. Under the collective BCs,

$$\alpha_a = \frac{\sinh(L_0 \beta - \beta w)}{\sinh(L_0 \beta)} \quad (\alpha_a(0) = 1, \alpha_a(L_0) = 0), \quad \alpha_L = \frac{r_{AL} \sinh(\beta w)}{\sinh(L_0 \beta)} \quad (\alpha_L(0) = 0, \alpha_L(L_0) = r_{AL}).$$

The harmonic-residual assertion $\text{Lop}[\alpha] = (-T_w \alpha'' + K_\eta \alpha)/\mu_\eta \equiv 0$ (a raise on failure) proves the profiles genuinely annihilate the operator. Its able-to-fail tooth uses a *non-kernel* corruption (a wrong wavenumber $\sinh(L_0 \beta - 2\beta w)$ or $\alpha + 1$) — a mere rescale or argument sign-flip stays in $\ker(\mathcal{L}_0)$ and leaves the residual 0, so it would be a non-tooth; the adversarial leg’s decisive ablation confirmed a kernel-preserving corruption is caught only by the *separate* BC assert.

M/K by real $\int dw$ operator projection. With $\langle f, g \rangle_\mu = 4\pi \int_0^{L_0} \mu_\eta f g dw$,

$$M_{AB} = 4\pi \int_0^{L_0} \mu_\eta \alpha_A \alpha_B dw, \quad K_{AB} = 4\pi \int_0^{L_0} [T_w \alpha'_A \alpha'_B + K_\eta \alpha_A \alpha_B] dw,$$

computed by `sp.integrate` / native `Integrate`. Closed forms (both engines agreeing): $K_{aa} = 4\pi T_w \beta / \tanh(L_0 \beta)$, $K_{aL} = -4\pi T_w \beta r_{AL} / \sinh(L_0 \beta)$, $K_{LL} = 4\pi T_w \beta r_{AL}^2 / \tanh(L_0 \beta)$, $\det K = 16\pi^2 T_w^2 \beta^2 r_{AL}^2$; and the M_{AB} forms with $[M_{AB}] = M$, $[K_{AB}] = M T^{-2}$. The genuineness spine (the anti-“typed M/K ” burden, since one could otherwise type the legacy static Hessian $\{\chi^2 \kappa + \sigma_A, -\chi \kappa, \kappa + \sigma_L\}$) is twofold: (i) an *independent re-integration* of the full integrands compared against the hardcoded report closed-forms (not against the same projection call — that would be a vacuous $X \equiv X$); dropping the $T_w \alpha' \alpha'$ gradient makes it mismatch and fires. (ii) the `forbidden_fit_flags` (`K_from_static_hessian`, `M_or_K_typed_from_legacy_values`, `full_matrix_fit`) computed from the *symbol names* of M_{AB}/K_{AB} — robust to SymPy’s assumption-fragility (`Symbol('kappa') \neq Symbol('kappa', positive=True)`), so an object-set intersection would silently miss): the names must lie in $\{\mu_\eta, T_w, \beta, L_0, r_{AL}, \pi\}$ and contain none of $\{\kappa, \chi, \sigma_A, \sigma_L\}$; typing K_{AB} from the legacy Hessian introduces those names, flips the flag, and is caught in both engines (`K_AB_provenance=operator_projection_not_static_Hessian`).

The dynamical-EOM left-hand side. The collective EOM LHS is $M_{AB} \ddot{Q}^B + K_{AB} Q^B = F_A^{(\text{HF})}$ with $Q = (\delta_a, \delta_L)$, emitted as the two expanded rows. This stage emits the LHS and EOM structure only; the right-hand side $F_A^{(\text{HF})}$ (the Hellmann–Feynman force) is a symbolic placeholder filled by Stage 015 (as is the HF two-route enforcement, the `pathA_31 v1 rig locus`). A native $M_{\text{det}} \rightarrow 0$ degenerate counterfactual ($\alpha_a \rightarrow 0$, recomputed via the same projection) is caught by a native `M_det \equiv 0` test, not the Stage-015 structure gate.

Consumed packet and the anti-tautology guard. The frozen wall packet $\{L_0 = 37/20, T_w = 1, K_\eta = 1, \beta = 1\}$ is cited from Gate-1 with two independently-corruptible relations: site A (constitutive) $\beta - \sqrt{K_\eta/T_w} \equiv 0$ and site B (branch) $\beta L_0 - 37/20 \equiv 0$ (the branch-determinable L/a), plus the frozen-export anchor. *The packet values are held as independent cited datums* (`K_eta_cited`, ...) — the guard does *not* re-expand K_η through the alias $K_\eta = T_w \beta^2$, which would collapse site A to $\beta - \beta \equiv 0$ (a disguised $X \equiv X$); a lone `K_eta_cited` corruption genuinely breaks site A. L_0 is already registered (Stage 011, `ACTION-geometry`) — not re-counted.

Calibration and register (the first Part-II counted knobs). The wall constitutive packet contributes **three counted CALIB knobs** $\{\mu_\eta$ (inertia, $M L^{-1}$), T_w (tension, $M L T^{-2}$), β (inverse-length scale, $L^{-1}\}$, with $K_\eta = T_w \beta^2$ a *derived* manifestation (edge R29) — so three counted, not four. β is counted, *not* dressed as geometry: the source is explicit that geometry alone does not derive the wall constants, and $\beta L_0 = 37/20$ is $\beta_{(\text{calibrated})} \cdot L_0$ (= the L/a geometry), not an independent branch-pin of β (the conservative, non-shrinking read). r_{AL} (dimensionless collective BC ratio) is a control-construction symbol, tracked but not counted. Edge R30 is the named-pending nonlinear-throat reduction that would derive $\{\mu_\eta, T_w, \beta\}$ from the medium (reduction debt).

Verification. Dual-engine, independent routes, zero file I/O — the source pair’s scratch-YAML/`_sympy_exprs.wl` export and the hybrid `.wl`’s `Get/sympy*-cross-check/Export` are retired; the `.wl` KEEPS its native `DSolveValue+Integrate` route but its asserts are *re-targeted* to its own native M/K (Mathematica-native `Csch/Coth` closed forms, algebraically equal to the SymPy `tanh/sinh` forms),

with a genuine arity self-check. The 3-way cut is clean: no Stage-014 truncation/generalized-eig/ β_{L_0} -sweep and no Stage-015 legacy Hessian / Hellmann–Feynman force / static-dynamic limit is computed. SymPy 78 PASS / Mathematica 84 PASS (the +6 = the .w1 arity block), exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend; Grok’s compute-verification pass caught a kernel-preserving residual-tooth defect (folded) plus hardening nits. Full tri-review on fresh agents: FIDELITY_CLEAN (every profile/ $M/K/\det$ independently re-derived via a different normalization path; `expand_hyperbolic_args` a math-preserving identity) and ADVERSARIAL_CLEAN (a ten-ablation matrix, all firing: the free-symbol-name ancestry, the non-kernel residual tooth, the `K_eta_cited` anti-tautology, the native degenerate- $M_{\det}$ guard, the corrupt- $[T_w]$ probe). One non-blocking nit (the baseline re-integration was value-wise $X \equiv X$) was remediated to compare against the hardcoded closed-forms, then REVERIFY_CLEAN. Six able-to-fail teeth, mutations on copies.

B.9 Stage 014: Truncation Consistency — Combined-Basis Generalized Eigenproblem, β_{L_0} Sweep, and N-Convergence (Check II-G2b)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 0$ breathing mode; the truncation-consistency component of the joint BREATHING_CALIBRATED, with the M/K -projection + (a, L) -closure = Stage 013 and the legacy-Hessian + Hellmann–Feynman force = Stage 015)

Claim status. NUMERICALLY LOCATED

Purpose. Certify that Stage-013’s two-mode collective closure $Q = (\delta_a, \delta_L)$ is a *faithful* truncation, not an artifact. Embed the two collective harmonic-lift profiles α_a, α_L in a larger Galerkin basis $B = \{\alpha_a, \alpha_L, g_1 \dots g_N\}$ (N sine lane modes $g_n = \sin((n - \frac{1}{2})\pi w/L_0)$), solve the combined-basis generalized eigenproblem $Kv = \omega^2 Mv$, and measure how much of each of the two lowest generalized modes lives in $\text{span}\{\alpha_a, \alpha_L\}$ — the modal overlaps o_1, o_2 — across a β_{L_0} sweep and as $N \rightarrow 16$. The first *numeric / float-bearing* stage of the rebuild.

Status. NUMERICALLY LOCATED — the generalized eigenproblem, the modal overlaps, the β_{L_0} validity window, and the N-convergence are numerically computed (`scipy.linalg.eigh+quad; Eigensystem+NIntegrate`). The verdict is the joint BREATHING_CALIBRATED; this stage carries its *truncation-consistency* component (2/3). It **adds no new counted knobs**: the underlying wall constants $\{\mu_\eta, T_w, K_\eta, \beta\}$ are Stage-013’s calibration inputs, and the numeric truncation controls are method/tolerance parameters. Stage-013’s collective profiles and the frozen packet are *consumed* with dual-site citation-integrity; c_S is *not* consumed (matter-sector deferred, $k\xi \ll 1$).

The combined-basis generalized eigenproblem. Over $B = \{\alpha_a, \alpha_L, g_1 \dots g_N\}$ with normalized $x = w/L_0 \in [0, 1]$, the mass/stiffness Gram is built by numeric quadrature, $M_{ij} = \int_0^1 \phi_i \phi_j$, $K_{ij} = \int_0^1 [\phi'_i \phi'_j + \beta^2 \phi_i \phi_j]$, rank-deficient rows are dropped, and $Kv = \omega^2 Mv$ is solved and sorted ascending. The two collective modes $\alpha_a = \sinh(\beta(1-x))/\sinh \beta$, $\alpha_L = \sinh(\beta x)/\sinh \beta$ are *cited* from Stage 013 (not re-derived); the numeric 2×2 collective block equals Stage-013’s M_{AB}/K_{AB} only up to the 4π , L_0 -Jacobian, and μ_η -weight prefactors, which cancel in the generalized eigenproblem and the mass-normalized overlaps — so the seam is guarded via the consumed profiles, *not* a naive M_{aa} equality.

The modal overlaps and the truncation floor. Projecting each of the two lowest generalized eigenvectors onto $\text{span}\{\alpha_a, \alpha_L\}$ via the mass-Gram gives $o_k = \|P_\alpha v_k\|_M / \|v_k\|_M$. The certificate is

$$\text{pass} \iff o_1 \geq \text{FLOOR} \wedge o_2 \geq \text{FLOOR} \wedge \min(\omega_1^2, \omega_2^2) > 0, \quad \text{FLOOR} = 0.9,$$

a genuine predicate on the *computed* overlaps. At the physical $\beta_{L_0} = 37/20$, $N = 16$: $o_1 = 0.99311$, $o_2 = 0.98776$, $\min(\omega^2) = 3.42252$, $\text{gap} = 2.22787$, $\text{pass} = \text{true}$. All three conjuncts are separately able-to-fail (the o_2 floor, the $\min(\omega^2) > 0$ positivity leg, and the window edge).

The computed validity window. Sweeping $\beta_{L_0} \in \{0.1, \dots, 3.0, 5, 10, 18, 30, 50\}$ — *spanning well beyond* the pass region (to $o_1 \approx 0.27$ at $\beta_{L_0} = 50$) — the certificate passes for $\beta_{L_0} \in \{0.1 \dots 3.0\}$ and **fails** for $\beta_{L_0} \geq 5$ ($o_1 = 0.860 < \text{FLOOR}$ already at $\beta_{L_0} = 5$). The window $\{\text{min}:0.1, \text{max}:3.0\}$ is *computed* from the passing set; it has a real upper edge. The honest caveat: clean two-mode truncation holds only for order-unity wall stiffness $K_\eta/T_w \lesssim 2.6$ (the emergent edge, since $\beta^2 = K_\eta/T_w$ and $(3/1.85)^2 \approx 2.63$) — an emergent number, not a typed criterion. The physical $\beta_{L_0} = 37/20$ is the *cited* Gate-1 anchor, not the β that maximizes overlap. *The pathA_31 v1 rejection was a gamed truncation threshold; the floor and window here are genuinely computed and able-to-fail.*

N-convergence and the counterfactual overlaps. At $\beta_{L_0} = 37/20$, o_1 is stable across $N \in \{4, 8, 12, 16\}$ ($0.99301 \rightarrow 0.99311$, $\max|\Delta| \approx 10^{-4}$), $\text{pass} = \text{true}$ at every N — so the certificate is not a small- N artifact (the mass condition growth $3.1\text{e}3 \rightarrow 1.9\text{e}5$ is a benign conditioning effect). Two counterfactual profiles pin the honest scope: the *degenerate* $\alpha_a \equiv 0$ collapses the collective span (rank deficient, $o_2 = 0.2227 < \text{FLOOR}$) and **fails** the floor; the *constant* $\alpha_a \equiv 1$ — a *wrong* profile — gives $o_1 = 1.0$, $o_2 = 0.9738 \geq \text{FLOOR}$ and **passes** the overlap. *The overlap therefore certifies truncation-consistency, not profile-correctness*: Stage 014 records that the constant profile passes and does *not* claim it is rejected — its rejection is Stage-015’s Hellmann–Feynman mismatch (and Stage-013’s $\mathcal{L}_0[\alpha] = 0$ residual).

Consumed closure and register. The collective profiles and the frozen packet $\{L_0 = 37/20, \beta = 1, \beta_{L_0} = 37/20\}$ are cited from Stage 013 with two independently-corruptible relations: site A (branch anchor) $\beta_{L_0} - 37/20 \equiv 0$, reading an independent cited datum; and site B (profile-consumption) the consumed profiles satisfy *both* the harmonic residual $-\alpha'' + \beta^2\alpha = 0$ *and* the collective BC values — the residual alone misses a kernel-preserving corruption (a rescale leaves the residual 0 but breaks the BC), so residual $\wedge$ BC are both able-to-fail. Stage 014 adds **zero new counted knobs**: the numeric controls $\{\text{FLOOR} = 0.9, N_{\text{final}} = 16, N_{\text{conv}}, \beta_{L_0}\text{-sweep}\}$ are method/tolerance parameters (tracked, not counted); the wall packet is consumed from Stage 013 (no double-count). One structural edge, R31 (the truncation validity window $K_\eta/T_w \lesssim 2.6$), discharges nothing.

Verification. Dual-engine, independent numeric routes, zero file I/O — the source pair’s numeric-WL scratch export and the hybrid .wl’s `Get/sympy*-float-diff/Export` are retired; the .wl **KEEPS** its native `NIntegrate+Eigensystem galerkinRow` route but its asserts are *re-targeted* to its own computed overlaps, floor, and window (it declares its own sweep grid and the cited profile closed forms natively, not via `DSolveValue`), with a genuine arity self-check. Agreement is transcript-level (both engines independently reach the same pass-pattern, window, and o_k to $\sim 10^{-13}$, tighter than the 0.9 margin). The 3-way cut is clean: no Stage-013 $\int dw$ projection and no Stage-015 Hellmann–Feynman force is computed. SymPy/SciPy 93 PASS / Mathematica 100 PASS (the +7 = the .wl arity/leakage block), exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (three BLOCKING folds: the site-B residual $\wedge$ BC, the $\min(\omega^2) > 0$ tooth, and the N-convergence

non-converging ablation; Grok’s compute-verification independently re-derived every anchor). Full tri-review on fresh agents: FIDELITY_CLEAN (an independent from-scratch Galerkin re-derived every value) and ADVERSARIAL_ISSUES — the anti-gaming floor/window proven clean (23/23 genuine ablations firing), but *two* able-to-fail teeth (the constant-passes-overlap honesty leg and the degenerate-bypass leg) were vacuous $x:=\text{const}$; $\text{expect_fail}(x)$ stamps in both engines. Remediated to a *shared predicate* read by both the baseline and the tooth’s mutated-copy assertion; a fresh-agent re-verify confirmed REVERIFY_CLEAN via the deciding coupling meta-test (corrupting the shared predicate makes the tooth stop firing, so the audit correctly fails). Nine able-to-fail teeth, mutations on copies.

B.10 Stage 015: Legacy-Hessian Structure Recovery & the Hellmann–Feynman Force (Check II-G2c)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 0$ breathing mode; the legacy-structure + Hellmann–Feynman-force component of the joint BREATHING_CALIBRATED, which it *completes* — with the M/K-projection + (a, L) -EOM-LHS = Stage 013 and the truncation consistency = Stage 014)

Claim status. EXACT WITHIN CLOSURE

Purpose. Complete the calibrated breathing sector. **(Q1)** Show Stage-013’s operator-projected closure M_{AB}, K_{AB} has the same structural signature as the legacy static energy Hessian $H_{\text{legacy}} = \partial^2 E_{\text{geom}} / \partial Q^2$ (the (a, L) closure *recovered*, not re-postulated). **(Q2)** Supply the EOM driving force $F_A^{(\text{HF})}$ — the right-hand side Stage 013 deferred — by *two genuinely different* constructions (a distributed source projection and a Hellmann–Feynman parametric derivative) that agree only after simplification. Then take the static-dynamic limit and land the completed joint verdict.

Status. EXACT WITHIN CLOSURE — the structure recovery and the Hellmann–Feynman force are exact closed-form algebra (`sp.integrate/FullSimplify`; `expect_zero` residuals, exact rationals $L_0 = 37/20$; the symbolic / float-free inverse of the numeric Stage 014). The verdict is the joint BREATHING_CALIBRATED; this stage carries its *legacy-structure + HF-force* component (3/3) and **completes** it, filling the EOM RHS $F_A^{(\text{HF})}$. It adds **one** new counted CALIB knob — the breathing driving-force scale V_{p0}/ℓ_c (with V_{p0} and the now-live ℓ_c its manifestations; ρ_* consumed) — making it the second calibration-adding Part-II stage after Stage 013. Stage-013’s profiles and M_{AB}/K_{AB} and Stage-014’s truncation certificate are *consumed* with dual-site citation-integrity; the legacy constants $\{\kappa, \chi, \sigma_a, \sigma_L\}$ are the legacy-Hessian pattern basis (not counted afresh); c_S is *not* consumed (matter-sector deferred, $k\xi \ll 1$).

The legacy energy and its Hessian. With $E_{\text{geom}} = \frac{1}{2}\kappa(\delta_L - \chi\delta_a)^2 + \frac{1}{2}\sigma_a\delta_a^2 + \frac{1}{2}\sigma_L\delta_L^2$,

$$H_{\text{legacy}} = \frac{\partial^2 E_{\text{geom}}}{\partial Q^2} = \begin{pmatrix} \chi^2\kappa + \sigma_a & -\chi\kappa \\ -\chi\kappa & \kappa + \sigma_L \end{pmatrix},$$

symmetric, negative off-diagonal ($-\chi\kappa < 0$ for $\chi, \kappa > 0$), $\det = \kappa\sigma_a + \kappa\chi^2\sigma_L + \sigma_a\sigma_L > 0$. The legacy constants enter *here*; a firewall echo confirms the cited M_{AB}/K_{AB} free symbols exclude them.

The structure recovery. The gate reads the cited M_{AB}/K_{AB} and matches the structural signature of H_{legacy} : M positive-definite by Sylvester — $M_{aa} > 0$ and $\det M > 0$ for $B = \beta L_0 > 0$ — discharged by *exact-identity certificates* (not `is_positive`, which returns `None` on the transcendentals, and *not* the $M_{aa} - m_{aa}^+ = 0$ form-equality, which is $X \equiv X$ once M is cited): $f_1 = \sinh B \cosh B - B$ has $f_1(0) = 0$, $f_1' = \cosh 2B - 1 = 2 \sinh^2 B \geq 0$; $f_2 = \sinh^2 B - B^2 = (\sinh B - B)(\sinh B + B) > 0$ via $g = \sinh B - B$, $g' = \cosh B - 1 = 2 \sinh^2(B/2)$. Then K symmetric, $K_{aL} < 0$, $\text{rank } K = \text{rank } H_{\text{legacy}} = 2$, and matching zero-pattern give $K_{\text{structure ok}}$; full matrix fit = false (the recovery is structural, $M_{aa} \neq H_{aa}$). The probes $M_{aa} \rightarrow -M_{aa}$ and $K_{aL} \rightarrow -K_{aL}$ flip the gate. *The operator-projected closure has the same structural form as the legacy energy Hessian.*

The Hellmann–Feynman force (both routes). With the confinement force density $\text{src} = \rho_* V_{p0}/\ell_c$ and the action measure $4\pi \int dw$:

$$\underbrace{F_A^{\text{dist}} = 4\pi \int_0^{L_0} \text{src } \alpha_A dw}_{\text{distributed projection}} = \underbrace{F_A^{\text{leg}} = -4\pi \int_0^{L_0} \rho_* \frac{\partial V_{\text{conf}}}{\partial q_A} \Big|_0 dw}_{\text{Hellmann–Feynman parametric}}$$

$V_{\text{conf}} = (V_{p0}/\ell_c)(r - R_0(w) - q_a \alpha_a - q_L \alpha_L)$, Route 2 a genuine $\partial/\partial q$ (the double-negative). They agree (hf force reduces = true, $F_a^{\text{dist}} = 4\pi V_{p0} \rho_* (\cosh L_0 \beta - 1)/(\beta \ell_c \sinh L_0 \beta)$), yet the *raw* unsimplified trees are computed-distinct: unsimplified routes identical = false ($+4\pi f(+\dots)$ vs $-4\pi f(-\dots)$). *This is the pathA_31 v1 rejection's other locus — the HF $x-x$ tautology — emitted here as two genuinely different constructions; the verdict rung places the anti- $x-x$ in the OR: typing Route 2 as a copy of Route 1 trips BREATHING_FAIL_HF_FORCE.* This fills the deferred RHS $F_A^{(\text{HF})}$.

Static-dynamic limit and the profile guard. $\ddot{Q} \rightarrow 0$ in $M_{AB}\ddot{Q} + K_{AB}Q = F_A$ gives $K_{AB}Q = F_A$, the static $\partial E_{\text{geom}}/\partial Q = 0$ limit (the exact $\ddot{Q} \rightarrow 0$ EOM residual identity, not a separate solve). The HF guard rejects both wrong profiles ($\alpha_a \equiv 0$: $F_a^{\text{dist}} = 0$; $\alpha_a \equiv 1$: $F_a^{\text{dist}} = 4\pi L_0 V_{p0} \rho_*/\ell_c$; both $\neq F_a^{\text{leg}}$, so hf mismatch = true). *The 014↔015 boundary:* the constant profile *passes* Stage-014's modal overlap but *fails* Stage-015's HF — the Hellmann–Feynman mismatch is the profile guard the overlap could not supply (with Stage-013's $\mathcal{L}_0[\alpha] = 0$ residual).

Consumed closure and register. Stage-013's profiles, M_{AB}/K_{AB} , and the frozen packet $\{L_0 = 37/20, \beta = 1, \beta L_0 = 37/20\}$ are cited with dual-site integrity: site A the branch anchor $\beta L_0 - 37/20 \equiv 0$; site B the profiles satisfy *both* the harmonic residual $-\alpha'' + \beta^2 \alpha = 0$ *and* the BC values (residual alone misses a kernel-preserving corruption); and the M/K citation covers every entry — two det-identities for the diagonals, and, because det carries M_{aL}^2 (blind to an off-diagonal sign flip), explicit sign checks $M_{aL} > 0$ (via $h = B - \tanh B$) and $K_{aL} < 0$. Register: **one** new counted CALIB knob, the driving-force scale V_{p0}/ℓ_c ; the legacy $\{\kappa, \chi, \sigma_a, \sigma_L\}$ are the pattern basis (not counted afresh). Two structural edges: R32 (the legacy-Hessian structure recovery, discharges nothing) and R33 (the confinement-drive reduction debt, a sibling of R30's wall-response debt, PENDING).

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's `sympy*` export and the hybrid `.wl's Get/checks/Export/digest` are retired; the `.wl` KEEPS its native HF `Integrate`, Hessian D, and structure `Det/MatrixRank`, re-targeted to its own values, citing the profiles and M/K as native literals (not `DSolveValue`, not `re-Integrate`) and building its own H_{legacy} , HF routes, and raw-tree anti- $x-x$ comparison, with an arity self-check. Agreement is transcript-level (the `.wl` $\tanh(\beta L_0/2)$ form equals the `.py` $(\cosh L_0 \beta - 1)/\sinh L_0 \beta$ via the half-angle identity). The 3-way cut is clean: no Stage-013 $\int dw$ projection, no Stage-014 generalized eig. SymPy 95 PASS / Mathematica

102 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (two BLOCKING folds: the det-blind off-diagonal sign hole $M_{aL} \rightarrow -M_{aL}$, added explicit sign checks; and the float-free positivity-certificate requirement — $M_{aa} > 0$ is not dischargeable by `is_positive`, so exact-identity certificates replace the vacuous form-equality). Full tri-review on fresh agents: **FIDELITY_CLEAN** (an independent read hand-verified all seven certificate identities and every value) and **ADVERSARIAL_ISSUES** — the three central burdens proven genuine by live copy-mutations, but *one* able-to-fail tooth (the HF profile-guard leg) was a vacuous $x-x$ self-compare in both engines. Remediated to read the real wrong-profile forces (with three lower-severity literal/scaffolding flags made-genuine or honestly de-counted); a fresh-agent re-verify confirmed **REVERIFY_CLEAN** via the coupling meta-test (each remediated construct fires under a mutation of its named object). Eight able-to-fail teeth, mutations on copies.

B.11 Stage 016: The $\ell = 2$ SO(3) Covariance Theorem (Check II-G3a)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ sector; the EARNED-FIRST leg of a two-way split of pathA_32 — the SO(3) covariance-theorem component of the joint **ISOTROPY_CALIBRATED**, with the grouped-P2 lane isotropy + the calibration partition = Stage 017)

Claim status. EXACT WITHIN CLOSURE

Purpose. Earn the $\ell = 2$ angular structure the grouped-lane isotropy rides. Show the five real $\ell = 2$ harmonics form one orthonormal SO(3)-irrep (Gram = I_5) with a single *computed* $-\Delta_{S^2}$ eigenvalue $\lambda_m = \ell(\ell + 1) = 6$ for every m (Laplace–Beltrami + Rayleigh quotient + eigenfunction residual), and that the K_2 angular stiffness $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$ uses that computed eigenvalue. Land the joint verdict as a partial component.

Status. EXACT WITHIN CLOSURE — the covariance theorem is exact symbolic algebra (genuine S^2 -integral Gram, the real $-\Delta_{S^2}$ operator, Rayleigh quotients, eigenfunction residuals, exact $\{L, M, T\}$ dimension vectors; `expect_zero/expect_bool` residuals, no `scipy/numpy/` floats/tolerances). The verdict is the joint **ISOTROPY_CALIBRATED**; this stage carries its *SO(3) covariance-theorem* component (1/2, the EARNED slice) and lands the joint as **partial** (016 earned, 017 pending) — not complete. It adds **zero** new counted knobs (a structural covariance edge, like Stages 011/012/014); the new symbols first appearing here — $T_\Omega/\tilde{T}_\Omega$ (the angular-stiffness density) and $\beta_2(w)$ (the frozen $\ell = 2$ radial profile) — have their counting *deferred* to Stage-017’s calibration partition. c_S is *not* consumed (the angular structure is speed-free).

The real basis and orthonormality. The five real $\ell = 2$ harmonics $Y_{20} = \sqrt{5/16\pi} (3 \cos^2 \theta - 1)$, $Y_{21c} = -\sqrt{15/4\pi} \sin \theta \cos \theta \cos \phi$, $Y_{21s} (\cos \phi \rightarrow \sin \phi)$, $Y_{22c} = \sqrt{15/16\pi} \sin^2 \theta \cos 2\phi$, $Y_{22s} (\cos 2\phi \rightarrow \sin 2\phi)$ are orthonormal on the sphere:

$$\text{Gram}_{ij} = \int_{S^2} Y_i Y_j d\Omega = \int_0^\pi \int_0^{2\pi} Y_i Y_j \sin \theta d\phi d\theta = \delta_{ij}, \quad \text{Gram} = I_5.$$

This is a computed integral — a wrongly-normalized harmonic breaks gram is identity and drives *covariant* to **FAIL_NOT_COVARIANT**.

The computed eigenvalue (the crux). With the real Laplace–Beltrami operator $-\Delta_{S^2} f = -[(1/\sin \theta)\partial_\theta(\sin \theta \partial_\theta f) + (1/\sin^2 \theta)\partial_\phi^2 f]$, two independent computations certify λ_m is *computed*, not typed: the Rayleigh quotient $\lambda_m = \int Y_m(-\Delta_{S^2})Y_m d\Omega / \int Y_m^2 d\Omega = 6$ for every m , *and* the eigenfunction residual $(-\Delta_{S^2})Y_m - \lambda_m Y_m = 0$ (proving each Y_m is a genuine eigenfunction). So $\lambda_m = \ell(\ell + 1) = 6$ is the same across all five m : the $\ell = 2$ sector is one 5-dimensional SO(3)-irrep with a single eigenvalue — the angular degeneracy that *is* the covariance.

The K_2 angular stiffness. The angular stiffness $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$ (with $M_2 = \tilde{M}$) is assembled with the *live* computed eigenvalue (`build_K2(lambdas)`, not a typed 6). Its computed-ness is able-to-fail through a residual on the assembled coefficient $((-\Delta_{S^2})Y - (\text{coeff in } K_2)Y = 0$, which a wrong typed coefficient breaks) *and* the *bare* forced eigenvalue probe (coefficient $2 \neq 6 \Rightarrow \text{FAIL_NOT_COVARIANT}$). The source’s $k_{\text{coeff equal}}$ *self-compare* $\lambda - \lambda \equiv 0$ *is vacuous and is de-counted*; its role is carried by these two genuine checks.

Angular dimensional consistency. On pathA_32’s own convention — volume densities on the wall measure $dV = a^2 dw d\Omega$ (dimension L^3), β_2 dimensionless, $\beta_2' = L^{-1}$, $[\mu_\eta] = ML^{-3}$, $[T_w] = ML^{-1}T^{-2}$, $[K_\eta] = [T_\Omega] = ML^{-3}T^{-2}$ — one finds $[M_2] = M$, each K_2 term MT^{-2} , $[K_2/M_2] = T^{-2}$. The dimensions are *sourced* from the density integrals (not back-solved). Corrupting the sourced $[T_\Omega]$ (and $\tilde{T}_\Omega$) by one power of L makes the K_2 sum inhomogeneous $\Rightarrow \text{FAIL_DIMENSIONAL}$; the same holds for a one-power corruption of $[\mu_\eta]$, $[T_w]$, or $[K_\eta]$.

The able-to-fail battery. The scoped verdict runs 016’s gates only (dimensional, covariant, tautology, able-to-fail; 017’s stability/denominator/lane/dynamic gates are not computed here). The aggregate battery is 016’s three probes — wrong eigenvalue (coeff $2 \neq 6 \Rightarrow \text{FAIL_NOT_COVARIANT}$), tautology hash collision (identical harmonic inputs $\Rightarrow$ non-distinct SHA-256 hashes $\Rightarrow \text{FAIL_TAUTOLOGICAL}$), and dimensional corrupt T_Ω ($\Rightarrow \text{FAIL_DIMENSIONAL}$) — each a computed conjunction of (its mutation fires its verdict) $\wedge$ (its self-ablation suppresses); *neutering any one flips the aggregate false*.

Honest scope. Angular earned / radial calibrated: 016 derives the angular structure, but the radial profile $\beta_2(w)$ and the radial/support scalars remain frozen calibration inputs (not derived from the Gate-1 R_0 support equation) — which is *why* the joint is ISOTROPY_CALIBRATED rather than ISOTROPY_PASS; the calibration partition is Stage-017’s. The 54/5 quadrupole normalization, outgoing odd- N coefficients, and the solved nonlinear branch ($G = \text{GENUINE_BLOCKED}$) are deferred Gate 4/5/6 work.

Consumed closure and register. *Provenance, not a cross-stage relation:* pathA_32’s volume-density / dimensionless- β_2 convention differs from Stage-013’s line-density / $\beta = L^{-1}$ convention (related by an $\int a^2 d\Omega \approx L^2$ bridge, not equal), so the Stage-013 relation $K_\eta = T_w \beta^2$ does *not* transfer. Hence μ_η, T_w are cited as the same physical wall constants counted at Stage 013 (provenance), the frozen radial reference and the Gate-1 D/N boundary (Stage 012 R28) are cited, and the genuine able-to-fail integrity is pathA_32’s *own* $[M_2] = M/[K_2] = MT^{-2}$ dimensional consistency. Register: **zero** new counted knobs; the new T_Ω/β_2 counting is deferred to Stage 017; structural edge **R34** (the $\ell = 2$ SO(3) covariance provenance — five real harmonics = one SO(3)-irrep with $\lambda_m = 6$; discharges nothing).

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair’s scratch-YAML engine-agreement handoff is severed; the .w1 KEEPS its native **Integrate/ D/Hash** route (its own `dimOf/scopedVerdict`), not a mirror, agreeing at the transcript level (Gram = I_5 , $\lambda_m = 6$, the three probe verdicts, the 016 aggregate), with an arity self-check. SymPy 82 PASS / Mathematica 91 PASS,

exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend; the Grok compute-verify pass caught the volume-vs-line dimensional-convention mismatch (the cross-stage dual-site was non-transferable → reframed to provenance + self-contained integrity), and strengthened the $k_{\text{coeff equal}}$ de-count to a residual on the assembled K_2 coefficient. Full tri-review on fresh agents: **FIDELITY_CLEAN** (an independent read hand-re-derived the harmonics, the Laplace–Beltrami operator, all Rayleigh $\lambda = 6$ with zero residuals, and the full $L/M/T$ dim walk) and **ADVERSARIAL_CLEAN** (per-tooth ablation: all 82 SymPy checks + 4 native .wl ablations fired at their own assert) — with one low-severity stamped literal (participates in verdict) remediated to a computed verdict-propagation check; a fresh-agent **REVERIFY_CLEAN** confirmed it fires under its own ablation. Symbolic per-tooth ablation, mutations on copies.

B.12 Stage 017: Grouped-P2 Lane Isotropy (Check II-G3b)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ sector; the COMPLETING leg of the two-way split of pathA_32 — the grouped-lane isotropy + calibration partition that consumes Stage 016’s covariance theorem, lands the joint **ISOTROPY_CALIBRATED**, and exports the $\ell = 2$ port kernel)

Claim status. EXACT WITHIN CLOSURE

Purpose. Show that the grouped $\{20, 21, 22\}$ $\ell = 2$ lanes, assembled from Stage 016’s consumed covariance ($\lambda_m = 6$, $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$), respond isotropically — the P_2 -projection raw-D defects vanish (primary) and the normalized-u response defects vanish (cross-check) — and that the response is genuinely able-to-fail (the six-probe battery). Land the joint verdict as complete and export the port kernel.

Status. EXACT WITHIN CLOSURE — mixed exact-symbolic + numeric-sample: the lane algebra, D-lanes, raw-D/normalized-u defects, and the c/s degeneracy are exact symbolic (`compact(expr)==0`, `expect_zero/expect_bool`); the response-probe “moves under ε ” tests and the stability/denominator windows are numeric over a calibration sample/window (tolerance-guarded, as the source). The verdict is the joint **ISOTROPY_CALIBRATED**; this stage carries its *grouped-lane isotropy* component (2/2) and lands the joint as **complete** (016 earned-cited $\wedge$ 017 earned-here). It is the *third* calibration-adding Part-II stage — it adds two counted **CALIB** knobs $\{T_\Omega, \beta_2\}$.

The grouped-lane assembly. Each channel is assembled from the consumed eigenvalue $\lambda_m = 6$ and the K_2 form:

$$M_2 = c_{\text{self}} \tilde{M}, \quad K_2 = c_{\text{self}} (\tilde{K} + \lambda_m \tilde{T}_\Omega), \quad c_{\text{self}} = \int Y_m^2 d\Omega = 1 \quad (\text{the Gram diagonal, cited from 016}),$$

with D-lanes $D_0 = K_2 - (\tilde{B}_0 + \tilde{Z}_0)$, $D_2 = -(\tilde{M} + \tilde{B}_2 + \tilde{Z}_2)$, $D_4 = -(\tilde{B}_4 + \tilde{Z}_4)$. The three grouped lanes are 20, 21 = avg(21c, 21s), 22 = avg(22c, 22s); this grouped $\{M_2, K_2, D_{0,2,4}, \tilde{B}/\tilde{Z}\}$ is the exported $\ell = 2$ port kernel. $\lambda_m = 6$ and the K_2 form are cited from Stage 016, not re-derived.

Raw-D defects = 0 — the primary isotropy test. With the P_2 -projection defect_pair(t) = $((2t_{20} - t_{21} - t_{22})/10, (t_{21} - t_{22})/2)$, the grouped D-lane defects vanish at every order, $\{a_{D0}, b_{D0}, a_{D2}, b_{D2}, a_{D4}, b_{D4}\} = 0$ (exact), with the within-group degeneracy $D_{21c} = D_{21s}$, $D_{22c} = D_{22s}$. This is the isotropy statement: the grouped $\ell = 2$ response is angularly degenerate because the lanes ride 016’s single eigenvalue $\lambda_m = 6$.

Normalized-u defects = 0 — the cross-check. The DtN-observable normalized responses $u_2 = -D_2/D_0$, $u_4 = (D_2^2 - D_0 D_4)/D_0^2$ also have vanishing P_2 -projection defects. This is a *cross-check*, not the primary test: a pure-prefactor anisotropy *moves* the raw-D defects but leaves the normalized-u defects zero (the pure prefactor probe), so normalized-u alone would miss a real anisotropy — raw-D is decisive.

The consumed covariance — a genuine cross-stage dual-site. Because Stages 016 and 017 share the same pathA_32 convention (volume densities, dimensionless β_2) — unlike 016’s provenance-only cite of Stage 013 — the cited $\lambda_m = 6$ and K_2 form a *checkable* relation. Site A: the λ_m fed into the lane assembly equals 016’s exported eigenvalue. Site B: the *assembled* lane K_2 ’s $\tilde{T}_\Omega$ coefficient equals the cited λ_m (reads the assembled lane, guarding the assembly formula). The λ -dimensionless trap: a corruption $\lambda : 6 \rightarrow 4$ is dimensionally silent, keeps the windows positive, and *keeps raw-D isotropy incidentally unbroken* (all-lanes $\lambda = 4$ gives raw-D= 0) — so an incidental gate cannot catch it. A one-site corruption fires an explicit integrity guard; the coordinated-both-sites corruption is caught by a single-harmonic $(-\Delta_{S^2})Y_{20} = \lambda Y_{20}$ echo. Stage 017 does not re-derive the covariance theorem.

The able-to-fail battery. 017’s aggregate is its six response probes, each a computed conjunction; *neutering any one flips* able to fail ok *false*: pure prefactor/sector selective/ m dependent ($\rightarrow$ FAIL_ANISOTROPIC_BRANCH, load-bearing on the computed move-flags — their forced verdict via `case_verdict(lane_collapse_ok=False)` is vacuous by design), degenerate β_2 ($\rightarrow$ FAIL_STABILITY), singular denominator ($\rightarrow$ FAIL_SINGULAR_RESPONSE), static drop inertia ($\rightarrow$ FAIL_STATIC_RESPONSE). The three 016-owned probes are cited via 016’s three-probe aggregate; the joint able to fail is 017’s six $\wedge$ 016’s cited three.

Honest scope and register. Calibrated not PASS: 017 earns the grouped-lane isotropy, but the wall profile $\beta_2(w)$, $R_0(w)$, the radial scalars $\tilde{M}/\tilde{K}/\tilde{T}_\Omega$, the support scalars $\tilde{B}/\tilde{Z}$, and the calibration window are frozen calibration inputs — *not* derived from the Gate-1 R_0 support equation (edge **R36**, the ISOTROPY_PASS target). Register: two new counted CALIB knobs $\{T_\Omega, \beta_2\}$ (the $\ell = 2$ angular-stiffness density — a genuinely new physical DOF absent from the $\ell = 0$ breathing packet — and the frozen $\ell = 2$ radial profile). The radial scalars $\tilde{M}/\tilde{K}/\tilde{T}_\Omega$ are *derived* manifestations $= \int \rho \beta_2^2 dV$ (edge **R35**, the $\ell = 2$ analogue of R29). The port-kernel support/Maxwell scalars $\{\tilde{B}, \tilde{Z}\}$ are frozen inputs first load-bearing here but tracked/downstream-pinned (the isotropy verdict is value-independent of them; the $\tilde{Z}$ Maxwell couplings pin at Stages 018–024). μ_η, T_w are cited provenance (Stage 013; $K_\eta = T_w \beta^2$ R29 non-transferable across the volume-vs-line convention); c_S is not consumed. Edge **R34** (016’s covariance provenance) is backfilled. Part-II counted CALIB set $= \{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$. Deferred (Gate 4/5/6): the 54/5 normalization, outgoing odd- N coefficients, and the solved nonlinear branch (G =GENUINE_BLOCKED).

Verification. Dual-engine, independent symbolic + native routes, zero file I/O — the source pair’s scratch-YAML handoff is severed; the .wl keeps its native Integrate/FullSimplify/ D/N route (its own `verdictFromGates/caseVerdict/dimOf`), not a mirror, agreeing at the transcript level (the grouped lanes, raw-D= 0, normalized-u= 0, the six probe verdicts, the joint ISOTROPY_CALIBRATED), with an arity + unevaluated-leakage self-check. SymPy 118 PASS / Mathematica 127 PASS, exit 0, CWD-independent. The directive design-review used the Codex $\rightarrow$ Grok $\rightarrow$ Codex bookend; the Grok compute-verify pass caught a blocking gap (the harmonics/intS2 infrastructure 017 needs for its anisotropy coefficients had been mislabeled for exclusion) plus two nits (the export coefficient pinned to $\tilde{K} + 6\tilde{T}_\Omega$; the three anisotropy probes’ forced verdict flagged vacuous so the load rides the move-flags), and independently confirmed that all-lanes $\lambda = 4$ keeps raw-D= 0 — proving the explicit dual-site necessary. Full tri-review on fresh agents:

FIDELITY_CLEAN (independent re-derivation of the anisotropy coefficients $\{2/7, 1/7, -2/7\}$, the Y_{20} echo, the pure-prefactor discriminator, the six probe verdicts, and the joint landing) and ADVERSARIAL_ISSUES (42 per-tooth ablations, 41 firing at their own assert) with one genuine finding — the .w1 Site-B (K_2 -form) residual reconstructed K_2 from `lambdaByChannel` instead of reading the assembled lane, so an assembly-formula corruption fired only downstream. Remediated: the .w1 Site-B now reads the assembled lane (mirroring SymPy), with matching assembly-formula ablation teeth in both engines. A fresh-agent REVERIFY_CLEAN confirmed the coupling meta-test (the fixed tooth fires at its own named assert on an assembly-formula corruption in both engines; neutering the fix makes it vacuous), with no regression. Symbolic/numeric per-tooth ablation, mutations on copies.

B.13 Stage 018: The Outgoing $\ell = 2$ DtN Hankel Fingerprint and χ_Q Sign (Check II-G4a)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative sector; the EARNED-FIRST leg of a four-way split of pathA_33 — the outgoing DtN Hankel-fingerprint + χ_Q -sign component of the joint QUAD_CALIBRATED, with the prefactor algebra = Stage 019, the $54/5 = 2 \cdot 27/5$ provenance partition + the CALIBRATED label = Stage 020, and the $\hat{\mu}_0$ -free dimensional closure = Stage 021)

Claim status. EXACT WITHIN CLOSURE

Purpose. Earn the exterior-wave radiative signature the quadrupole normalization rides. Show the outgoing $\ell = 2$ DtN response $\hat{Y}_2^{\text{out}}(z) = -3/\Lambda_2^{\text{out}}(z)$, $\Lambda_2^{\text{out}} = z h_2^{(1)'} / h_2^{(1)}$, $z = a\omega/c_s$, yields the DERIVED rational fingerprint $u_2 = a^2/9c_s^2$, $u_4 = 4a^4/81c_s^4$, $v_5 = a^5/27c_s^5$ (the 27 computed from $v_5^z = 1/27$), and the sign $\chi_Q = +1$ outgoing / -1 incoming computed from $j_2 \pm i y_2$. Land the joint verdict as a partial component.

Status. EXACT WITHIN CLOSURE — the fingerprint is exact symbolic algebra (the outgoing $\ell = 2$ spherical-Hankel logarithmic derivative and its small-frequency series are exact expansions; `expect_zero/expect_bool/expect_fail` residuals, no `scipy/numpy/floats/tolerances`). The verdict is the joint QUAD_CALIBRATED; this stage carries its *outgoing-fingerprint + χ_Q -sign* component (1/4, the EARNED-FIRST slice) and lands the joint as **partial** (018 earned; 019/020/021 pending) — not complete. It adds **zero** new counted knobs (an EARNED/structural fingerprint slice, like Stages 016/011/012/014); c_s is a units carrier only (the earned rationals and χ_Q are c_s -free), cited from Stage-005 R1 as provenance, not a consumed value.

The outgoing DtN log-derivative. The exterior $\ell = 2$ outgoing solution is the spherical Hankel function $h_2^{(1)}(z) = j_2(z) + i y_2(z)$, $z = a\omega/c_s$, with $j_2 = (3/z^3 - 1/z) \sin z - (3/z^2) \cos z$, $y_2 = (1/z - 3/z^3) \cos z - (3/z^2) \sin z$ built from explicit rational $\cdot \sin / \cos$ (no stored artifact). The DtN eigenvalue and normalized response are

$$\Lambda_2^{\text{out}}(z) = z \frac{h_2^{(1)'}(z)}{h_2^{(1)}(z)}, \quad \hat{Y}_2^{\text{out}}(z) = -\frac{3}{\Lambda_2^{\text{out}}(z)},$$

where $-3 = -(\ell + 1)$ for $\ell = 2$ is the outgoing static DtN slot (self-derived). This is self-contained exterior-wave algebra: it does *not* consume the interior port kernel (that enters Stage-019's prefactor).

The fingerprint (the EARNED headline). Expanding $\hat{Y}_2^{\text{out}}$ about $z = 0$,

$$\hat{Y}_2^{\text{out}}(z) = 1 + \frac{1}{9}z^2 + \frac{4}{81}z^4 + i\frac{1}{27}z^5 + O(z^6),$$

so $u_2^z = 1/9$, $u_4^z = 4/81$, $v_5^z = 1/27$ (the odd z^1, z^3 vanish; the z^5 term is purely imaginary — the radiating part, $\text{coeff}(z, 5)/i$). Restoring units through $z = a\omega/c_s$: $u_2 = a^2/9c_s^2$, $u_4 = 4a^4/81c_s^4$, $v_5 = a^5/27c_s^5$. The 27 is EARNED as $1/v_5^z$ — the radiative-slot denominator, series-derived, *not* a typed literal (this is the `derived_in_gate` 27 that Stage-020's $54/5 = 2 \cdot 27/5$ partition rides). The coefficients are read off the *actual* symbolic series of $-3/(z h_2^{(1)'}/h_2^{(1)})$ with $h_2^{(1)}/\Lambda_2^{\text{out}}/\hat{Y}_2^{\text{out}}$ emitted — not hardcoded and compared. Perturbing the typed target (u_2^z expected $1/9 \rightarrow 1/8$) fires the assert, and mutating the *derivation* ($h \cdot (1 + z^4) \rightarrow u_4^z = 112/81$, leaving u_2^z, v_5^z) fires exactly that coefficient's `expect_fail` — a genuine per-coefficient firewall.

The χ_Q sign. Against the canonical radiative slot $a^5/(27c_s^5)$, $\chi_Q = v_5/(a^5/27c_s^5) = +1$ outgoing ($h_2^{(1)}$) and -1 incoming ($h_2^{(2)} = j_2 - i y_2$); the sign *emerges* from $j_2 \pm i y_2$ (the z^5 coefficient flips), never a typed $+1$. The standing branch j_2 gives $\Lambda_{\text{stand}}(0) = +2$ ($= +\ell$, not the outgoing -3), $\hat{Y}_{\text{stand}}(0) = -3/2 \neq 1$, and no radiating term — proving the $+1/27$ rational is outgoing-BC-selected, not universal.

Passivity and the units-restored dim leg. The radiating $v_5\omega^5$ arises from the genuine outgoing BC: the imposed-dissipation probe replaces it with a phenomenological sink and is rejected (`FAIL_NOT_OUTGOING`) by source-label provenance, with a genuine *dynamic* self-ablation re-run (with $\neq$ without, not a constant — the `pathA_33 v1` trip-up avoided); the static $\leftrightarrow$ dynamic consistency is the fingerprint's own static slot $\hat{Y}_2^{\text{out}}(\omega \rightarrow 0) = 1$ (*not* the prefactor $P_0 = N_0/D_0$, which is 019's). The units-restored dim leg runs on the real physical coefficients ($[u_2] = T^2$, $[u_4] = T^4$, $[v_5] = T^5$, with $[a] = L$, $[c_s] = LT^{-1}$); corrupting an a/c_s power fires `FAIL_DIMENSIONAL`. This is *not* the $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ gate (that is 021's) — the 018 slice contains no $\hat{\mu}_0/N_0/D_0/P_0/\text{build_dimensions}$ construct, keeping the back-solved- $\hat{\mu}_0$ v1 rig out.

Honest scope. Earned exterior-wave signature / calibrated magnitude: 018 derives the fingerprint shape ($1/9, 4/81, 1/27$) and the $\chi_Q = +1$ sign, but the $54/5$ normalization magnitude and G are calibrated ($G = \text{GENUINE_BLOCKED}$), landed at Stage 020; the prefactor algebra is Stage 019; the $\hat{\mu}_0$ -free dimensional closure is Stage 021. The actual-branch a -scaling and the numerical port scalars N_n/D_n are deferred Gate 4/5/6 work. The χ_Q reconciliation (`pathA_33 +1` vs `pathA_22b` ≈ 0.712 — same name, different computations) is a tracked Part-VII item, not merged here.

Consumed closure and register. *Provenance only, no dual-site:* 018 is self-contained exterior-wave algebra, so — unlike Stage-017's genuine cross-stage dual-site — there is no checkable relation to guard. 017's $\ell = 2$ port kernel (the interior wall mode; literal $N_n/D_n = 019$'s), 009/010's bulk Helmholtz mode, and c_s (Stage-005 R1) are cited as narrative provenance with no guard on any object the fingerprint never uses; c_s is an inert units carrier, and c_S (the frozen-wall speed) is not a live symbol here. Register: **zero** new counted knobs; structural edge **R37** (the outgoing-DtN $\ell = 2$ Hankel-fingerprint provenance — $u_2 = a^2/9c_s^2$, $u_4 = 4a^4/81c_s^4$, $v_5 = a^5/27c_s^5$ with the 27 computed from v_5^z , and $\chi_Q = +1$ outgoing / -1 incoming; discharges nothing). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed; the `.w1` KEEPS its native `j2/y2/Series/ Coefficient` route (its own `dim0f`),

not a mirror, agreeing at the transcript level ($\Lambda_2^{\text{out}}/\hat{Y}_2^{\text{out}}$, the derived $u_2^z = 1/9/u_4^z = 4/81/v_5^z = 1/27$, $\chi_Q = +1/\text{incoming } -1$, the standing static slot, the PARTIAL landing), with an arity + unevaluated-leakage self-check. SymPy 59 PASS / Mathematica 65 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (both engines compute-verified the series $1 + z^2/9 + 4z^4/81 + iz^5/27$, the incoming $\chi_Q = -1$, and the standing $\Lambda_{\text{stand}}(0) = +2$; three non-blocking clarity nits folded — the P_0 name collision, the passivity scope, the 3a/3b self-ablation scoping). Full tri-review on fresh agents: FIDELITY_CLEAN (an independent read hand-re-derived j_2/y_2 , Λ_2^{out} , $\hat{Y}_2^{\text{out}}$, the full series, $\chi_Q = \pm 1$, the standing slot) and ADVERSARIAL_CLEAN (per-tooth ablation: 30/32 targeted mutations fired at their own assert, the 2 non-fires harness artifacts re-covered by corrected runs; no vacuous tooth) — no remediation required. Symbolic per-tooth ablation, mutations on copies.

B.14 Stage 019: The Squared-Denominator Prefactor Algebra $P(\omega) = D_0 N/D^{\text{cons}^2}$ (Check II-G4b)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative sector; the SECOND leg of a four-way split of pathA_33 — the squared-denominator prefactor-algebra component of the joint QUAD_CALIBRATED, with the DtN Hankel fingerprint = Stage 018 (done), the $54/5 = 2 \cdot 27/5$ provenance partition + the CALIBRATED label = Stage 020, and the $\hat{\mu}_0$ -free dimensional closure = Stage 021)

Claim status. EXACT WITHIN CLOSURE

Purpose. Earn the prefactor structure the quadrupole normalization rides. Show the squared-denominator prefactor $P(\omega) = D_0 N(\omega)/D^{\text{cons}}(\omega)^2$, $N = N_0 + N_2\omega^2 + N_4\omega^4$, $D^{\text{cons}} = D_0 + D_2\omega^2 + D_4\omega^4$, Taylor-expands about $\omega = 0$ to $P_0 = N_0/D_0$, $P_2 = (D_0N_2 - 2D_2N_0)/D_0^2$, $P_4 = (D_0^2N_4 - 2D_0(D_2N_2 + D_4N_0) + 3D_2^2N_0)/D_0^3$ — with the $-2D_2N_0$ factor-of-2 the signature of the squared denominator (plain N/D gives only $-D_2N_0$, provably wrong). Land the joint verdict as a partial component.

Status. EXACT WITHIN CLOSURE — the prefactor algebra is exact symbolic: the coefficients are exact rational functions of the abstract port scalars, read off a symbolic **series**(...) of D_0N/D^{cons^2} (**expect_zero**/ **expect_bool**/**expect_fail** residuals, no **scipy**/**numpy**/ floats; the light integer sample substitution is a transcript-level cross-check only). The verdict is the joint QUAD_CALIBRATED; this stage carries its *squared-denominator prefactor-algebra* component (2/4) and lands the joint as **partial** (018 done, 019 earned here; 020/021 pending) — not complete. It adds **zero** new counted knobs (an EARNED/structural prefactor-algebra slice, like Stages 018/016/011/012/014); the abstract D_0, D_2, D_4 are Stage-017's exported D-lanes and the N_0, N_2, N_4 are the deferred port N -moments, cited as provenance, not consumed values.

The squared-denominator prefactor object. The port numerator and the conservative (“consumption”) denominator are

$$N(\omega) = N_0 + N_2\omega^2 + N_4\omega^4, \quad D^{\text{cons}}(\omega) = D_0 + D_2\omega^2 + D_4\omega^4,$$

and the prefactor is the *squared-denominator* object

$$P(\omega) = \frac{D_0 N(\omega)}{D^{\text{cons}}(\omega)^2} \quad (\text{not the plain } N/D).$$

The D_n are Stage-017's exported $\ell = 2$ port-kernel D-lanes and the N_n are the concrete port N -moments of **build_port_moments**; here they are carried as abstract, port-agnostic symbols — the algebra below

holds for *any* nonzero $D_0 \dots N_4$, so no port value is consumed (the numerical (D_n, N_n) are deferred Gate-6 branch data).

The prefactor algebra (the EARNED headline). Expanding $P(\omega)$ about $\omega = 0$ and reading the coefficients off the actual series,

$$P_0 = \frac{N_0}{D_0}, \quad P_2 = \frac{D_0 N_2 - 2D_2 N_0}{D_0^2}, \quad P_4 = \frac{D_0^2 N_4 - 2D_0(D_2 N_2 + D_4 N_0) + 3D_2^2 N_0}{D_0^3}.$$

The $-2D_2 N_0$ term is the *signature* of the squared denominator: it arises only from the $-2x$ term of $(1+x)^{-2}$ with $x = (D_2/D_0)\omega^2 + (D_4/D_0)\omega^4$. The coefficients are $\text{coeff}(\omega, n)$ off the *actual* symbolic series of $D_0 N/D^{\text{cons}^2}$ — with the object and its series emitted — and are checked against an *independent* typed reference (a genuine derive-then-check that could disagree), not hardcoded and compared. Mutating the object (e.g. to plain N/D , or perturbing a D_n/N_n power) changes the extracted coefficient and fires exactly that coefficient's `expect_fail`; a swapped-in *correct* object flips the discriminator to true — a genuine positive control.

The N/D self-check. The plain (non-squared) object exposes the factor of two directly: N/D^{cons} series-expands to $P_2^{\text{plain}} = (D_0 N_2 - D_2 N_0)/D_0^2$, so the computed gap

$$P_2^{\text{plain}} - P_2 = \frac{D_2 N_0}{D_0^2} \neq 0,$$

i.e. $-D_2 N_0$ versus the correct $-2D_2 N_0$. The gap is *computed* (not asserted); the counterfactual probe `3g_wrong_prefactor_object` replaces $D_0 N/D^{\text{cons}^2}$ with plain N/D , so `plain_equals_correct_P2 = False` and the probe fires `FAIL_PREFACTOR_ALGEBRA`, while the correct object does *not* fire. Its self-ablation is a genuine *dynamic* re-run of an 019-local verdict gate (the prefactor match $\wedge$ the N/D self-check; with $\neq$ without, never a constant, and never the joint `base_verdict`) — the pathA_33 v1 trip-up avoided.

Provenance-only consumption and units-free scope. This is where Stage-018's deferred literal N_n/D_n port-kernel consumption lands — at the *provenance* level, not a value-consumption dual-site. Because the algebra is port-agnostic and `build_port_moments` is emitted-but-never-checked (deferred Gate-6 branch data), there is no checkable cross-stage relation to guard: Stage-017's D-lanes, the concrete N -moments, and Stage-018's fingerprint context are cited as narrative provenance with no guard/dual-site on any of them (the same provenance-only landing as Stage 018; a guard on the unused/deferred moments would be a vacuous tooth). The stage is *units-free*: it introduces no $c_s/a/G/\hat{\mu}_0$ and has no dimensional leg (the $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ closure is Stage-021's). A runtime free-symbol guard asserts the earned expressions' symbols are exactly $\{\omega, D_0, D_2, D_4, N_0, N_2, N_4\}$.

Honest scope. Earned prefactor structure / calibrated magnitude: 019 derives the port-agnostic algebra (P_0, P_2, P_4 and the N/D factor-of-two self-check), but the $54/5 = 2 \cdot 27/5$ normalization magnitude and G are calibrated ($G = \text{GENUINE_BLOCKED}$), landed at Stage 020, and the numerical port scalars (D_n, N_n) from the solved nonlinear branch remain deferred Gate-6 work. The DtN fingerprint is Stage 018; the $\hat{\mu}_0$ -free dimensional closure is Stage 021.

Consumed closure and register. *Provenance only, no dual-site:* the abstract D_0, D_2, D_4 are Stage-017's exported D-lanes and the N_0, N_2, N_4 are **build_port_moments**' concrete port N -moments (deferred Gate-6, asserted against nothing); Stage-018's fingerprint context is a narrative cite. Register: **zero** new counted knobs; structural edge **R38** (the squared-denominator prefactor-algebra provenance — $P_0 = N_0/D_0$, $P_2 = (D_0N_2 - 2D_2N_0)/D_0^2$, $P_4 = (D_0^2N_4 - 2D_0(D_2N_2 + D_4N_0) + 3D_2^2N_0)/D_0^3$ with the $-2D_2N_0$ factor-of-2 the squared-denominator signature; discharges nothing). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed; the .wl KEEPS its native **serW/Coefficient/FullSimplify** route on its own $P(\omega)$ /plain objects (typing its own expected $P_0/P_2/P_4$), not a mirror, agreeing at the transcript level (the series-extracted coefficients, the plain $P_2 = (D_0N_2 - D_2N_0)/D_0^2$, and **plainEqualsCorrectP2=False**), with an arity + unevaluated-leakage self-check. SymPy 18 PASS / Mathematica 24 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (both engines compute-verified the series $N_0/D_0 + (D_0N_2 - 2D_2N_0)\omega^2/D_0^2 + \dots$, the plain N/D gap D_2N_0/D_0^2 , and the emitted-but-never-checked reading of **build_port_moments**; four nits folded — a line-ref fix, the D-lane/ N -moment split, the sibling-symbol wording, and the $D_0..N_4$ -only sample substitutions). Full tri-review on fresh agents: **FIDELITY_CLEAN** (an independent read hand-re-derived $P_0/P_2/P_4$ and the N/D gap D_2N_0/D_0^2) and **ADVERSARIAL_ISSUES** (per-tooth ablation: the central firewall, N/D self-check, and dynamic 019-local ablation all genuine; two subsumed/mirror teeth flagged — one made-genuine as a swapped-in correct-object positive control, one honestly de-counted — plus a de-obfuscation of narrative tokens and a legibility fix, all re-verified **REVERIFY_CLEAN** by the coupling meta-test). Symbolic per-tooth ablation, mutations on copies.

B.15 Stage 020: The $54/5 = 2 \cdot 27/5$ Provenance Partition and the CALIBRATED Verdict Label (Check II-G4c)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative sector; the THIRD leg of a four-way split of pathA_33 — the $54/5 = 2 \cdot 27/5$ provenance-partition + the CALIBRATED-verdict-label component of the joint **QUAD_CALIBRATED**, with the DtN Hankel fingerprint = Stage 018 (done), the squared-denominator prefactor algebra = Stage 019 (done), and the $\hat{\mu}_0$ -free dimensional closure = Stage 021 (the completing leg))

Claim status. EXACT WITHIN CLOSURE

Purpose. Land the CALIBRATED headline. Show the assembled $\ell = 2$ quadrupole magnitude $54Gc_s^5/(5a^5c^5)$ decomposes as $54/5 = 2 \cdot 27/5$, with the 27 earned (**derived_in_gate**, Stage-018's fingerprint $1/v_5^z$) and the $2/5 + G$ calibrated (**external_bridge_input**, the GR Burke–Thorne $2G/5c^5$, $G = \text{GENUINE_BLOCKED}$); a four-way provenance partition classifies the assembled $54/5$ as **external_bridge_input** (external dominates), so the source-faithful verdict lands **QUAD_CALIBRATED** (not **QUAD_PASS**) — driven by provenance, not by $G \rightarrow \lambda G$ invariance.

Status. EXACT WITHIN CLOSURE — units-bearing but exact symbolic/rational for the earned content: the a^{-5} scaling (exact a -power via Stage-018's frozen slot), the Γ_5/χ_Q equivalence (**compact(...)** = 0 residuals), the $54/5 = 2 \cdot 27/5$ SymPy rational identity (bound to **target_rhs/v5_slot**), and the four-way provenance partition (exact class comparison via tag dominance); **expect_zero/expect_bool/**

`expect_fail` residuals, no floats/tolerances in the earned content. The verdict is the joint `QUAD_CALIBRATED`; this stage carries its *provenance-partition* + *CALIBRATED-label* component (3/4) and lands the joint as **partial** (018/019 done, 020 earned here; 021 the completing leg) — and it is the leg where the *CALIBRATED classification* is determined. It adds **zero** new counted knobs (a classification slice, like Stages 018/019).

The assembled magnitude and the Γ_5/χ_Q equivalence. The $\ell = 2$ quadrupole normalization the gravity sector must match is the Burke–Thorne form

$$\hat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5} \iff \gamma_{\text{quad}}^{\text{eff}} = \frac{2G}{5c^5}.$$

The equivalence is a self-contained algebraic bridge. With $P_0^{\text{phys}} = (c_s/a)^2 (N_0/D_0)$ (Stage-019’s prefactor) and Stage-018’s outgoing $\chi_Q = +1$, the forward residual

$$\text{forward} = \frac{54 G c_s^5}{5 a^5 c^5} \cdot \chi_Q \cdot \frac{a^5}{27 c_s^5} - \frac{2G}{5c^5} = \frac{2G(\chi_Q - 1)}{5c^5}$$

vanishes *iff* $\chi_Q = 1$ and $54/27 = 2$ (the 27 is Stage-018’s radiative slot $1/v_5^z$). So the outgoing sign Stage 018 derived is exactly the sign a positive quadrupole demands; a wrong $2/5 \rightarrow 3/5$ breaks the bridge (3e→FAIL_EQUIVALENCE).

The $54/5 = 2 \cdot 27/5$ decomposition (the headline — a bound SymPy identity). The assembled coefficient factors as

$$\frac{54}{5} = 2 \cdot \frac{27}{5},$$

with the 27 = `derived_in_gate` (Stage-018’s fingerprint $1/v_5^z$) and the $2/5+G$ = `external_bridge_input` (the GR Burke–Thorne $2G/5c^5$). This is a SymPy-verified rational identity *bound to the stage’s own assembled expressions* — not the source’s typed string “ $54/5 = 2 \cdot 27/5$ ”, and not a bare-literal `Rational(54,5) - 2 * Integer(27)/5` arithmetic tautology. The magnitudes are extracted: `mag = compact(target_rhs/(G*c_s^5/(a^5*c^5))) = 54/5` and `27_slot = compact(a^5/(c_s^5 * v5_slot)) = 27` (from Stage-018’s frozen slot `v5_slot = a^5/27c_s^5`), then `compact(mag - 2 * 27_slot/5) = 0`. Mutating `target_rhs` (54 → 55) or the identity ($2 \cdot 26/5$) fires the assert.

The a^{-5} target scaling. The assembled magnitude carries a^{-5} as a *derived* consequence of a -cancellation, not a typed target power: the Burke–Thorne coefficient $2G/5c^5$ is a -free, and Stage-018’s radiative slot `v5_slot = a^5/27c_s^5` supplies the a^5 , so `derived_power = a-power( $\gamma_{\text{target}}$ ) - a-power(v5_slot) = 0 - 5 = -5`, and the reconstructed $\gamma_{\text{target}}/(\chi \cdot \text{v5_slot}) = \text{target_rhs}$ (a -power -5). The a^5 is typed *once*; the wrong-scaling probe ($a^5 \rightarrow a^4$) fires at the assembled residual (3c→FAIL_SCALING). The actual branch a -scaling (from N_0/D_0) is deferred Gate-6.

The four-way provenance partition and the PROVENANCE-driven verdict. Every quantity is classified from its provenance *tags* by a tag-dominance rule `deferred > external > derived > convention`: `fingerprint_27` → `derived_in_gate`, `PN_2_over_5` → `external_bridge_input`, G → `external_bridge_input`, and the assembled $54/5$ (mixing the earned 27 tag with the external $2/5$ tag) → `external_bridge_input` because *external dominates*. The 020-local verdict then reads the classes with the source-faithful rule

```
if g_class = derived & mag_class = derived: QUAD_PASS; else: QUAD_CALIBRATED,
```

so with G and the assembled 54/5 both external, the verdict lands **QUAD_CALIBRATED**: the PDE delivers the form/branch (the 27), not Newton's G ($G = \text{GENUINE_BLOCKED}$). The classifier's dominance rule is proven by a truth-table + key-class + tag-mutation tooth (the source **partition_ok** alone is trivially true), and the verdict is proven to *read* provenance by a **QUAD_PASS** positive control (a coherent all-derived partition) *and* a required mixed control (one external, one derived $\rightarrow$ **QUAD_CALIBRATED**, the case an inverted rule would mislabel).

The verdict is provenance-driven, not $G \rightarrow \lambda G$ -invariance-driven. A separate, non-verdict-driving diagnostic makes the point sharp: 54/5 is a pure number, hence G -invariant ($G \rightarrow \lambda_G G$ leaves it unchanged), yet it is **external_bridge_input** by provenance. An invariance-only test would *mislabel* the calibrated 54/5 as earned (**invariance_only_trap_catches_54_over_5** = True); the classification is set by the provenance tags, and **3f_partition_mislabel** fires **FAIL_PROVENANCE_PARTITION** in both directions (G external $\rightarrow$ derived and 27 derived $\rightarrow$ external).

Units-bearing scope (no dimensional-homogeneity gate). Unlike Stage 019 (units-free), Stage 020 is units-bearing — $\{c_s, a, c, G\}$ live in the scaling/equivalence algebra — but it does algebra + provenance, *not* the dimensional homogeneity gate $[P_0^{\text{phys}}] = 1$ (that is Stage-021's). The cut is by operation, enforced by a runtime free-symbol allowlist (no $\hat{\mu}_0$ symbol; the source's legacy **gamma_quad_eff** $\hat{\mu}_0$ string is dropped), and a structural no-dimensional-dependency cut (no **dim_of** construct; a structural assertion that the 020-local verdict's signature/bytecode carries no dim parameter/dependency — the **.wl** checks its definition arity/text).

Honest scope. Earned classification / calibrated magnitude: 020 earns the a^{-5} scaling, the Γ_5/χ_Q equivalence, the 54/5 = 2·27/5 decomposition, and the four-way provenance partition; the 54/5 magnitude and G are calibrated ($G = \text{GENUINE_BLOCKED}$). The c^5 is the GR/λ_γ units bridge ($P_0 \propto c_s^5/c^5 = 1/\lambda_\gamma^5$) — c is the light cone c_γ in its GR-bridge role, a benchmark carrier, not EM propagation and not a fresh knob. The actual branch a -scaling and the numerical (D_n, N_n) port scalars remain deferred Gate-6. The dimensional closure $[P_0^{\text{phys}}] = 1$ is Stage-021's.

Consumed closure and register. *Provenance, no dual-site:* Stage-018's $\chi_Q = +1$ and 27 ($= 1/v_5^2$) genuinely enter 020's self-contained equivalence bridge (a corrupt cited value breaks **forward** — the equivalence tooth itself, not a manufactured cross-stage guard); Stage-019's $P_0 = N_0/D_0$ appears only in the Γ_5 definition, absent from the equivalence ok residual (a $\{N_0, D_0\} \subseteq \Gamma_5 \wedge \notin$ residual firewall guards this); Stage-017's D-lanes are the provenance of the abstract port scalars. Register: **zero** new counted knobs; structural edge **R39** (the 54/5 = 2·27/5 provenance-partition + the **CALIBRATED**-verdict provenance; discharges nothing — a classification landing, not a reduction). $G = \text{GENUINE_BLOCKED}$ /external (already registered); c = the c_γ GR-units-bridge (cited benchmark); the 2/5 = GR bridge; the 27 = Stage-018's derived fingerprint. Part-II counted **CALIB** set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. Unlike Stages 018/019 (which kept the **.wl**'s already-native blocks), the source **.wl** had *zero* 020 content, so the 020 **.wl** is a genuinely fresh independent authoring: native **Exponent/Together/Cancel/FullSimplify** for the scaling + the Γ_5 bridge, native **Rational/Simplify** for the identity bound to its own **targetRHS/ v5Slot**, a native **Association**-based provenance classifier using a rank-based **MaximalBy** (a genuinely different mechanism from the **.py** if-chain, same dominance), and **expr/.G** $\rightarrow \lambda_G G$ for the g -invariance trap — no **Series**, no **.py** mirroring — agreeing at the

transcript level (the a^{-5} , **forward** = 0, the bound $54/5 = 2 \cdot 27/5$, the provenance classes, the CALIBRATED verdict with the QUAD_PASS and mixed controls, both 3f directions), with an arity + unevaluated-leakage self-check. SymPy 74 PASS / Mathematica 82 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded four blocking genuineness gaps; Grok's compute-verify then caught the CALIBRATED verdict-rule *inversion* (the source default is **else**→QUAD_CALIBRATED; the directive had inverted it to **PASS** unless both external — a shippable rig, fixed by a required mixed control) plus a bare-literal-identity nit; a Codex confirm then swept two consistency gaps, all folded to DIRECTIVE_CLEAN. Full tri-review on fresh agents: FIDELITY_CLEAN (an independent read hand-re-derived the Γ_5 bridge, the bound identity, the classify-provenance dominance, and the g-invariance trap) and ADVERSARIAL_ISSUES (per-tooth ablation: the four key genuineness risks — the bound identity, the mixed control catching the inverted rule, the proven classifier, the genuinely-authored .w1 — all clean; three low-severity vacuous/subsumed teeth flagged, all remediated make-genuine with no de-counts — the structural dimensional cut, the $\{N_0, D_0\} \subseteq \Gamma_5$ before/after firewall, the independently-classified baseline-external side — re-verified REVERIFY_CLEAN by the coupling meta-test). Symbolic per-tooth ablation, mutations on copies.

B.16 Stage 021: The $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ Dimensional Closure (Check II-G4d)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative sector; the FOURTH, completing leg of a four-way split of pathA_33 — the $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ dimensional-closure component of the joint QUAD_CALIBRATED, with the DtN Hankel fingerprint = Stage 018 (done), the squared-denominator prefactor algebra = Stage 019 (done), and the $54/5 = 2 \cdot 27/5$ provenance partition + the CALIBRATED verdict label = Stage 020 (done))

Claim status. EXACT WITHIN CLOSURE

Purpose. Complete the joint QUAD_CALIBRATED. Certify the assembled $\ell = 2$ quadrupole prefactor $P_0^{\text{phys}} = (c_s/a)^2(N_0/D_0)$ is dimensionless from the sourced port dimensions $[N_0] = L^{-1}M$, $[D_0] = L^{-1}T^{-2}M$ — a $\hat{\mu}_0$ -FREE gate that catches the natural-units trap (the handoff's $P_0 = N_0/D_0$ silently drops $(c_s/a)^2$) and reads the sourced port dimensions (corrupt- $[N_0]$ fires FAIL_DIMENSIONAL) rather than a back-solved free carrier $\hat{\mu}_0$ (corrupt- $[G]$ does not fire).

Status. EXACT WITHIN CLOSURE — exact symbolic (L, M, T) -triple dimensional-vector algebra, float-free: the $\hat{\mu}_0$ -free gate ($\text{dim_of}(P_0^{\text{phys}}) = \text{ZERO_DIM}$), the natural-units-trap catch (dropping the frequency normalization leaves $T^2 \neq 1$), the corrupt-dim scope truth-table, and the anti-v1 back-solve-tautology demotion; **expect_zero/expect_bool/expect_fail** residuals, no floats/tolerances. The verdict is the joint QUAD_CALIBRATED; this stage carries its *dimensional-closure* component (4/4) and lands the joint **complete** (018/019/020 done, 021 earned here). It adds **zero** new counted knobs (a dimensional-closure slice, like Stages 018/019/020). $\hat{\mu}_0$ is a free-carrier non-verdict diagnostic, not a counted knob.

The $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ gate. With the sourced port dimensions $[N_0] = L^{-1}M$ (the density/continuity-port numerator) and $[D_0] = L^{-1}T^{-2}M$ (the carried reduced static conservative denominator $D_0 = K - B_0 - Z_0$), and the frequency normalization $(c_s/a)^2$,

$$[P_0^{\text{raw}}] = [N_0/D_0] = T^2, \quad [(c_s/a)^2] = T^{-2}, \quad [P_0^{\text{phys}}] = [(c_s/a)^2 N_0/D_0] = 1,$$

so `dimensional_ok = True`. The verdict-bearing check is `dimensional_ok = (dim_of( $P_0^{\text{phys}}$ ) = ZERO_DIM)` — a computed (L, M, T) -triple equality reading the sourced dims, not a typed "1". Crucially $\hat{\mu}_0$ does *not* enter the verdict: it is absent from `raw_symbol_dims` (the gate's dimension map), and the verdict's computed read-set excludes $\hat{\mu}_0/\mu_{\text{dim}}/\text{homogeneity_pass}$.

The natural-units-trap catch (the dimensional milestone). The handoff's $P_0 = N_0/D_0$ silently drops the $(c_s/a)^2$ frequency-normalization factor (a natural-units artifact where c_s/a was set to 1). The gate catches it: $[P_0^{\text{raw}}] = T^2 \neq 1$, so the 3d drop-normalization probe fires `FAIL_DIMENSIONAL`. The frequency normalization is required for $[P_0^{\text{phys}}] = 1$.

The corrupt-dim scope truth-table. The $\hat{\mu}_0$ -free gate fires precisely on the dims that enter P_0^{phys} :

corrupt $[N_0] \rightarrow 0$	$[P_0^{\text{raw}}] = (1, -1, 2)$, $[P_0^{\text{phys}}] = (1, -1, 0) = L M^{-1} \neq 1$	$\rightarrow \text{FAIL_DIMENSIONAL (3d')}$
corrupt $[D_0] \rightarrow 0$	$[P_0^{\text{phys}}] = (-1, 1, -2) \neq 1$	$\rightarrow \text{FAIL_DIMENSIONAL}$
corrupt $[c_s] \rightarrow 0$	$[P_0^{\text{phys}}] = (-2, 0, 2) \neq 1$	$\rightarrow \text{FAIL_DIMENSIONAL}$
corrupt $[G] \rightarrow 0$	$[P_0^{\text{phys}}] = 1$ still ($G \notin \text{free}$)	$\rightarrow \text{NO_FAIL (scope)}$
correct	$[P_0^{\text{phys}}] = 1$	$\rightarrow \text{NO_FAIL}$

The corrupt- $[N_0]$ result is $[P_0^{\text{phys}}] = (1, -1, 0)$ — the $(c_s/a)^2$ factor *remains* (it is not $-[D_0] = (1, -1, 2)$, which is $[P_0^{\text{raw}}]$). The gate is G-independent; the corrupt- $[G]$ row is a dependency-scope diagnostic, asserted via the runtime test $G \notin \text{free_symbols}(P_0^{\text{phys}})$.

The $\hat{\mu}_0$ homogeneity diagnostic is NON-verdict — the anti-v1 discriminator. The full closure $\hat{m}_0^2 P_0 = 54 G c_s^5 / (5 a^5 c^5)$ can be made dimensionally homogeneous by back-solving the free carrier $[\hat{\mu}_0] = L^{-1} T^{-1} M^{-1/2}$ from $(\text{rhs_dim} - \text{p0_dim})/2$, giving $[\text{lhs}] = [\text{rhs}] = L^{-2} T^{-2} M^{-1}$, `homogeneity_pass = True`. This is a *tautology*: re-solving $[\hat{\mu}_0]$ after *any* corruption keeps `homogeneity_pass = True` (the back-solve absorbs the change), so it can never fail — exactly the v1-rejected rig, correctly demoted here to a labelled diagnostic (`participates_in_verdict = False`, computed from the verdict read-set). The decisive anti-v1 tooth is two-part: (a) a computed read-set assertion that the verdict excludes $\hat{\mu}_0/\mu_{\text{dim}}/\text{homogeneity_pass}$ (and $\hat{\mu}_0 \notin \text{raw_symbol_dims}$), and (b) a deliberately-wired mutant verdict = the back-solved `homogeneity_pass`, re-run over all four corruptions and shown to stay `NO_FAIL` on all of them — proving it is vacuous, so the real verdict must be the $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = \text{ZERO}$ gate (which fires on three of the four). The gate's $[N_0]/[D_0]/[c_s]$ FAIL rows, not the $[G]$ row, are what reject the all-`NO_FAIL` back-solved gate.

The $\hat{Y}$ dimensionless check. The outgoing physical expansion $\hat{Y} = 1 + u_2 \omega^2 + u_4 \omega^4 + i v_5 \omega^5$ with Stage-018's frozen fingerprint $u_2 = a^2/9c_s^2$, $u_4 = 4a^4/81c_s^4$, $v_5 = a^5/27c_s^5$ (a provenance-frozen local fixture, not the Stage-018 machinery) is dimensionless (each term $T^2 \cdot T^{-2} = 1$, etc.), so $[\hat{Y}] = 1$. The check is wrapped in a structured `try_dim_of/Catch` so a corrupted ω -power ($u_2 \omega^3$) fires the named $\hat{Y}$ -dimensionless assert (a caught mismatch), not an uncaught `DimError` that would bypass the tooth.

The completing landing. Stage 021 lands the joint `QUAD_CALIBRATED complete` (018 fingerprint $\wedge$ 019 prefactor $\wedge$ 020 provenance-partition + `CALIBRATED` $\wedge$ 021 dim closure). **Complete $\neq$ pass:** the joint token stays `QUAD_CALIBRATED` (calibrated, not `QUAD_PASS`) — Stage 020 already determined the `CALIBRATED` classification and $G = \text{GENUINE_BLOCKED}$; 021 supplies the dimensional closure that makes the assembled magnitude dimensionally sound, but does not flip the verdict. The PDE delivers the form/branch (the 27 fingerprint + the dimensionally-sound closure), not Newton's G .

Honest scope. Earned closure / calibrated magnitude: 021 earns the $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ closure, the natural-units-trap catch, and the corrupt- $[N_0]/[G]$ discriminator (the gate reads the sourced port dims, not a back-solved $\hat{\mu}_0$); the 54/5 magnitude and G are calibrated ($G = \text{GENUINE_BLOCKED}$). 021 is the other half of the 020/021 operation-level cut — Stage 020 did algebra + provenance, Stage 021 does the $[\cdot]$ dimensional-homogeneity gate, the only pathA_33 leg with a dimensional gate. The sourced port dims $[N_0] = L^{-1}M$ (pathA_43 density-port numerator) and $[D_0] = L^{-1}T^{-2}M$ (the carried reduced static conservative denominator) are sourced dimensional inputs (Cluster C stages 024/027 + pathA_34 not yet built in build order); they genuinely enter the gate (so the corrupt- $[N_0]$ tooth is genuine), but 021 does not re-derive them. The actual branch a -scaling and the numerical (D_n, N_n) port scalars remain deferred Gate-6.

Consumed closure and register. *Provenance, no dual-site:* the sourced port dims (pathA_43 numerator + the carried conservative denominator), Stage-018's $u_2/u_4/v_5$ (a local frozen fixture for the $\hat{Y}$ check, not `build_fingerprint`), Stage-019's $P_0 = N_0/D_0$ (entering P_0^{raw}), and Stage-020's assembled `target_rhs` (the $\hat{\mu}_0$ diagnostic's rhs only). The sourced port dims genuinely enter the $\hat{\mu}_0$ -free gate (the corrupt- $[N_0]$ probe is the genuine able-to-fail tooth). Register: **zero** new counted knobs; structural edge **R40** (the $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ dimensional closure; discharges nothing — a dimensional-consistency closure, not a reduction; records that 021 completes the joint `QUAD_CALIBRATED`, which stays `CALIBRATED` not `PASS`). $\hat{\mu}_0$ is a free-carrier non-verdict diagnostic; the sourced port dims are dimensional provenance; c/G are already registered. Part-II counted `CALIB` set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. Like Stages 018/019 (and unlike Stage 020's fresh authoring), the source `.wl` has the 021 dimensional block, so the 021 `.wl` keeps its native, already-independent Wolfram dimensional-vector engine (a `Which`-based `dimOf` over `Times/Power/Plus`, native $\langle |\dots| \rangle$ `rawDims`, `Series`-free dimension algebra) and severs only the YAML machinery; the Stage-018 fingerprint (j_2/y_2) and Stage-019 prefactor blocks are removed and the $\hat{Y}$ coefficients become local `u2Sourced/u4Sourced/v5Sourced` fixtures. It adds (the source `.wl` lacked both) the corrupt- $[G]$ -does-not-fire scope control and the dynamic 021-local self-ablation; the `.wl`'s local pass-guard uses a split-out `yhatOk` and excludes `homogeneityPass` (a print-only diagnostic), with an arity + unevaluated-leakage self-check. Agreement is at the transcript level ($[P_0^{\text{raw}}] = T^2$, $[(c_s/a)^2] = T^{-2}$, $[P_0^{\text{phys}}] = 1$, `3d/3d'` `FAIL_DIMENSIONAL`, corrupt- $[G]$ `NO_FAIL`, `homogeneity_pass = True` diagnostic). SymPy 42 `PASS` / Mathematica 50 `PASS`, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded six blocking findings — most notably that the originally-proposed corrupt- $[G]$ anti-v1 discriminator was *backwards* (a back-solved $\hat{\mu}_0$ gate re-solves μ_{dim} after every mutation, so it fires on nothing, and it is the $\hat{\mu}_0$ -free gate's $[N_0]/[D_0]/[c_s]$ `FAIL` rows that reject it) — then a Grok compute-verify pass independently confirmed the $\hat{\mu}_0$ -free gate dims, the corrupt-dim truth-table, the back-solve-is-a-tautology crux, and the $\hat{Y}$ /structured-catch, and a final Codex confirm closed it to `DIRECTIVE_CLEAN`. Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read hand-re-derived the $\hat{\mu}_0$ -free gate dims, the corrupt-dim truth-table, and the back-solve-tautology table, confirming no 018/019/020 leakage and a genuine independent `.wl`) and `ADVERSARIAL_ISSUES` (per-tooth ablation: the two key teeth — the anti-v1 read-set/wired-mutant and the corrupt- $[G]$ scope control — genuine; five low-severity stamped/subsumed teeth flagged, remediated two make-genuine (`rerun_gate_logic` derived from the actual re-run, `participates_in_verdict` derived from the computed read-set) and three de-count (the `QUAD`-landing tautology and two subsumed summaries retained as labeled prints) — re-verified `REVERIFY_CLEAN` by the coupling meta-test, tallies 45/53→42/50 from the honest de-counts). Symbolic

per-tooth ablation, mutations on copies.

B.17 Stage 022: The Cross- $\ell - (\ell + 1)/\Lambda_\ell$ Fingerprints + the Gate-4 Non-Regression (Check II-G5a)

Part / anchor. Part II – Gravity (the frozen-throat cross- ℓ radiative sector; the EARNED-FIRST leg of a two-way split of pathA_34 — the cross- ℓ outgoing DtN $-(\ell+1)/\Lambda_\ell$ fingerprints ($\ell = 0, 1, 2$) + the Gate-4 quadrupole non-regression component (1/2) of the joint FAIL_UNDERDETERMINED_NOT_PREDICTIVE). The native nullspace underdetermination departure — dim-8 / return-nullity-2, the residuals vs pathA_29, the $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ Gate-6 selector need — that *delivers* the FAIL is Stage 023 (II-G5b).

Claim status. EXACT WITHIN CLOSURE

Purpose. Generalize Stage 018's $\ell = 2$ -only DtN Hankel fingerprint across ℓ : certify that the outgoing DtN responses $\hat{Y}_\ell^{\text{out}} = -(\ell + 1)/\Lambda_\ell$ for $\ell = 0, 1, 2$ — computed from the spherical Hankel h_ℓ — series-expand to the radiative fingerprint $\{1, 1/2, 1/27\}$ at leading orders $\{\omega^1, \omega^3, \omega^5\}$, with the static-DtN slot $\Lambda_\ell(z \rightarrow 0) = -(\ell + 1)$ derived from the Hankel logarithmic derivative and the radiative order verified by scanning; and that the $\ell = 2$ leg reproduces the completed pathA_33 quadrupole fingerprint $\{u_2, u_4, v_5\} = \{1/9, 4/81, 1/27\}$ (the Gate-4 non-regression — the cross- ℓ generalization does not regress the earned quadrupole).

Status. EXACT WITHIN CLOSURE — exact symbolic, float-free, dimensionless z -space ($z = a\omega/c_s$): the cross- ℓ spherical-Hankel logarithmic derivatives and their small-frequency series are exact `sympy.series` / native `Series` expansions; `bool_zero/expect_bool/expect_fail` residuals, no floats/tolerances, no units-restoration leg. The verdict is the joint FAIL_UNDERDETERMINED_NOT_PREDICTIVE; this stage carries its *fingerprint + non-regression* component (1/2) and lands it as an **EARNED-FIRST PARTIAL** (022 earned; 023 pending and it is 023 that delivers the FAIL). It adds **zero** new counted knobs (an earned exterior-wave signature slice, like Stages 016/018). c_s is an R1-derived units carrier, not a counted knob; the $\ell = 0/1$ stiffnesses and the return admittances $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ are Stage 023's, not counted here.

Two distinct verdicts, both printed. The script emits an *022-local* exit-0 gate and, separately, the joint physics label:

```
LOCAL_AUDIT_VERDICT: CROSS_L_FINGERPRINT_OK (fingerprints + non-regression; the exit-0 gate)
JOINT_LANDING_LABEL (PARTIAL; FAIL not evaluated by 022): FAIL_UNDERDETERMINED_NOT_PREDICTIVE
```

022's script passes (exit 0) — the earned content is correct and every tooth fires; the FAIL token is the joint gate's physics label carried as a 1/2 partial. 022 is the earned half of a gate that ultimately fails (cf. Stage 003 = earned transverse photons + the characterized FAIL_CAUCHY departure). The 022-local verdict's computed read-set is exactly `{cross_l_fingerprints, ell2_non_regression}` and provably excludes any nullspace / return / `base_verdict` object (none is built here) — so the default-verdict-is-PASS trip-up is avoided: 022 does not compute or print a FAIL from any rigged mechanism.

The cross- ℓ outgoing DtN fingerprints (the earned headline). For each ℓ the exterior outgoing solution is the spherical Hankel function of the first kind $h_\ell^{(1)} = j_\ell + i y_\ell$ ($z = a\omega/c_s$); the normalized

radiative response is $\hat{Y}_\ell^{\text{out}}(z) = -(\ell + 1)/\Lambda_\ell(z)$ with $\Lambda_\ell(z) = z h_\ell^{(1)'} / h_\ell^{(1)}$. Series-expanding about $z = 0$,

$$\begin{aligned} \ell = 0 : \quad \hat{Y}_0 &= 1 + i z + \dots & \Rightarrow \text{radiative_coeff} &= 1 \text{ at order } \omega^1, \\ \ell = 1 : \quad \hat{Y}_1 &= 1 + \frac{1}{2}z^2 + i \frac{1}{2}z^3 + \dots & \Rightarrow \text{radiative_coeff} &= 1/2 \text{ at order } \omega^3, \\ \ell = 2 : \quad \hat{Y}_2 &= 1 + \frac{1}{9}z^2 + \frac{4}{81}z^4 + i \frac{1}{27}z^5 + \dots & \Rightarrow \text{radiative_coeff} &= 1/27 \text{ at order } \omega^5. \end{aligned}$$

The leading radiative coefficient is $\{1, 1/2, 1/27\}$ at order $2\ell + 1 = \{\omega^1, \omega^3, \omega^5\}$, extracted as the imaginary coefficient $\text{coeff}(z, 2\ell + 1)/i$ and checked derive-vs-typed — the derived side is the actual series of $-(\ell + 1)/(z h'/h)$, emitted, not hardcoded. The static-DtN slot $\Lambda_\ell(z \rightarrow 0) = -(\ell + 1)$ is *derived* from the Hankel logarithmic derivative ($h_\ell \sim z^{-(\ell+1)}$ as $z \rightarrow 0$), read off `lam_series.coeff(z, 0)` — not the hand-set numerator (the source’s set-then-compare is de-rigged). The radiative order is verified by *scanning* the imaginary series for its first nonzero power and asserting all lower imaginary coefficients vanish (not the source’s dodgeable “preselected $2\ell + 1$ coefficient is nonzero”: a $+iz$ corruption of the $\ell = 2$ series leaves z^5 nonzero yet the true leading imaginary order becomes 1; the scan catches it). The static value $\hat{Y}_\ell(0) = 1$ is a de-counted printed diagnostic ($= -(\ell + 1)/\Lambda_{\text{static}}$, subsumed by the derived Λ_{static}).

The incoming-sign-flip (the genuine sign tooth). The incoming branch $h_\ell^{(2)} = j_\ell - i y_\ell$ flips *only* the radiative (imaginary) sign — `radiative_coeff` $\rightarrow \{-1, -1/2, -1/27\}$ with every lower real coefficient unchanged (`bool_zero(incoming + outgoing) = 0`). This computed sign-flip is the genuine sign content. The $\chi_Q = \text{radiative_coeff}/(\text{canonical slot}) = +1$ equality is a de-counted printed diagnostic — algebraically subsumed by the $\ell = 2$ non-regression ($27v_5 - 1 \equiv 27(v_5 - 1/27)$), so once the $\ell = 2$ slot passes $\chi_Q = 1$ is automatic; not an independent tooth.

The Gate-4 quadrupole non-regression. The $\ell = 2$ leg $(-(\ell + 1)/\Lambda_\ell|_{\ell=2} = -3/\Lambda_2)$ reproduces Stage 018’s earned quadrupole fingerprint $u_2 = 1/9$, $u_4 = 4/81$, $v_5 = 1/27$. This is a derive-vs-typed non-regression: 022 re-derives the $\ell = 2$ tuple from its own Hankel and checks it against Stage 018’s independently-earned typed literals (not a same-machinery reconstruction of the typed side — no subsumed $X \equiv X$). The cross- ℓ $-(\ell + 1)$ normalization gives the right -3 at $\ell = 2$, so the general machinery does not regress the quadrupole. Each of u_2, u_4, v_5 has its own coefficient-isolating mutant (the **3e** incoming mutation flips only v_5 , so u_2/u_4 need their own derivation-copy mutants — the Stage 018 pattern); breaking the $\ell = 2$ branch to incoming fires `FAIL_QUAD_REGRESSION`, whose self-ablation is a dynamic two-verdict re-run (with $\neq$ without) re-scoped to the 022-local verdict.

Honest scope. Earned cross- ℓ signature + non-regression / the FAIL is 023’s: 022 derives the outgoing cross- ℓ fingerprints and the Gate-4 quadrupole non-regression — the report’s earned fingerprint / raw-order / Gate-4 subset. The rest of the report’s earned content (the residual form/sign/order, conditional on a positive bounded branch) plus the FAIL-delivering native nullspace underdetermination (dim-8 / return-nullity-2, ϵ_{eff} left free at the linear Gate-5 level) and the deferred scalar/dipole magnitude/nonzero prediction belong to Stage 023. The joint FAIL is deferred to 023, not evaluated here (the `JOINT_LANDING_LABEL` is a printed string; 022 builds no nullspace and does not back-solve ϵ_{eff}/Z — the `FAIL_TAUTOLOGICAL` firewall, 023’s, is not reintroduced). The fingerprints are dimensionless z -space rationals; the $z = a\omega/c_s$ units realization and the residual dimensional gate are 023’s. The $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ selector (the first concrete Gate-6 input) is the deferred nonlinear solve — 023’s characterized departure.

Consumed closure and register. The one *checkable* consumption is Stage-018’s fingerprint $\{u_2, u_4, v_5\} = \{1/9, 4/81, 1/27\}$, which 022’s re-derived $\ell = 2$ fingerprint reproduces (a per-slot / **3e**-tooth-backed derive-vs-typed non-regression; the second site is Stage-018’s independently-earned literal,

not a same-machinery reconstruction). *Provenance / context (cited, not re-derived)*: Stage 019/020/021 + the completed pathA_33 QUAD_CALIBRATED joint (the prefactor $P(\omega) = D_0 N / D_{\text{cons}}^2$, the 54/5 magnitude, the $\hat{\mu}_0$ -free dimensional closure); Stage 008's raw $\ell = 0, 1, 2$ outgoing amplitudes; Stage 009/010's bulk Helmholtz mode; and c_s (the density sound speed, Stage 005 R1 $c_s^2 = 5K\rho^4/m$) as the units symbol in $z = a\omega/c_s$ (distinct from the frozen-wall Helmholtz speed c_S , 011–017). Register: **zero** new counted knobs; structural edge **R41** (the cross- ℓ outgoing-DtN $-(\ell + 1)/\Lambda_\ell$ fingerprint provenance — the $\ell = 0, 1, 2$ radiative signature $\{1, 1/2, 1/27\}$ at $\{\omega^1, \omega^3, \omega^5\}$ with $\Lambda_{\text{static}} = -(\ell + 1)$ derived + the scanned order, plus the Gate-4 quadrupole non-regression reproducing the Stage-018 fingerprint component of the now-completed pathA_33 fold; discharges nothing — earned exterior-wave structure + a consistency non-regression, not a reduction). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}$ (013) + $\{V_{p0}/\ell_c\}$ (015) + $\{T_\Omega, \beta_2\}$ (017) = 6.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. The .w1 is a genuinely independent route, *re-authored* (not a kept transliteration): it uses Wolfram's built-in `SphericalHankelH1/SphericalHankelH2 + SeriesCoefficient` — a materially different construction from the .py's hand-built `sin/cos jℓ/yℓ + Series/Coefficient` (the source .w1's `branchData` was a near-transliteration of the .py `dtN_branch`, so it was discarded — the mirror-policy transliteration screen). Agreement is at the transcript level (the derived $\{1, 1/2, 1/27\}$ at $\{\omega^1, \omega^3, \omega^5\}$, $\Lambda_{\text{static}} = \{-1, -2, -3\}$ with $h_{\text{mut}} = zh$ firing each and $H1 \leftrightarrow H2$ shown inert, the scanned order, the $\ell = 2$ non-regression, the per-slot mutants, the incoming sign-flip, and **3e→FAIL_QUAD_REGRESSION**); SymPy 55 PASS / Mathematica 64 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded five blocking findings (the .w1 must be re-authored independent, not the transliterated `branchData`; the normalization and raw-order teeth must be genuinely derived — a Λ_{static} from the log-derivative + a coefficient scan replacing the source's $X \equiv X$ set-then-compare and preselected-order check; the $\ell = 2$ tuple was double-counted and χ_Q was subsumed, so the duplicate was removed, χ_Q de-counted, and per-slot u_2/u_4 mutants added; the earned/deferred partition is a subset relation with an explicit local/joint output; the checkable consumption is Stage-018's fingerprint), then a Codex confirm-pass caught a further directive-level reasoning error (the proposed Λ_{static} ablation “outgoing→incoming” is *inert* — both branches give $-(\ell + 1)$ — so the mutant must be a pole-order corruption $h_{\text{mut}} = zh$; and Λ_{static} 's subsumed $\hat{Y}_\ell(0) = 1$ is de-counted). A Grok-4.5 compute-verify of the folded directive returned **DIRECTIVE_CLEAN**, independently confirming the cross- ℓ series, the pole-order-vs-inert mutant, the scan-vs-preselect counterexample, the χ_Q subsumption identity, and the u_2/u_4 isolation; a final Codex confirm closed the bookend. Full tri-review on fresh agents (arbiter re-run reproduced the build SymPy 56 / Mathematica 65, exit 0, CWD-independent): **FIDELITY_CLEAN** (an independent read hand-re-derived the cross- ℓ fingerprints, the pole-order Λ_{static} mutant vs the inert branch swap, the scan-vs-preselect order, the χ_Q subsumption, and the per-slot isolation — all computed not stamped, no 019/023 leakage, a genuine built-in `SphericalHankelH1 .w1`) and **ADVERSARIAL_ISSUES** (both confirm-pass reasoning catches held live-proven, all core physics teeth genuine; two low-severity redundancy teeth — a vacuous **3e rerun_gate_logic** constant-length check and a subsumed read-set-exclusion tooth — were remediated, the first make-genuine [compare the two traced verdicts] and the second an honest de-count [the adjacent exact-equality tooth retains the guard], plus two .w1 nits [a symbol typo and a mutation-aware parity fix]), re-verified **REVERIFY_CLEAN** by the coupling meta-test (each remediated construct fires under a mutation of its object and goes vacuous when neutered; no regression, both engines). Tallies 56/65 → 55/64 (net −1 per engine from the honest de-count). Symbolic per-tooth ablation, mutations on copies.

B.18 Stage 023: The Native Nullspace Underdetermination Departure (Check II-G5b)

Part / anchor. Part II – Gravity (the frozen-throat cross- ℓ return sector; the COMPLETING, FAIL-delivering leg of a two-way split of pathA_34 — Stage 022 earned the cross- $\ell - (\ell + 1)/\Lambda_\ell$ fingerprints + the Gate-4 non-regression (the EARNED-FIRST 1/2), and this stage builds the native nullspace underdetermination that *delivers* the joint FAIL_UNDERDETERMINED_NOT_PREDICTIVE (2/2)). This COMPLETES the pathA_34 fold (022^023) and closes the Gate-1–5 gravity ladder; the only remaining gravity item, Gate 6 (the nonlinear return-selector), is sim-deferred (a Part-VII open-item, R42).

Claim status. OPEN

Purpose. Certify that the linear Gate-5 return law is genuinely underdetermined: over 11 genuine generator dofs $[\Omega_U, \Omega_W, R_{\text{mix}}, g_U, g_W, D_0, K_{0c}, K_\eta, T_\Omega, Z_0^{\text{ret}}, Z_1^{\text{ret}}]$ the three collected named constraints $\{P_0^{\text{raw}}, K_{0c}, K_\eta + 2T_\Omega\}$ have Jacobian rank 3 $\Rightarrow$ native nullspace dimension 8, and augmenting with the return-transfer gradients $\{\partial T_0, \partial T_1\}$ gives return-augmented rank 5 $\Rightarrow$ *return-moving nullity 2*: the return admittances $\{Z_0^{\text{ret}}, Z_1^{\text{ret}}\}$ survive every collected constraint (Z is a pathA_29 premise) $\Rightarrow$ FAIL_UNDERDETERMINED_NOT_PREDICTIVE; and that a derived counterfactual selector collapses the return-moving nullity to 0 $\Rightarrow$ CROSS_L_RESIDUAL_PREDICTION (the able-to-fail witness).

Status. OPEN. FAIL-headline stage (cf. Stage 003 = earned transverse photons + the characterized FAIL_CAUCHY departure): the top-line verdict is a FAIL_*, but the earned content — a genuinely-computed native nullspace and the able-to-fail selector control — is the headline, and the FAIL is the first-class characterized landing (no softening). The computation is exact symbolic, float-free: the constraint Jacobian, its rank/nullspace, the return-transfer gradients, the A_0/A_1 residuals, and the (L, M, T) -triple dimensional algebra are exact `sympy` (`sp.Matrix(rows).rank()`) / native Wolfram (`NullSpace`, `MatrixRank`) computations; `expect_zero/expect_bool/expect_fail` residuals, no floats/tolerances. It adds **zero** new counted CALIB knobs (the set stays 6) and $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ add zero new free dofs (aliases of the existing $\varepsilon_0/\varepsilon_1$ debt); the effective $\ell = 0/1$ stiffnesses K_{0c}/K_1 add *counted* FREE-UNREDUCED PENDING reduction-debt.

Two distinct labels, both printed. The script emits a script/teeth status separately from the earned physics verdict:

AUDIT_STATUS=PASS (the script ran, all teeth fired, exit 0)

PHYSICS_VERDICT=FAIL_UNDERDETERMINED_NOT_PREDICTIVE (2/2, COMPLETING) (the earned characterized-FA)

The script PASSES (exit 0) by correctly *computing* the earned FAIL — the FAIL is the earned physics, not an execution failure.

The native nullspace (a genuine rank, not a rigged construction). The three collected named constraints are the $\ell = 2$ port kernel $P_0^{\text{raw}} = (\Omega_U^2 g_W + R_{\text{mix}} g_U)^2 / (\Omega_U^2 \Omega_W^2 - R_{\text{mix}}^2) / D_0$ (handoff §10.2–10.3), the $\ell = 0$ collective stiffness K_{0c} (§8.2 / Gate-2), and the $\ell = 1$ harmonic stiffness $K_1 = K_\eta + 2T_\Omega$ (§9.4). Their Jacobian J over the 11 dofs has

$\text{rank}(J) = 3$ (a genuine `sp.Matrix(rows).rank()` on symbolic ∂ -rows — no zero-padding, no hardcoded 8/2),

Augmenting J with the return-transfer gradients $\partial T_0, \partial T_1$ — where $T_\ell = K_\ell / (K_\ell + Z_\ell^{\text{ret}})$ is the DC return transmission — gives

$$\text{rank}(J \cup \{\partial T_0, \partial T_1\}) = 5, \quad \text{return_moving_nullity} = 5 - 3 = 2 \Rightarrow \text{FAIL_UNDERDETERMINED_NOT_PREDICTIVE}.$$

The two return admittances $\{Z_0^{\text{ret}}, Z_1^{\text{ret}}\}$ are untouched by every collected constraint ($\partial c / \partial Z_\ell^{\text{ret}} = 0$) yet move T_0/T_1 ; this is witnessed constructively — the unit vectors $e_{Z_0^{\text{ret}}}, e_{Z_1^{\text{ret}}}$ each lie in the constraint nullspace yet give $\Delta T_0 = -K_{0c} / (K_{0c} + Z_0^{\text{ret}})^2 \neq 0$ and $\Delta T_1 = -(K_\eta + 2T_\Omega) / (\cdots)^2 \neq 0$. Why the return survives: pathA_29 records Z as a *premise* ($Z_is_premise=\text{True}$, $boundary_dof=\text{none}$) — the linear theory supplies no equation to fix it.

The counterfactual selector control (the able-to-fail). Adding the two selector rows $\{Z_0^{\text{ret}} = K_{0c}, Z_1^{\text{ret}} = K_\eta + 2T_\Omega\}$ raises the constraint rank $3 \rightarrow 5$ (native nullity $8 \rightarrow 6$) and makes $T_0 = T_1 = 1/2$ (constants, zero gradient), so $\text{return_moving_nullity}$: $2 \rightarrow 0$ and the verdict flips to **CROSS_L_RESIDUAL_PREDICTION**. Because base_verdict defaults to the predictive token and returns the FAIL only when the computed return-moving nullity exceeds 0, **the FAIL is earned from the computed nullity, not baked**. $\{Z_0^{\text{ret}} = K_{0c}, Z_1^{\text{ret}} = K_\eta + 2T_\Omega\}$ is a *counterfactual rank-collapse witness* ($\text{derived_from_named_pde}=\text{False}$, $\text{control_only}=\text{True}$), *not* a proven Gate-6 selector; the earned export is the *need* — Gate 6 must supply two independent equations fixing the two return directions (or an equivalent return law).

The A_0/A_1 residuals (forward-built; consuming Stage-022's $\{1, 1/2\}$). The $\ell = 0/1$ return residual amplitudes are the cross- ℓ /port continuation of Stage 009's **RETURN_RESIDUAL_PREDICTION** ($A_{\text{res}} \propto \varepsilon_\ell / (1 + \varepsilon_\ell)$), built *forward* from Stage 022's earned coefficients $v_0 = 1$, $v_1 = 1/2$ and the transmission $(1 - T_\ell)$:

$$A_0 = i v_0 (a\omega/c_s) M_0 (1 - T_0), \quad A_1 = i v_1 (a\omega/c_s)^3 D_1 (1 - T_1), \quad \varepsilon_\ell = Z_\ell^{\text{ret}} / K_\ell,$$

with $1 - T_\ell = \varepsilon_\ell / (1 + \varepsilon_\ell)$ *forward* from the transfer definition — never back-solved from a residual. They are checked against the *independent* pathA_29 form expected $A_0 = i a \omega M_0 \varepsilon_0 / (c_s (1 + \varepsilon_0))$, expected $A_1 = i a^3 \omega^3 D_1 \varepsilon_1 / (2c_s^3 (1 + \varepsilon_1))$, and $A_0 - \text{expected } A_0 = 0$, $A_1 - \text{expected } A_1 = 0$ at orders ω^1, ω^3 . The consumption is checkable: $A_1 - \text{expected } A_1 = i(a^3 \omega^3 / c_s^3) D_1 (\varepsilon_1 / (1 + \varepsilon_1)) (v_1 - 1/2)$, zero iff $v_1 = 1/2$ — the $2c_s^3$ encodes the consumed $1/2$, so corrupting v_1 ($1/2 \rightarrow 1/3$) fires the A_1 form residual. The residual class (form/sign/order) is earned; the magnitude is deferred because the nullspace leaves ε_{eff} free.

The dimensional gate and the anti-back-solve firewall. A recursive (L, M, T) -triple dim_of (raising on a sum-mismatch) certifies, from the sourced dims, $[A_0] = (0, 1, -1)$, $[A_1] = (1, 1, -1)$, $[T_\ell] = [\varepsilon_\ell] = [P_0^{\text{phys}}] = (0, 0, 0)$ with $[P_0^{\text{phys}}] = (c_s/a)^2 [P_0^{\text{raw}}]$; corrupting a sourced dim ($[M_0]$) fires **FAIL_DIMENSIONAL** while the unconstrained free carrier q_{free} gives **NO_FAIL** (the free-carrier-independence control). The **FAIL_TAUTOLOGICAL** firewall classes the ε_{eff} magnitude **deferred_branch_data** while underdetermined (so emitting it as derived is forbidden); a dedicated **emit_epsilon_magnitude_as_derived** mutation (and **assert_not_derive** on the genuinely-023-derived forward T_0/T_1 map) fire **FAIL_TAUTOLOGICAL** — the guard against the v_1 rejection's $\varepsilon_{\text{eff}}/Z$ back-solve.

Honest scope. The FAIL is the earned physics: the genuinely-computed native nullspace (dim 8 / return-moving nullity 2) shows the linear theory cannot pin the $\ell = 0/1$ return; Gate 6's nonlinear closure (sim-deferred) must supply two independent return equations (or an equivalent return law). The raw nullity 8 is partly bookkeeping — two of the three constraints (K_{0c}, K_1) self-pin their own stiffness, so

P_0^{raw} alone gives rank 1 / nullity 10, but the return-moving nullity is still 2; the verdict-bearing quantity is `return_moving_nullity`, the raw nullity a reported diagnostic. The residual *class* (form/sign/order) is earned; the *magnitude* is deferred (ε_{eff} left free), never back-solved.

Consumed closure and register. Two checkable relations: Stage-022's $\{1, 1/2\}$ (corrupting the cited v_1 fires A_1) and the pathA_29 residual form (Stage 009/010, $A_{\text{res}} \propto \varepsilon_\ell / (1 + \varepsilon_\ell)$, from which expected A_0/A_1 is built independently). *Provenance (cited, not re-derived)*: pathA_29's `Z_is_premise`; Stage 008's $R_0 = -M_0 / R_1 = -D_1$ targets and the M_0/D_1 moments; Stage 017's port kernel P_0^{raw} ; Stage 013/017 as context for the effective stiffnesses; and c_s (Stage 005 R1 $c_s^2 = 5K\rho^4/m$) as the units symbol in $z = a\omega/c_s$ (distinct from the frozen-wall c_S , 011–017). Register: **zero** new counted CALIB knobs (set stays 6); $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ add **zero** new free dofs; but K_{0c}/K_1 add *counted FREE-UNREDUCED PENDING* reduction-debt. The return admittances $\{Z_0^{\text{ret}}, Z_1^{\text{ret}}\}$ are coordinate *aliases* of the existing $\varepsilon_0/\varepsilon_1$ FREE-UNREDUCED debt ($\varepsilon_\ell = Z_\ell^{\text{ret}}/K_\ell$ invertible — no double-count, no new dof); the effective stiffnesses K_{0c} and the $\ell = 1$ sector $\{K_\eta, T_\Omega\}$ (via $K_1 = K_\eta + 2T_\Omega$, dims MT^{-2}) are FREE-UNREDUCED PENDING scalar-reduction *counted as reduction debt* (pending debt stays counted until derived), *not* DERIVED, *not* CALIB — their dims do not match registered Stage 013 $K_\eta = T_w\beta^2$ ($ML^{-1}T^{-2}$) or Stage 017 T_Ω ($ML^{-3}T^{-2}$), which Stage 017 records as non-transferable across the volume-vs-line convention (the Stage-016 lesson), so they are not identified with the raw densities (an explicit scalar-reduction to the wall packet would discharge the debt). New structural/obligation edge **R42** (the cross- ℓ nullspace underdetermination departure + the sharpened Gate-6 return-selector obligation; it sharpens the existing $\varepsilon_0/\varepsilon_1$ R24-family debt and adds no free dofs). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}$ (013) + $\{V_{p0}/\ell_c\}$ (015) + $\{T_\Omega, \beta_2\}$ (017) = 6. This stage completes the pathA_34 fold (022^023) and closes the Gate-1–5 gravity ladder.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. The `.w1` is a genuinely independent route, *re-authored* (the source `rankAuditFor` block was structurally parallel to the `.py`'s `augRank-rank0`, so it was discarded per the mirror-policy screen): it proves the nullspace constructively — `Length[NullSpace[J_base]] = 8`, `MatrixRank[basis.GreturnT] = 2`, the explicit Z_ℓ^{ret} unit directions lie in `NullSpace[J_base]`, and (with the selector) `Length[NullSpace[J_sel]] = 6` with the original return gradients giving `Greturn.NullSpace[J_sel]T = 0` — a materially different algorithm from the `.py` rank-subtraction. Agreement is transcript-level (`native_nullspace_dimension=8`, `return_moving_nullity=2`, `baseline_verdict=FAIL_UNDERDETERMINED_NOT_I`, `selector_control_verdict=CROSS_L_RESIDUAL_PREDICTION`, the A_0/A_1 residual-zero, the dim gate, and the probe tokens); SymPy 111 PASS / Mathematica 117 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend on the pathA_34 v1 rejection locus (the sharpest in Part II — v1 was rejected for a rigged zero-padded nullity, flag-driven probes, and a headline-only `.w1`): Codex folded seven blocking findings (the `.w1` decisive re-author; the “selector collapses the return-moving nullity, not the nullity” math fix — native nullity $8 \rightarrow 6$, rank $3 \rightarrow 5$ — with corrected isolated ablation teeth; the counterfactual-witness-not-proven-selector relabel; the provenance-cut fix — $\{1, 1/2\}$ as `cited_earned_input`, `assert_not_derive` rewired to the 023-derived T_0/T_1 map, the `gate4_prefactor` tag dropped; the strengthened firewall — the `emit_epsilon` tooth, the de-counted `rerun_gate_logic`, the fixed `able_to_fail_bad`; the K_{0c}/K_1 PENDING-not-DERIVED register correction (dims $(0, 1, -2) \neq 013/017$ — the Stage-016 convention trap); and the $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ aliases-not-new-dofs register correction), with confirm-passes folding the consistency-sweep gaps. A Grok-4.5 compute-verify returned `DIRECTIVE_CLEAN`, independently confirming (with its own SymPy) the rank/nullity/return-moving, the selector flip, $A_1 - \text{expected } A_1 = 0$ iff $v_1 = 1/2$, the dims, the register conventions, and the `.w1` constructive identity, plus one honest-scope note (the raw nullity 8

includes K_{0c}/K_1 self-constraint bookkeeping; the verdict rides return-moving 2); a closing Codex confirm closed the bookend. Full tri-review on fresh agents (arbiter re-run reproduced the build SymPy 116 / Mathematica 123, exit 0, CWD-independent): **FIDELITY_CLEAN** (an independent read hand-re-derived the rank audit, the selector flip, the A_0/A_1 forward-build + the $v_1 = 1/2$ consumption, and the dims — all faithful, genuine rank not zero-padded, no ε back-solve, a materially-different constructive .w1) and **ADVERSARIAL_CLEAN** (per-tooth ablation, 15 mutations across both engines: hardcoding the rank, zero-padding the Jacobian, and faking the .w1 **NullSpace** basis all make the audit FAIL; the four KEY anti-rig properties confirmed; four non-blocking de-count nits). Codex remediated two make-genuine (the witness-preservation assert recomputes each Jacobian-row dot product from the stored witness vector; the neutralized-mutation meta-test uses a cache-distinct inert context with an independence check) and two honest de-counts (the provenance/premise documentation asserts $\rightarrow$ labeled prints; the T/ε identity $\rightarrow$ a self-consistency check), re-verified **REVERIFY_CLEAN** by the coupling meta-test (each made-genuine tooth fires under a mutation of its object and goes vacuous when neutered; no regression, both engines). Tallies 116/123 $\rightarrow$ 111/117 (net $-5/-6$ per engine from the honest de-counts). Symbolic per-tooth ablation, mutations on copies. With this stage the pathA_34 fold is complete (022^023) and the Gate-1–5 gravity ladder is closed.

B.19 Stage 024: The Density-Native $\ell = 2$ Quadrupole Two-Port Derivation (Check II-P1)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative-port sector, Cluster C; the EARNED-FIRST derivation leg of a four-way split of pathA_43 — the density-native $\ell = 2$ quadrupole two-port derivation component of the joint **DENSITY_PORT_HOSTED**, with the **COMPUTED** vector-freedom taint = Stage 025, the continuity-moment lineage token-check = Stage 026, and the able-to-fail port checks + closure overlay (the **COMPLETING** leg) = Stage 027)

Claim status. EXACT WITHIN CLOSURE

Purpose. Earn the density-native coupling **STRUCTURE** the quadrupole port rides. Show the reduced $\ell = 2$ two-port over (q_2 wall, Φ_2 bulk-density), with static operator $M = \begin{pmatrix} \varpi_{q_2} & -\lambda_c \\ -\lambda_c & \varpi_{\Phi_2} \end{pmatrix}$ and source $g_{\text{base}}(\eta_q, \eta_\phi)$, $g_{\text{base}} = \sqrt{\rho_{\text{eff}}} c_s^2 I_{25} \Xi_Q / a^{7/2}$, yields the load-bearing Φ_2 -response $(M^{-1} \text{source})[\Phi_2] = P_{\text{den}} / \Delta$ and the port numerator $N_{0,\text{den}} = P_{\text{den}}^2 / \Delta^2 = I_{25}^2 \Xi_Q^2 c_s^4 \rho_{\text{eff}} (\eta_\phi \varpi_{q_2} + \eta_q \lambda_c)^2 / (a^7 (\lambda_c^2 - \varpi_{\Phi_2} \varpi_{q_2})^2)$, density-hosted and vector-free. Export $N_{0,\text{den}}$; land the joint verdict as a partial component.

Status. EXACT WITHIN CLOSURE — the derivation is exact symbolic algebra (the 2×2 static-operator inverse and the resulting rational port numerator are exact **Matrix.inv()**/**Solve** algebra; **expect_zero/expect_bool** residuals, no **scipy/numpy/floats/tolerances**). The verdict is the joint **DENSITY_PORT_HOSTED**; this stage carries its *density-native two-port derivation* component (1/4, the EARNED-FIRST slice) and lands the joint as **partial** (024 earned; 025/026/027 pending) — not complete. **DENSITY_TWO_PORT_DERIVED.** The LOCAL audit verdict (exit-0 gate): $N_{0,\text{den}}$ is genuinely built from the inverse/elimination, the factorization cross-check holds, the $\Delta \neq 0$ guard + the coupling-vanishes + the density-only host-set teeth fire, and the two routes agree. It adds **zero** new counted knobs (a **DERIVATION**/structure slice, like Stages 018/016/022); the operator entries $\varpi_{q_2}/\varpi_{\Phi_2}/\lambda_c$, the moment I_{25} , Ξ_Q , η_q/η_ϕ , and ρ_{eff} are **SIM_DEFERRED/GAP** tracked-not-counted.

The density two-port operator. The reduced, mass-normalized static Lagrangian of the $\ell = 2$ sector is a symmetric 2×2 quadratic form over (q_2, Φ_2) (the wall angular amplitude and the bulk-density amplitude), driven by a continuity/interface source:

$$M = \begin{pmatrix} \varpi_{q_2} & -\lambda_c \\ -\lambda_c & \varpi_{\Phi_2} \end{pmatrix}, \quad (g_q, g_\phi) = g_{\text{base}}(\eta_q, \eta_\phi), \quad g_{\text{base}} = \frac{\sqrt{\rho_{\text{eff}}} c_s^2 I_{25} \Xi_Q}{a^{7/2}},$$

with a negative off-diagonal $-\lambda_c$ (the wall $\leftrightarrow$ bulk mixing). The operator entries are abstract symbols whose literal throat values are `SIM_DEFERRED` — the two-port structure holds for any admissible $\varpi_{q_2}, \varpi_{\Phi_2}, \lambda_c$. This is the $\ell = 2$ continuation of the consumed bulk mode (009/010) and wall mode (016/017).

The port numerator (the EARNED headline). The radiative amplitude is the Φ_2 component of the inverted response. With determinant Δ and adjugate-source numerator P_{den} ,

$$\Delta = \varpi_{q_2} \varpi_{\Phi_2} - \lambda_c^2, \quad P_{\text{den}} = \varpi_{q_2} g_\phi + \lambda_c g_q, \quad (M^{-1} \text{source})[\Phi_2] = \frac{P_{\text{den}}}{\Delta},$$

$$N_{0,\text{den}} = \left(\frac{P_{\text{den}}}{\Delta} \right)^2 = \frac{I_{25}^2 \Xi_Q^2 c_s^4 \rho_{\text{eff}} (\eta_\phi \varpi_{q_2} + \eta_q \lambda_c)^2}{a^7 (\lambda_c^2 - \varpi_{\Phi_2} \varpi_{q_2})^2}.$$

The host-set is $\subset \{q_2, \Phi_2, c_s, a, \rho_{\text{eff}}, I_{25}, \Xi_Q, \eta_q, \eta_\phi, \varpi_{q_2}, \varpi_{\Phi_2}, \lambda_c\}$ and contains *none* of the retired vector symbols $\{A_w, F_{\mu w}, J^w, U, W, \Omega_U, \Omega_W, R_{\text{mix}}, g_U, g_W\}$. $N_{0,\text{den}}$ is computed via `Matrix.inv()` (SymPy) / `Solve+eliminate` (WL), not typed.

The load-bearing inverse and the two routes. The two engines derive the same $N_{0,\text{den}}$ by materially different algorithms: SymPy takes the full `static_operator.inv()` and reads response = $(M^{-1} \text{source})[\Phi_2]$; Mathematica keeps a native Green-DtN elimination — solve the wall row for q_2 ($q_2 \rightarrow (\lambda_c \Phi_2 + g_q)/\varpi_{q_2}$), substitute into the bulk row, `Solve` the reduced scalar interface equation, then square. In the current `pathA_43` source the SymPy inverse was *decorative* ($N_{0,\text{den}}$ assembled from a typed $P_{\text{den}}^2/\Delta^2$); the reshape makes it `LOAD-BEARING` — $N_{0,\text{den}} = \text{make_N0}(\text{response})$ with `make\_N0(r) := r^2`, so the dataflow is response $\rightarrow N_{0,\text{den}}$, with P_{den}/Δ retained only as the trace form guarded by the factorization cross-check `compact(response - Pden/Δ) = 0`. Because a typed $P_{\text{den}}^2/\Delta^2$ passes the cross-check and prints an identical transcript, a code-read cannot catch a decorative inverse; the enforcement is the `RUNTIME` dataflow probe `make\_N0(response + δprobe) ≠ make\_N0(response)`, which fires on a response-independent builder (the analog of Stage-023's genuine `Matrix.rank()`). The two-route agreement is the `ARBITER`'s oracle — `simplify(N0,denpy - N0,denwl) = 0`, canonical not raw-string — neither engine reads the other.

Nonsingular guard, coupling boundary, host-set. The inversion is defined only where $\Delta = \varpi_{q_2} \varpi_{\Phi_2} - \lambda_c^2 \neq 0$ (positivity of the entries does not imply it: $\lambda_c^2 = \varpi_{q_2} \varpi_{\Phi_2}$ is admissible), so a named $\Delta \neq 0$ guard runs before the inversion/`Solve` in both engines, with a singular control ($\lambda_c^2 \rightarrow \varpi_{q_2} \varpi_{\Phi_2}$) that fails at that guard; the stronger $\Delta > 0$ stability reading is `SIM_DEFERRED`, not gated. The derivation boundary is the coupling: $g_{\text{base}} \rightarrow 0 \Rightarrow N_{0,\text{den}} = 0$ `COMPUTED` (the zero-control recomputes with $g_{\text{base}} = 0$ and yields exactly 0), while baseline $N_{0,\text{den}} \neq 0$. The density-only host-set is a manifest-construction check ($N_{0,\text{den}}.\text{free_symbols} \subset$ the density set, $\cap$ the vector set = $\emptyset$), not the computed taint graph (that is 025's). Per-tooth ablation: teeth A (inverse factorization + dataflow probe), B ($\Delta \neq 0$ guard), C (coupling-vanishes, two named asserts), D (density-only host-set), E (`.wl` arity + unevaluated-leakage) each fire at their own assert.

Honest scope. Earned structure / SIM_DEFERRED magnitude: 024 derives the reduced coupling STRUCTURE (the two-port form, its host-set, the inverse/elimination), but the literal magnitudes (I_{25} , Ξ_Q , η_q , η_ϕ , the λ_c throat value) are SIM_DEFERRED, and G , 2/5, 54/5 are calibrated ($G = \text{GENUINE_BLOCKED}$) — the 54/5 partition + closure overlay + PN match-back are Stages 027/028, not 024. c_s is the density/sound (ripple) speed — the analog “speed of light” of the gravitational sector (Stage-005 R1) — and a is the CONV pin. ρ_{eff} is the effective reduced-3D mass density ($[ML^{-3}]$) that mass-normalizes (q_2, Φ_2); it is *not* Stage-005’s ρ_0 (a 4D number density $[L^{-4}]$), and the reduction $\rho_{\text{eff}} \leftarrow \{\rho_0, m, \text{geometry}\}$ is SIM_DEFERRED/GAP.

Consumed closure and register. *Provenance only, no cross-stage dual-site:* 024’s genuineness check is intra-stage (the two routes + the load-bearing factorization cross-check). The **physical_relations** trace exhibits $\varpi_{q_2} = K_2/M_2 = (c_s/a)^2 \kappa_q$ (pathA_32 wall; 016/017), $\varpi_{\Phi_2} = (c_s/a)^2 (6 + (ma)^2)$ (pathA_29 bulk, $6 = \ell(\ell + 1)$; 009/010/016), and $\lambda_c = (c_s/a)^2 \hat{\lambda}_Q$ (projected continuity/interface); the COMPUTED taint-set proof that these tags are genuine is 025’s. I_{25} is a typed $\ell = 2$ continuity-moment input (lineage validated in Stage 026, a forward ref). Register: **zero** new counted knobs; $\varpi_{q_2}/\varpi_{\Phi_2}/\lambda_c$, $I_{25}/\Xi_Q/\eta_q/\eta_\phi$, and ρ_{eff} are SIM_DEFERRED/GAP tracked-not-counted; structural edge **R43** (the density-native $\ell = 2$ two-port algebraic derivation provenance, scoped to 024’s algebra — *not* “the vector scaffold retired,” which is the joint result proven by 025/027; discharges nothing). Symbol dims recorded as provenance ($[I_{25}] = L^{5/2}$, $[c_s] = LT^{-1}$, $[\rho_{\text{eff}}] = ML^{-3}$, $[a] = L$, $[\varpi_{q_2}] = [\varpi_{\Phi_2}] = [\lambda_c] = T^{-2}$, $[\eta_q] = [\eta_\phi] = [\Xi_Q] = 1 \Rightarrow [N_{0,\text{den}}] = L^{-1}M$); the units-restored dim CHECK is 027’s. Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$. $N_{0,\text{den}}$ is exported to 025 (taint) / 026 ($I_{25} \in \text{free_symbols}$) / 027 (dimension, a-scaling, sign, closure).

Verification. Dual-engine, independent symbolic routes, zero file I/O — pathA_43 was already contract-clean (no Export/Put/Import; the **results.yaml** was written by an external orchestrator), so the reshape “sever the bridge” step is a no-op (the lightest reshape in Part II); the work was decomposition + the genuineness upgrade + the LOCAL/PARTIAL framing. The .w1 KEEPS its native Green-DtN Solve+eliminate- q_2 route (its own clean/fmt, not a mirror), agreeing at the transcript level ($N_{0,\text{den}}$ canonical factored, the Φ_2 -response = P_{den}/Δ , the $\Delta \neq 0$ guard status, the **physical_relations** provenance, the density-only host-set, the LOCAL verdict, the JOINT PARTIAL), with an arity + Module-result-shape + unevaluated-leakage self-check. SymPy 7 PASS / Mathematica 10 PASS, both exit 0, CWD-independent (repo root + /tmp). The directive design-review used the Codex→Grok→Codex bookend (a Codex xhigh pass folded 4 BLOCKING — the $\Delta \neq 0$ guard + singular control, the $\rho \rightarrow \rho_{\text{eff}}$ mis-provenance correction, the per-tooth independent-oracle restructure, R43 scoped to 024’s algebra; a confirm-pass folded 2 further BLOCKING — tooth A’s runtime dataflow probe and tooth C’s two named asserts; both engines compute-verified $N_{0,\text{den}}$, response $\equiv P_{\text{den}}/\Delta$, $[N_{0,\text{den}}] = L^{-1}M$, and the a-powers $-3 \rightarrow -5$). Full tri-review on fresh agents: FIDELITY_CLEAN (an independent read re-derived $N_{0,\text{den}}$ by a third linsolve route, diff = 0) and ADVERSARIAL_CLEAN (per-tooth ablation: every tooth A–E fired at its own named assert, the decorative-inverse rig caught at runtime by the δ_{probe} in both engines; no vacuous tooth) — zero remediation required, the contract-clean lightest-reshape stage. Symbolic per-tooth ablation, mutations on copies.

B.20 Stage 025: The Density-Port Vector-Freedom Taint Proof (Check II-P2)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative-port sector, Cluster C; the ANTI-RIG PROOF leg of a four-way split of pathA_43 — the vector-freedom taint proof component of the joint

DENSITY_PORT_HOSTED, with the density two-port DERIVATION = Stage 024, the continuity-moment lineage token-check = Stage 026, and the able-to-fail port checks + closure overlay (the COMPLETING leg) = Stage 027)

Claim status. EXACT WITHIN CLOSURE

Purpose. Prove Stage-024’s exported $\ell = 2$ density-port numerator $N_{0,\text{den}}$ is computationally VECTOR-FREE — that it cannot hide a disguised copy of the retired EM vector port $P_A = \Omega_{U}^2 g_W + R_{\text{mix}} g_U$. Compute the provenance taint over $N_{0,\text{den}}.\text{free_symbols}$ via a fixed tag map and show it equals exactly the density ancestry $\{\text{continuity_interface}, \text{pathA_29_bulk}, \text{pathA_32_wall}\}$ with **vector_port** $\notin$ it, source-map complete, and no vector-host symbol. The decisive check is the COMPUTED taint over provenance TAGS — never a name-check — which catches density-LOOKING relabels a name-check misses. Export the vector-freedom verdict; land the joint as a partial component.

Status. EXACT WITHIN CLOSURE — exact symbolic set-algebra (the taint is a union of provenance tags over the actual $N_{0,\text{den}}.\text{free_symbols}$; **expect_bool** residuals, no **scipy/numpy/floats/tolerances**). The verdict is the joint DENSITY_PORT_HOSTED; this stage carries its *vector-freedom taint proof* component (2/4) and lands the joint as **partial** (024/025 earned; 026/027 pending) — not complete. **DENSITY_PORT_VECTOR_FREE.** The LOCAL audit verdict (exit-0 gate, CONDITIONAL on the cited **moment_valid** = True that Stage 026 earns): the computed taint is exactly the three density tags (P1), and source-map completeness $\wedge$ no vector-host $\wedge$ **vector_port** $\notin$ taint (P2). It adds **zero** new counted knobs (a proof/taint slice, like Stages 018/016/022); the vector/relabel/free-carrier symbols are control fixtures, tracked-not-counted.

The subject and the consumption oracle. Stage 025 CITES Stage-024’s factored export (it does not re-derive the inverse — that would blur the 024/025 cut):

$$N_{0,\text{den}} = \frac{I_{25}^2 \Xi_Q^2 c_s^4 \rho_{\text{eff}} (\eta_\phi \varpi_{q_2} + \eta_q \lambda_c)^2}{a^7 (\lambda_c^2 - \varpi_{\Phi_2} \varpi_{q_2})^2}.$$

The consumption-integrity oracle computes $N_{0,\text{den}}.\text{free_symbols}$ off the actual cited expression and asserts it equals the exported 10-symbol host contract $\{a, c_s, \rho_{\text{eff}}, I_{25}, \Xi_Q, \eta_q, \eta_\phi, \varpi_{q_2}, \varpi_{\Phi_2}, \lambda_c\}$ (the coordinate hosts q_2, Φ_2 are projected out of the numerator — they extend the *allowable* density-host universe to 12 but are not in $N_{0,\text{den}}.\text{free_symbols}$). A dropped/renamed symbol breaks the contract (tooth I).

The computed taint and the two separated predicates. Each symbol carries a fixed provenance tag set (faithful to the pathA_43 source with the 024 renames): $c_s, \rho_{\text{eff}}, \varpi_{\Phi_2} \rightarrow \{\text{pathA_29_bulk}\}$; $\varpi_{q_2} \rightarrow \{\text{pathA_32_wall}\}$; $\Xi_Q, \eta_q, \eta_\phi \rightarrow \{\text{continuity_interface}\}$; $a \rightarrow \{\text{pathA_29_bulk}, \text{pathA_32_wall}\}$; $\lambda_c \rightarrow$ all three; $I_{25} \rightarrow \{\text{continuity_interface}, \text{pathA_32_wall}\}$ (installed from **moment_valid**); the vector, relabel, and σ_{hidden} symbols $\rightarrow \{\text{vector_port}\}$; **free_carrier** $\rightarrow \emptyset$. The taint is COMPUTED — the union of **tag_map[s]** over $s \in N_{0,\text{den}}.\text{free_symbols}$ — not typed. Two predicates are separated: **P1 baseline_ancestry_ok** (the genuine numerator’s taint equals *exactly* the three density tags, baseline only) and **P2 vector_free** (**source_map_complete** $\wedge$ **vector_host_symbols** = $\emptyset \wedge$ **vector_port** $\notin$ taint). $N_{0,\text{den}}$ satisfies both; **LOCAL** = **P1** $\wedge$ **P2** $\wedge$ **moment_valid**.

Why the taint is over TAGS (the two caught rigs), and the de-counted witness. The pathA_43 history holds two rigs a name-check would pass by fiat: (Rig 1) a density-LOOKING two-port

built from $\{\omega_{\text{wall}}, \omega_{\rho}, r_{\text{mix}}, g_{\rho}, g_{q,\text{old}}\}$ — symbols not in `VECTOR_SYMBOLS` but tagged `vector_port`; and the hidden-vector rig $N_{0,\text{den}} \cdot \sigma_{\text{hidden}}$ (σ_{hidden} tagged `vector_port`, not in the name set). A `free_symbols` $\cap$ `VECTOR_SYMBOLS` name-check passes both; the `COMPUTED` taint catches them (`vector_port` $\in$ taint). The expression-level ablation (substitute vector-tainted symbols $\rightarrow 0$, compare) is `RETAINED` as a printed witness but is *de-counted*: once P2 holds the ablation set is empty so it cannot independently fail (a subsumed guard; also `nan` on the singular P^2/Δ^2 vector-port fixtures). The decisive gate is the taint-set (P1/P2) + `source_map_complete`. (The self-asserted-continuity_interface-tag rig, “Rig 2,” is Stage-026’s lineage token-check; 025 cites `moment_valid`.)

The able-to-fail battery (per-tooth ablation, coupling meta-test). Each rig fires at its OWN named assert and goes vacuous when neutered: A relabel and B hidden-vector via the *computed-taint* assert (a name-check misses them — the anti-fiat core); C vector-injection and G raw-vector-port via the *vector-host* assert; D provenance-less and E missing-symbol via the two halves (empty / missing) of *source-map completeness*; F tagged-carrier PASSES P2 with a legitimate four-tag taint (not the exact-three-tag P1) and FLIPS to fail when its tag is neutered (the reversibility control); I subject-integrity via the exact host-contract assert; and the two `.wl` runtime scanners (a call-arity mismatch, a leaked unevaluated authored-helper head). The adversarial re-review proved the decisive property load-bearing: downgrading the machinery to a name-check fires the relabel/hidden asserts in both engines.

Honest scope, consumed, register. 025 proves the vector-freedom `STRUCTURE`, `CONDITIONAL` on Stage 026 earning `moment_valid`; the EM $A_w/U, W$ scaffold `RETIREMENT` is the joint 025 $\wedge$ 026 $\wedge$ 027 result, recorded at Stage 027 (if 027 fails dimension/scaling/sign/closure, the port has not displaced the scaffold). Magnitudes ($\Xi_Q, \lambda_c, \rho_{\text{eff}}$ throat values; $G, 2/5, 54/5$) are `SIM_DEFERRED/CALIBRATED` downstream. *Consumed*: Stage-024’s $N_{0,\text{den}}$ (the subject; the checkable host contract) + Stage-026’s `moment_valid` (typed forward ref) + the density symbols’ provenance tags (024’s `physical_relations` + Stage-005 $c_s + a$ `CONV`). Register: **zero** new counted knobs; new structural edge **R44** — the density-port vector-freedom `PROOF` (the computed taint = the density ancestry, source-map complete, no vector host), recording only the vector-freedom *conjunct* (the retirement is the joint 025/027 result, 027 records it); discharges nothing, like R43/R41/R39. No new dims (the $[N_{0,\text{den}}] = L^{-1}M$ check is 027’s). Part-II counted `CALIB` set unchanged at $\{\mu_{\eta}, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_{\Omega}, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — pathA_43 was already contract-clean, so the reshape “sever the bridge” step is a no-op; the work was decomposition + the `.wl` `RE-AUTHOR` + the `LOCAL/PARTIAL` framing. *Unlike Stage 024 (which kept its .wl native for the derivation lane), Stage 025 RE-AUTHORS its .wl*: the source `.wl` taint machinery was a transliteration of the `.py`, so the mirror policy required a genuinely independent computation — the decisive taint + source-map completeness use a native directed `Graph` of `DirectedEdge`["sym:" $\rightarrow$ "tag:"] + `VertexOutComponent` reachability (not `Variables+Lookup+Union`), and the ablation witness uses $D[\text{expr}, v]$ partial-derivative independence. Transcript-level agreement, neither engine reading the other. SymPy 18 PASS / Mathematica 18 PASS, both exit 0, CWD-independent (repo root + `/tmp`). The directive review used the Codex $\rightarrow$ Grok $\rightarrow$ Codex bookend (a Codex xhigh pass folded 5 `BLOCKING` — most importantly that the expression ablation is logically subsumed by the taint-set gate, reframed to a de-counted witness with the computed taint-set identity + source-map completeness the decisive gate; the two-predicate P1/P2 split; R44 recording only the vector-freedom conjunct; the cite-the-factored consumption; the tooth-H split into a review-acceptance and two runtime scanners — then Grok-4.5 compute-verified the subsumption, the relabel/hidden taint vs the name-check, and the reversibility control `DIRECTIVE_CLEAN`, a Codex confirm-pass folded 4 consistency-sweep gaps, and a Codex final re-

confirm returned `DIRECTIVE_CLEAN`). Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read re-derived $N_{0,\text{den}}$ from Stage-024's 2×2 inverse, $\text{diff} = 0$; confirmed the tag map, the P1/P2 separation, the de-counted witness, the genuine independent `.wl`) and `ADVERSARIAL_CLEAN` (per-tooth ablation on both engines; the name-check-dodge test proved the tag-taint load-bearing) — two [**minor**] non-blocking checks that could not independently fail (a subsumed universe-membership assert; a masked `moment_valid` assert) were de-counted to labeled diagnostic prints (tally $20 \rightarrow 18$ per engine, no other check touched), then fresh-agent `REVERIFY_CLEAN`. Symbolic per-tooth ablation, mutations on copies.

Appendix C

Part III Stage Cards

C.1 Stage 003: Transverse Photon Pair and the Stray Longitudinal Mode

Part / anchor. Part III – Light

Claim status. OPEN

Purpose. Test whether the shear-surface brane carries light of the Maxwell kind. The transverse question earns a clean result – two massless photons at $c_\gamma^2 = \mu_R/\rho_{br}$; the longitudinal question earns a characterized departure – one stray second-class mode where Maxwell has a gauge-removed zero, with the tuned Maxwell locus reachable only by tuning.

Status. OPEN. FAIL-headline stage: the top-line verdict is a **FAIL_***, but the earned content is the headline and the FAIL is the first-class characterized landing (no softening). *Earned headline:* the brane carries *light* – two massless transverse photons, with (for the first time in the program) $c_\gamma^2 = \mu_R/\rho_{br}$; verdict **PASS_TRANSVERSE_UNDISTURBED**. *Top-line departure:* **FAIL_CAUCHY_STRAY_LONGITUDINAL** – on the provenance-fixed single-medium branch the $(\nabla \cdot u) + \theta$ sector carries one stray propagating longitudinal DOF, a Dirac–Bergmann *second-class* pair rather than a Maxwell first-class Gauss gauge-removal. *Reachability:* the tuned Maxwell locus ($K_\theta = C_J^2/\rho_{br}$, $B_{\text{eff}} = 0$, $m_\theta^2 = 0$) is reachable and emits **C5_RESOLVED_MAXWELL_BY_TUNING** – so the FAIL is not hardcoded – but **BY_TUNING**, not **WITH_PROVENANCE**.

Derivation. The primitive brane Lagrangian (no pre-completed square used as input) is

$$L = \frac{1}{2}\rho_{br}(\partial_t u)^2 - \frac{1}{2}\mu_R(\nabla \times u)^2 - \frac{1}{2}B(\nabla \cdot u)^2 + J(\partial_t \theta) \delta \rho_B + \frac{1}{2}K_\theta(\nabla \theta)^2.$$

Transverse (earned). For a polarization $\perp \mathbf{k}$, $\nabla \cdot u = 0$ and the θ couplings vanish, so $L_T = \frac{1}{2}\rho_{br}(\partial_t u_T)^2 - \frac{1}{2}\mu_R k^2 u_T^2$, giving

$$\omega^2 = \frac{\mu_R}{\rho_{br}} k^2, \quad c_\gamma^2 = \frac{\mu_R}{\rho_{br}},$$

two polarizations $\Rightarrow$ `physical_dof` = 2, massless. The speed is able-to-fail: $\epsilon = 2\rho_{br}$ shifts it to $\mu_R/(2\rho_{br})$ (**FAIL_TRANSVERSE_DISTURBED**). **Longitudinal (departure).** Number conservation slaves $\delta \rho_B = -\rho_{B0} \nabla \cdot u$; integration by parts gives the Maxwell-form cross-term with the *derived* sign $C_J = -J\rho_{B0}$, and the finite conjugate-density stiffness the Cauchy modulus $B_{\text{eff}} = \rho_{B0}^2/\chi_c$. Since $\partial_t \theta$ enters

only through the first-order cross-term, the momenta are $p_u = \rho_{br} \partial_t u_L$, $\pi_\theta = Jk\rho_{B0}u_L$, forcing the primary constraint $\Phi_1 = \pi_\theta - Jk\rho_{B0}u_L$; preservation gives $\Phi_2 = -k(Jp_u\rho_{B0} + k\kappa_{\text{phase}}\rho_{br}\theta)/\rho_{br}$, and

$$\{\Phi_1, \Phi_2\} = \frac{k^2(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})}{\rho_{br}} \neq 0$$

$\Rightarrow$ second-class pair (0 first-class, 2 second-class), $N_{\text{phys}} = (4 - 2)/2 = 1$. The finite-stiffness pole $\omega^2 = k^2\kappa_{\text{phase}}\rho_{B0}^2/[\chi_c(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})]$ has positive residue and a bounded reduced Hamiltonian: a real stray longitudinal mode, not a ghost and not a gauge-removal.

Physical use, fidelity, and the departure. The brane carries light: two transverse photons at $c_\gamma^2 = \mu_R/\rho_{br}$ from the medium's own moduli (light = in-plane brane shear). Honest scope: the order-parameter phase θ does *not*, on its frozen definitions, supply a gauge-removed Maxwell scalar – it leaves one stray second-class longitudinal mode; the obstruction is pinned to the Cauchy modulus $B_{\text{eff}} = \rho_{B0}^2/\chi_c$ plus the conventional phase-gradient sign $K_\theta = -\kappa_{\text{phase}} < 0$ (vs the electric sign $K_\theta = C_J^2/\rho_{br} > 0$). Fidelity: the decisive sign $C_J = -J\rho_{B0}$ is *derived* here by an in-script integration by parts (the earned source had asserted it), closing that stage's known **FIDELITY_ISSUES**; the conventional K_θ sign is a labeled postulate whose verdict-dependence is exercised by the reachable Maxwell-locus control. Resolution of the stray mode is calibrated / sim-deferred; μ_R, ρ_{br} are calibrated inputs.

Appendix D

Reproducibility Map

D.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's reproducibility rules.

D.2 Reproducibility Layers

Layer	What it controls	Minimum evidence required
No reproducibility layers populated yet.		

D.3 Canonical Source Registry

Placeholder for source artifacts.

D.4 Primary Symbolic-Audit Registry

Placeholder for symbolic audit artifacts.

D.5 Repository Freeze Protocol

Placeholder for future release-freeze requirements.

Appendix E

Source File Index

E.1 Purpose of This Appendix

Placeholder for the rebuilt ledger’s source-file index.

E.2 Source Classes

Placeholder for source classes.

E.3 Generated Ledger Files

File family	Role
<code>paper/frontmatter/*.tex</code>	Placeholder governance chapters.
<code>paper/parts/*.tex</code>	Future theorem-block chapters.
<code>paper/stages/stage_NNN.tex</code>	Future per-stage cards.
<code>scripts/ledger_stageNNN*.py</code>	Future SymPy audit scripts.
<code>mathematica/ledger_stageNNN*.wl</code>	Future Mathematica audit scripts.

E.4 Precedence and Conflict-Resolution Rules

Placeholder for source precedence and conflict-resolution rules.

Appendix F

Layer-2 Fit-Insertion Index

F.1 Generated Coverage Summary

Item	Count
Phase-A candidates	0
Audited candidates	0

F.2 Phase-A Candidate Rows

No candidate rows exist in the empty ledger skeleton.

F.3 Visible Grouping and Exclusion Accounting

No grouping or exclusion accounting exists yet.

Appendix G

Layer-2 Parameter-Provenance Audit

G.1 Generated Coverage Summary

Item	Count
Top-level provenance parameter records	0
Canonical candidates	0

G.2 Published-Target Benchmark Sources

No benchmark rows exist in the empty ledger skeleton.

G.3 Parameter Genealogy Rows

No parameter genealogy rows exist yet.

G.4 Visible Alias Accounting

No alias accounting exists yet.

Appendix H

Fill Workflow and Editorial Rules

H.1 Purpose of This Appendix

Placeholder for the rebuilt ledger fill workflow.

H.2 What Filled Means

Placeholder for completion criteria.

H.3 Governing Maps Used During Filling

Placeholder for status, notation, anchor, dependency, source, and reproducibility controls.

H.4 Stage Appendix Fill Protocol

Placeholder for future stage-card authoring rules.

H.5 Theorem-Block Fill Protocol

Placeholder for future Part authoring rules.

H.6 Minimal Build and Static-Check Workflow

1. Confirm every retained `\input` target exists.
2. Run the SymPy and Mathematica audit runners.
3. Build `paper/pde_ledger.tex`.